Hippocampal patterns and associative memory: Distinct intracranial EEG temporal encoding patterns support memory

Alice M. Xue (alicexue@stanford.edu), Shaw Hsu, Karen F. LaRocque,
Omri Raccah, Alex Gonzalez, Josef Parvizi, & Anthony D. Wagner

Contents

Behavior **1**

dprime figure . . . 3
Fig 1 . . . 6
Fig S2 . . . 7
Fig S3 . . . 9
Fig S4 . . . 19
Fig S5 . . . 24

HFB **25**

Study . . . 25
Test cue . . . 44
Test response . . . 59
Fig 2 . . . 73
Fig S6 . . . 75
Fig S7 . . . 76
Pattern similarity . . . 77
~ Fig 3A . . . 77
~ Fig 3B . . . 80
Fig 3 . . . 83

Interaction: TPS x memory outcome **83**

Exclude S5 **85**

~ Fig 3A . . . 85
~ Fig 3B . . . 87
Interaction: TPS x memory outcome . . . 89

Exclude S2 and S5 **90**

~ Fig 3A . . . 90

~ Fig 3B . . . 92

Interaction: TPS x memory outcome . . . 94

Fig S8 . . . 96

Fig S9 . . . 99

Fig 11 . . . 99

~ Fig S11A . . . 100

~ Fig S11B . . . 100

Fig S11 . . . 100

Alternative memory coding **100**

HFB . . . 100

Fig S10 . . . 133

Pattern similarity . . . 135

~ Fig S11A . . . 135

~ Fig S11B . . . 138

Fig S11 . . . 140

Interaction: TPS x memory outcome . . . 141

~ Fig S12A . . . 143

~ Fig S12B . . . 145

Fig S12 . . . 147

Interaction: TPS x memory outcome . . . 148

Behavior

Recognition dprime

d' = Z(total hit rate) – Z(false alarm rate), where the total hit rate was calculated as the sum of the number of item hits and associative hits divided by the total number of studied items and the false alarm rate was calculated as the number of false alarms divided by the total number of lure items

subj	d.prime	assoc.hit.rate	assoc.error.rate
S1	1.789	0.231	0.014
S2	1.564	0.088	0.000
S3	2.591	0.358	0.000
S4	2.267	0.328	0.023
S5	2.143	0.083	0.000
S6	2.538	0.283	0.023
S7	2.666	0.171	0.000

perf_type	mean.perf	sd.perf
d.prime	2.223	0.421
assoc.hit.rate	0.220	0.111
assoc.error.rate	0.009	0.011

```
##
##  One Sample t-test
##
## data:  dprime.data$d.prime
## t = 13.98, df = 6, p-value = 8.351e-06
## alternative hypothesis: true mean is not equal to 0
## 95 percent confidence interval:
##  1.833632 2.611677
## sample estimates:
## mean of x
##  2.222654
```

```
## [1] "adapt_delta: 0.8"
```

```
## [1] "max_treedepth: 10"
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: d.prime ~ 1
##    Data: dprime.data (Number of observations: 7)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Regression Coefficients:
##           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept     2.22      0.21     1.80     2.64 1.00     1898     1580
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.53      0.20     0.29     1.03 1.00     1582     1819
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

effect	component	group	term	estimate	std.error	conf.low	conf.high
fixed	cond	NA	(Intercept)	2.220112	0.2065002	1.7962076	2.641273
ran_pars	cond	Residual	sd___Observation	0.534625	0.1961034	0.2886927	1.028564

Table 4: posterior mean [95% HDI]

1	emmean	lower.HPD	upper.HPD

overall	2.22	1.813	2.651

```
## [1] "adapt_delta: 0.8"
```

```
## [1] "max_treedepth: 10"
```

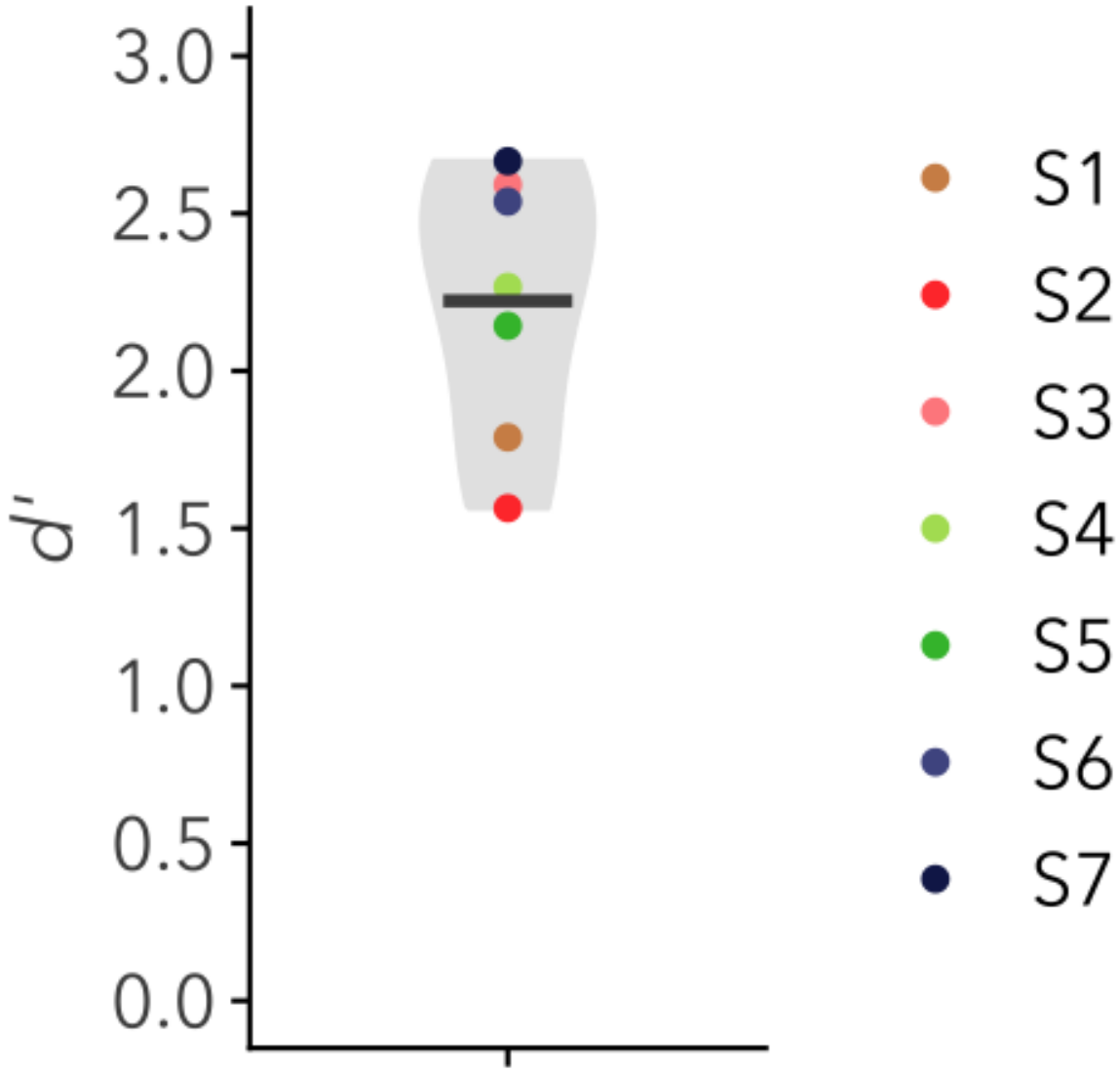
```
##  Family: gaussian
##   Links: mu = identity
## Formula: assoc.hit.rate ~ 1
##    Data: dprime.data (Number of observations: 7)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Regression Coefficients:
##           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept     0.22      0.06     0.11     0.33 1.00     1859     1414
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.14      0.06     0.07     0.30 1.00     1489     1599
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

effect	component	group	term	estimate	std.error	conf.low	conf.high
fixed	cond	NA	(Intercept)	0.2181936	0.0557730	0.1051450	0.3290048
ran_pars	cond	Residual	sd__Observation	0.1437130	0.0576621	0.0747866	0.2982737

Table 6: posterior mean [95% HDI]

1	emmean	lower.HPD	upper.HPD
overall	0.218	0.104	0.328

dprime figure




```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: d.prime ~ 1 + runNum + (1 | subj)
##    Data: dprime.block.data (Number of observations: 81)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
```

```
## Multilevel Hyperparameters:
## ~subj (Number of levels: 7)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.31      0.18     0.05     0.76 1.01      637      789
##
## Regression Coefficients:
##           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept     1.96      0.18     1.60     2.31 1.00     1843     2228
## runNum        0.01      0.02    -0.02     0.05 1.00     3209     2414
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.59      0.05     0.50     0.69 1.00     2548     2850
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).

## [1] "adapt_delta: 0.99"

## [1] "max_treedepth: 10"

## d.prime ~ 1 + runNum + (1 | subj)

## d.prime ~ 1 + (1 | subj)

##            elpd_diff se_diff elpd_loo se_elpd_loo p_loo se_p_loo looic se_looic
## mod.simple   0.0       0.0   -73.4      5.4         6.3   0.8    146.8  10.8
## mod1        -1.3       0.8   -74.7      5.5         7.3   1.0    149.5  11.0

## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'mod.simple' is the best model (ELPD =
## -73.40), followed by 'mod1' (diff-ELPD = -1.33 +- 0.84, p = 0.112)

## [1] "adapt_delta: 0.8"

## [1] "max_treedepth: 10"

##  Family: gaussian
##   Links: mu = identity
## Formula: assoc.hit.rate ~ 1 + runNum + (1 | subj)
##    Data: dprime.block.data (Number of observations: 81)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 7)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.13      0.06     0.06     0.29 1.01      782     1456
##
## Regression Coefficients:
##           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
```

```
## Intercept     0.21      0.06     0.08     0.33 1.00     1018     1128
## runNum        0.00      0.00    -0.00     0.01 1.00     4060     3018
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.13      0.01     0.11     0.15 1.00     2649     2215
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
## [1] "adapt_delta: 0.8"
```

```
## [1] "max_treedepth: 10"
```

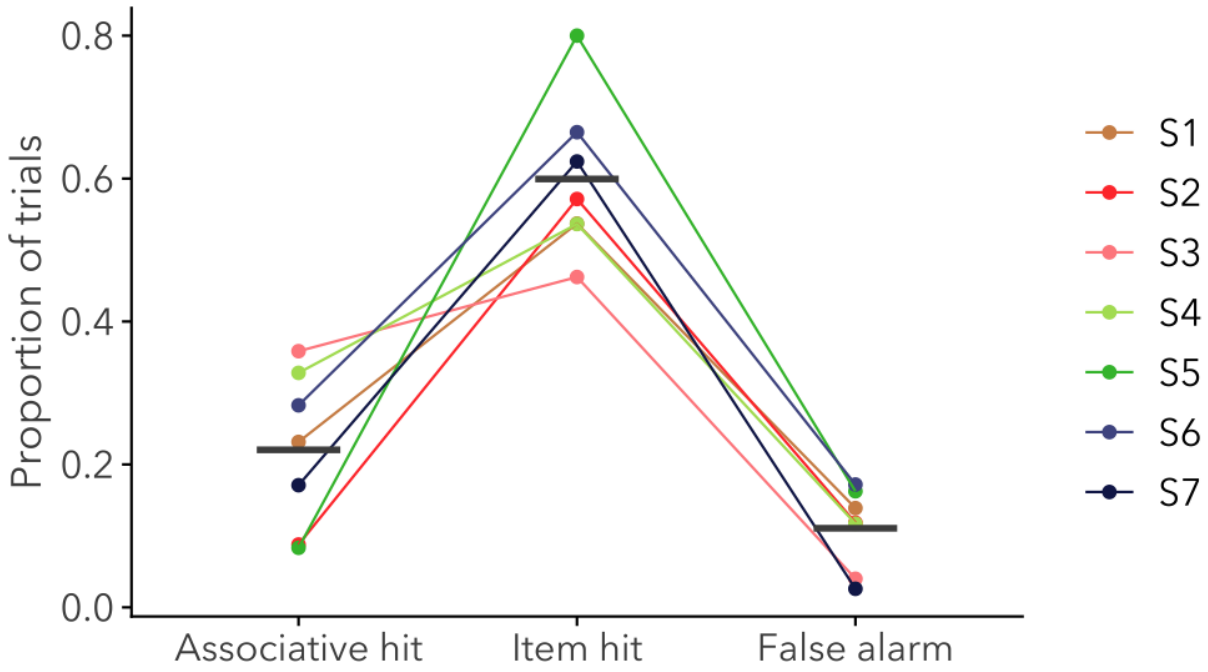
```
## assoc.hit.rate ~ 1 + runNum + (1 | subj)
```

```
## assoc.hit.rate ~ 1 + (1 | subj)
```

```
##            elpd_diff se_diff elpd_loo se_elpd_loo p_loo se_p_loo looic se_looic
## mod.simple   0.0       0.0    49.2      5.9         7.0   1.0    -98.4  11.8
## mod1        -0.9       0.8    48.4      5.9         7.9   1.1    -96.7  11.7
```

```
## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'mod.simple' is the best model (ELPD =
## 49.22), followed by 'mod1' (diff-ELPD = -0.86 +- 0.82, p = 0.291)
```

Fig 1



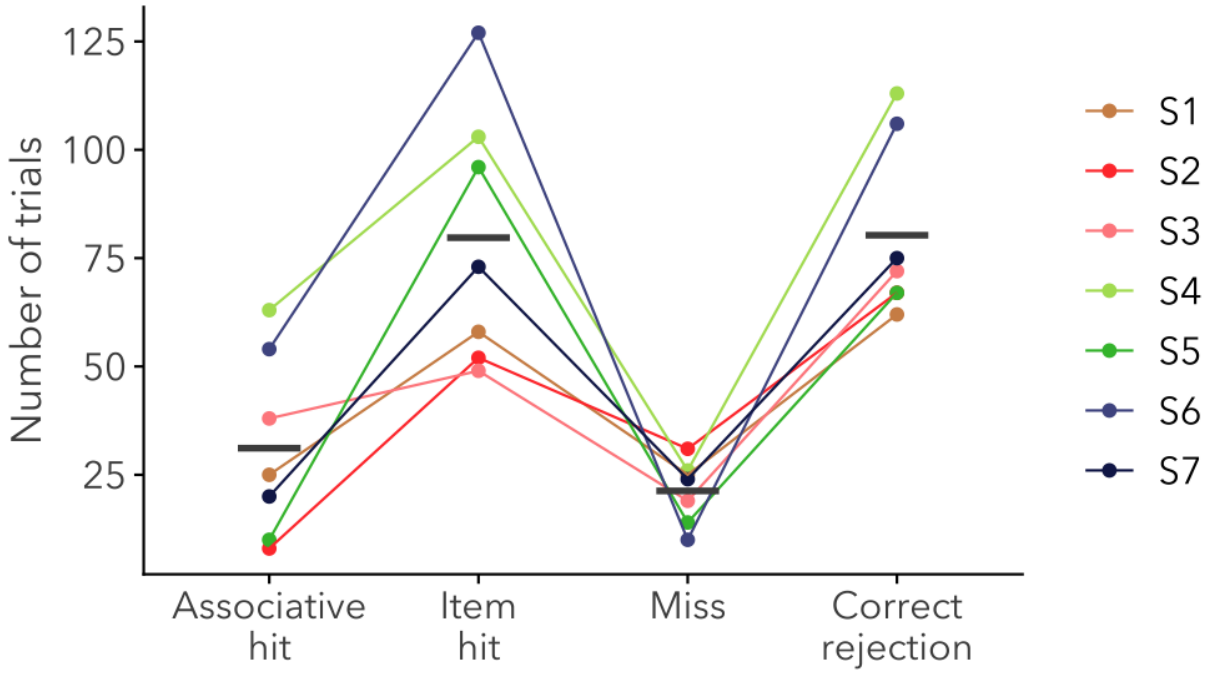
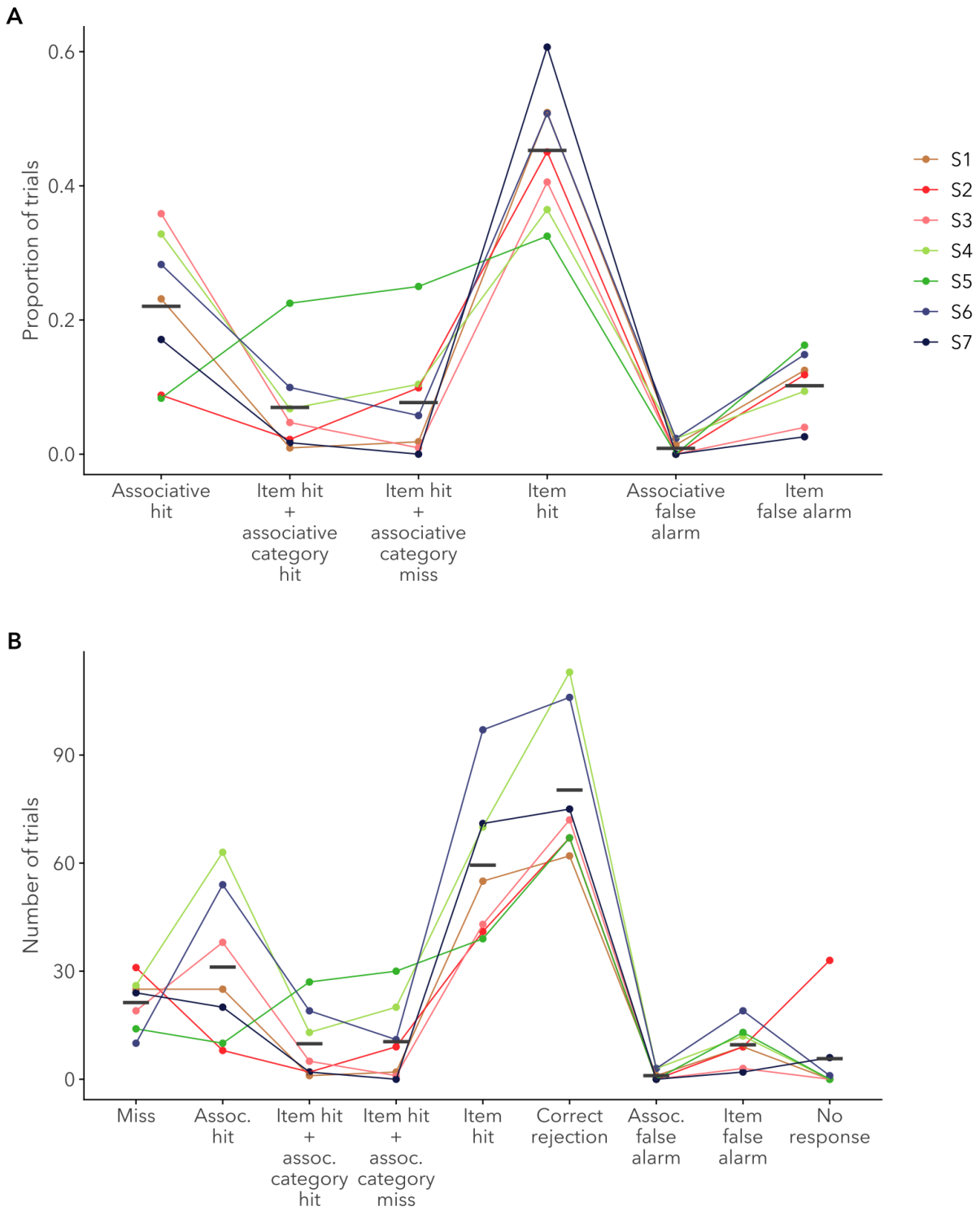
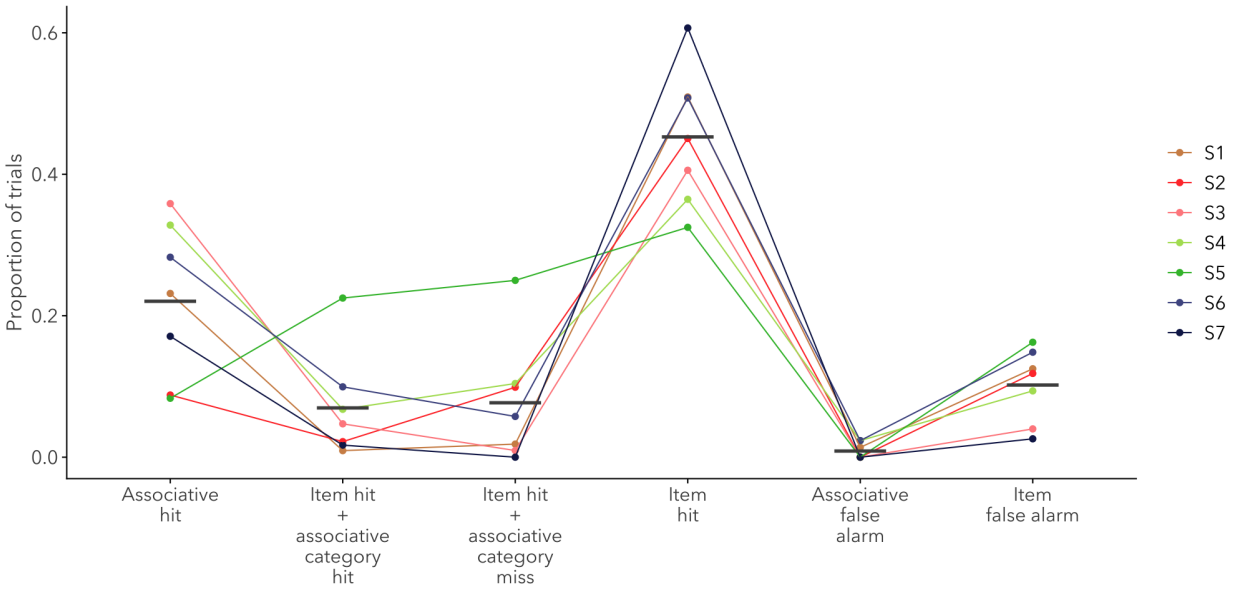

Number of trials
125
100
75
50
25
Associative hit
Item hit
Miss
Correct rejection
S1
S2
S3
S4
S5
S6
S7

Fig S2


A
Proportion of trials
0.6
0.4
0.2
0.0
Associative hit
Item hit + associative category hit
Item hit + associative category miss
Item hit
Associative false alarm
Item false alarm
S1
S2
S3
S4
S5
S6
S7
B
Number of trials
90
60
30
0
Miss
Assoc. hit
Item hit + assoc. category hit
Item hit + assoc. category miss
Item hit
Correct rejection
Assoc. false alarm
Item false alarm
No response


Proportion of trials
0.6
0.4
0.2
0.0
Associative hit
Item hit + associative category hit
Item hit + associative category miss
Item hit
Associative false alarm
Item false alarm
S1
S2
S3
S4
S5
S6
S7

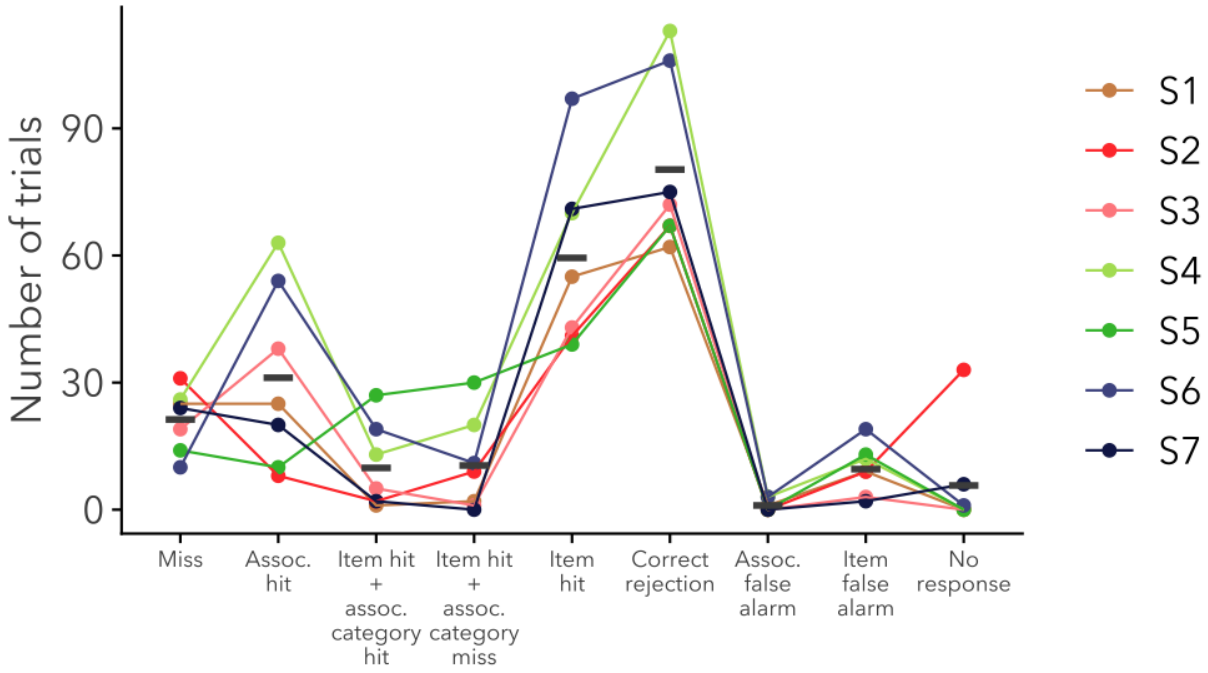
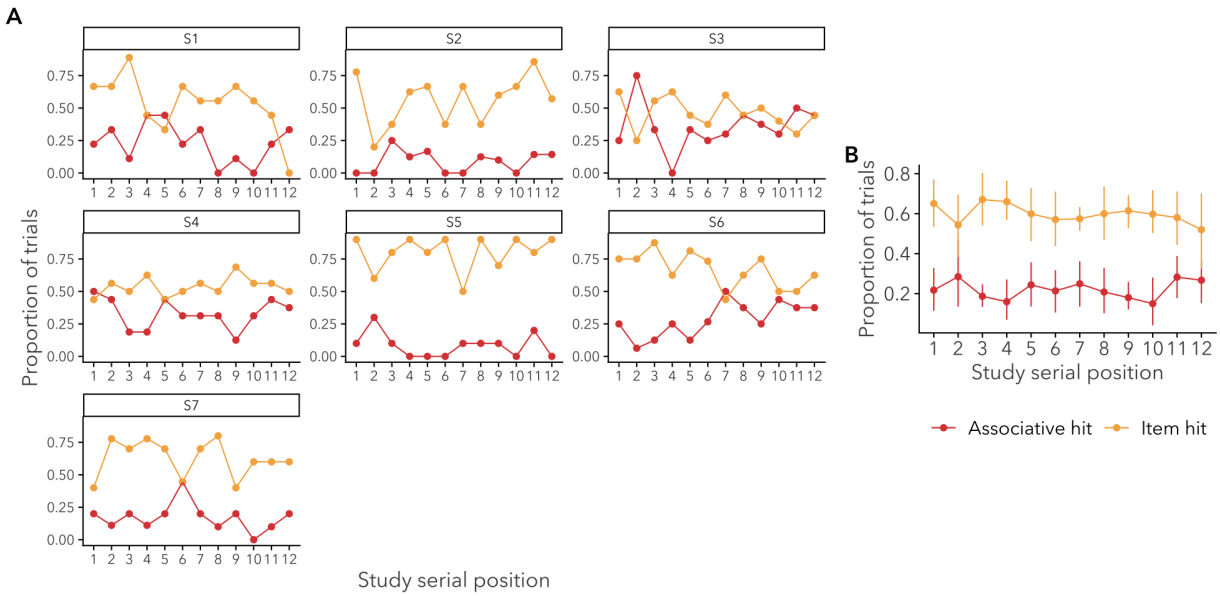

Number of trials
90
60
30
0
Miss
Assoc. hit
Item hit + assoc. category hit
Item hit + assoc. category miss
Item hit
Correct rejection
Assoc. false alarm
Item false alarm
No response
S1
S2
S3
S4
S5
S6
S7

Fig S3



Recognition

```
## [1] "adapt_delta: 0.8"

## [1] "max_treedepth: 10"

##  Family: bernoulli
##   Links: mu = logit
## Formula: is_miss ~ 1 + studyNum + (1 | subj)
##    Data: mod.data (Number of observations: 925)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 7)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.84      0.35     0.40     1.74 1.01      772     1060
##
## Regression Coefficients:
##            Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept     -2.08      0.52    -3.15    -1.09 1.00      727      996
## studyNum2      0.16      0.55    -0.89     1.27 1.00     1053     1428
## studyNum3      0.27      0.51    -0.67     1.30 1.00      962     1423
## studyNum4      0.52      0.51    -0.40     1.56 1.00      976     1226
## studyNum5      0.34      0.53    -0.64     1.42 1.00     1007     1357
## studyNum6      0.62      0.50    -0.33     1.68 1.01      967     1386
## studyNum7      0.43      0.51    -0.55     1.48 1.00     1023     1348
## studyNum8      0.48      0.51    -0.49     1.52 1.00     1010     1368
## studyNum9      0.62      0.50    -0.31     1.66 1.00      953     1153
## studyNum10     0.87      0.48    -0.02     1.88 1.00      904     1093
## studyNum11     0.16      0.53    -0.86     1.25 1.00      982     1160
```

```
## studyNum12     0.61      0.50    -0.35     1.63 1.01      941     1205
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
## is_miss ~ 1 + studyNum + (1 | subj)
```

```
## is_miss ~ 1 + (1 | subj)
```

```
##            elpd_diff se_diff elpd_loo se_elpd_loo p_loo  se_p_loo looic
## mod.simple    0.0       0.0  -391.5     18.4         6.5    0.4    783.0
## mod1         -8.8       2.8  -400.3     19.2        18.4    1.2    800.6
##            se_looic
## mod.simple   36.8
## mod1         38.3
```

```
## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'mod.simple' is the best model (ELPD =
## -391.52), followed by 'mod1' (diff-ELPD = -8.77 +- 2.76, p = 0.002)
```

Cued Recall

```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
##  Family: bernoulli
##   Links: mu = logit
## Formula: was_recalled ~ 1 + studyNum + (1 | subj)
##    Data: mod.data (Number of observations: 925)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 7)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.88      0.36     0.43     1.76 1.00      860     1401
##
## Regression Coefficients:
##            Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept     -1.31      0.46    -2.22    -0.39 1.00      906     1320
## studyNum2      0.22      0.38    -0.51     0.99 1.00     1194     1475
## studyNum3     -0.40      0.41    -1.21     0.38 1.00     1265     1788
## studyNum4     -0.46      0.42    -1.29     0.32 1.00     1229     2011
## studyNum5      0.02      0.39    -0.74     0.78 1.00     1185     1881
## studyNum6     -0.09      0.40    -0.86     0.70 1.00     1239     1926
## studyNum7      0.22      0.38    -0.52     0.97 1.00     1117     1701
## studyNum8     -0.06      0.39    -0.84     0.70 1.00     1168     1761
## studyNum9     -0.39      0.41    -1.22     0.42 1.00     1301     2135
## studyNum10    -0.35      0.41    -1.17     0.43 1.00     1308     2079
## studyNum11     0.34      0.37    -0.40     1.07 1.00     1198     1568
```

```
## studyNum12     0.23      0.39    -0.50     0.98 1.00     1146     1902
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
## [1] "adapt_delta: 0.99"
```

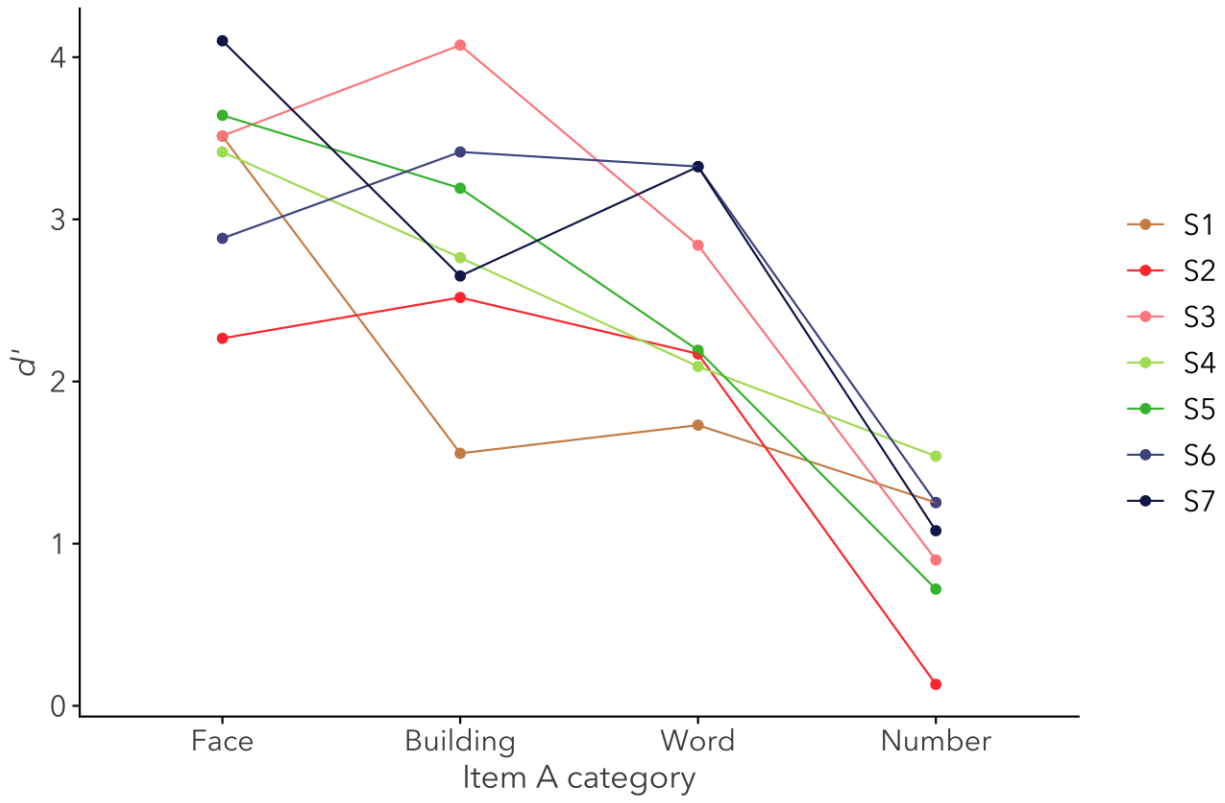
```
## [1] "max_treedepth: 10"
```

```
## was_recalled ~ 1 + studyNum + (1 | subj)
```

```
## was_recalled ~ 1 + (1 | subj)
```

```
##             elpd_diff se_diff elpd_loo se_elpd_loo p_loo  se_p_loo looic
## mod.simple    0.0       0.0  -484.8     15.6         6.6    0.4    969.7
## mod1         -6.1       3.5  -490.9     16.3        18.1    0.9    981.8
##             se_looic
## mod.simple   31.2
## mod1         32.6
```

```
## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'mod.simple' is the best model (ELPD =
## -484.83), followed by 'mod1' (diff-ELPD = -6.06 +- 3.47, p = 0.081)
```



```
## [1] "adapt_delta: 0.8"
```

```
## [1] "max_treedepth: 10"
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: dprime ~ 1 + cue1_category + (1 | subj)
##    Data: behavior.cat (Number of observations: 28)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 7)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.35      0.24     0.02     0.94 1.00      754      871
##
## Regression Coefficients:
##                       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## Intercept                 3.32      0.27     2.78     3.87 1.00     1318
## cue1_categoryBuilding    -0.44      0.33    -1.10     0.22 1.00     2757
## cue1_categoryWord        -0.79      0.33    -1.46    -0.12 1.00     2551
## cue1_categoryNumber      -2.33      0.34    -3.00    -1.65 1.00     2561
##                       Tail_ESS
## Intercept                  647
## cue1_categoryBuilding     2427
## cue1_categoryWord         2141
## cue1_categoryNumber       2159
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.61      0.11     0.44     0.84 1.00     1737     2378
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
## [1] "adapt_delta: 0.8"
```

```
## [1] "max_treedepth: 10"
```

```
## dprime ~ 1 + cue1_category + (1 | subj)
```

```
## dprime ~ 1 + (1 | subj)
```

```
##            elpd_diff se_diff elpd_loo se_elpd_loo p_loo se_p_loo looic se_looic
## mod          0.0       0.0   -29.5      3.1         7.4   1.3     59.0   6.2
## mod.simple -14.9       4.1   -44.4      2.8         3.1   0.6     88.8   5.7
```

```
## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'mod' is the best model (ELPD = -29.52),
## followed by 'mod.simple' (diff-ELPD = -14.86 +- 4.10, p < .001)
```

Table 7: emmeans

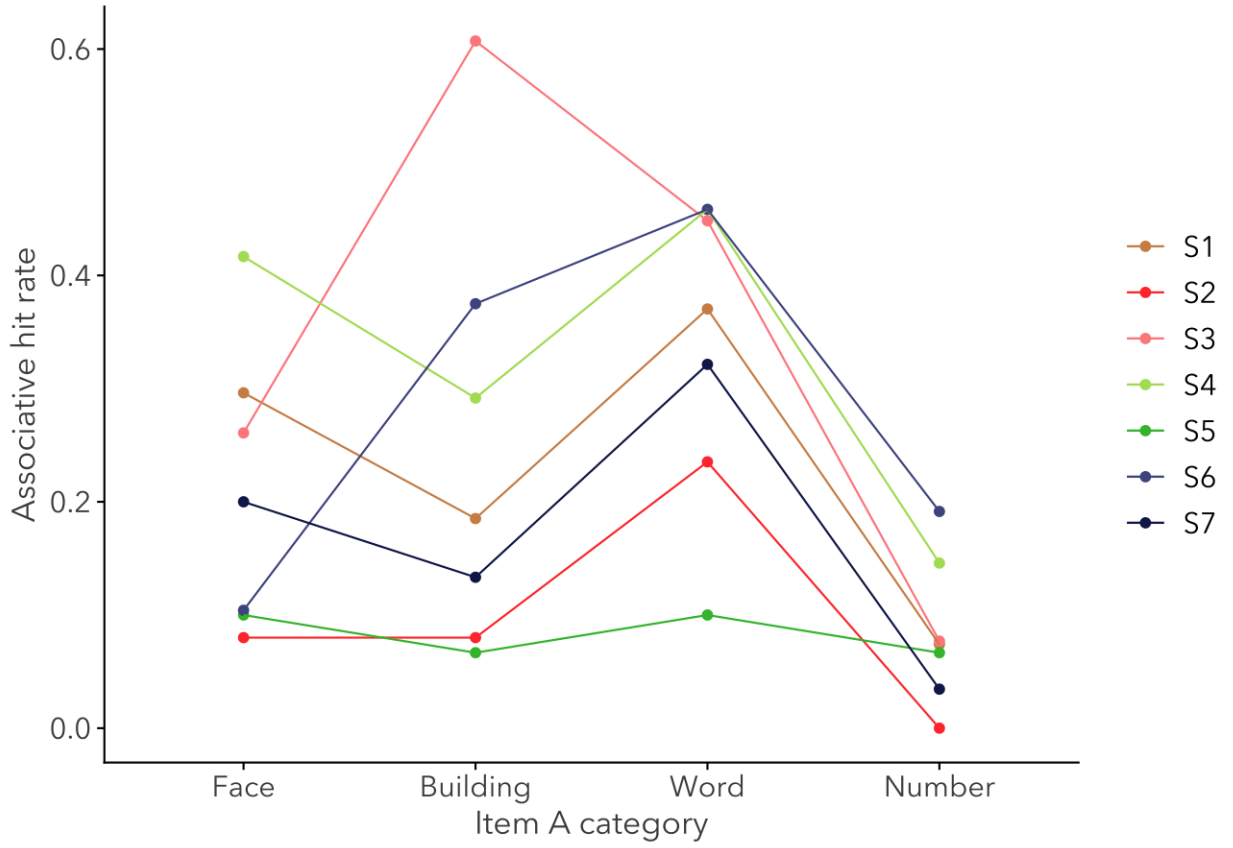
cue1_category	estimate	lower.HPD	upper.HPD
Face	3.328	2.763	3.843
Building	2.881	2.328	3.440
Word	2.530	1.991	3.118
Number	0.987	0.473	1.625

Table 8: emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
cue1_category	Face - Building	0	0.434	-0.217	1.103
cue1_category	Face - Word	0	0.793	0.078	1.410
cue1_category	Face - Number	0	2.337	1.683	3.031
cue1_category	Building - Word	0	0.355	-0.291	1.015
cue1_category	Building - Number	0	1.895	1.228	2.537
cue1_category	Word - Number	0	1.541	0.877	2.238

Table 9: emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
cue1_category	Building - Face	0	-0.434	-1.103	0.217
cue1_category	Word - Face	0	-0.793	-1.410	-0.078
cue1_category	Word - Building	0	-0.355	-1.015	0.291
cue1_category	Number - Face	0	-2.337	-3.031	-1.683
cue1_category	Number - Building	0	-1.895	-2.537	-1.228
cue1_category	Number - Word	0	-1.541	-2.238	-0.877




```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: prop ~ 1 + cue1_category + (1 | subj)
##    Data: behavior.cat (Number of observations: 28)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 7)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.12      0.06     0.03     0.27 1.00      852      860
##
## Regression Coefficients:
##                       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## Intercept                 0.21      0.07     0.08     0.34 1.00      993
## cue1_categoryBuilding     0.04      0.06    -0.08     0.16 1.00     2063
## cue1_categoryWord         0.13      0.06     0.01     0.25 1.00     2233
## cue1_categoryNumber      -0.12      0.06    -0.24    -0.00 1.00     2234
##                       Tail_ESS
## Intercept                 1302
## cue1_categoryBuilding     2170
```

```
## cue1_categoryWord         2000
## cue1_categoryNumber       2485
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.11      0.02     0.08     0.16 1.00     1302     1730
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
## prop ~ 1 + cue1_category + (1 | subj)
```

```
## prop ~ 1 + (1 | subj)
```

```
##            elpd_diff se_diff elpd_loo se_elpd_loo p_loo se_p_loo looic se_looic
## mod          0.0       0.0    16.5      5.6         9.8   3.4    -32.9  11.3
## mod.simple  -5.7       3.8    10.8      3.2         5.3   1.2    -21.5   6.5
```

```
## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'mod' is the best model (ELPD = 16.46),
## followed by 'mod.simple' (diff-ELPD = -5.70 +- 3.84, p = 0.137)
```

Table 10: emmeans

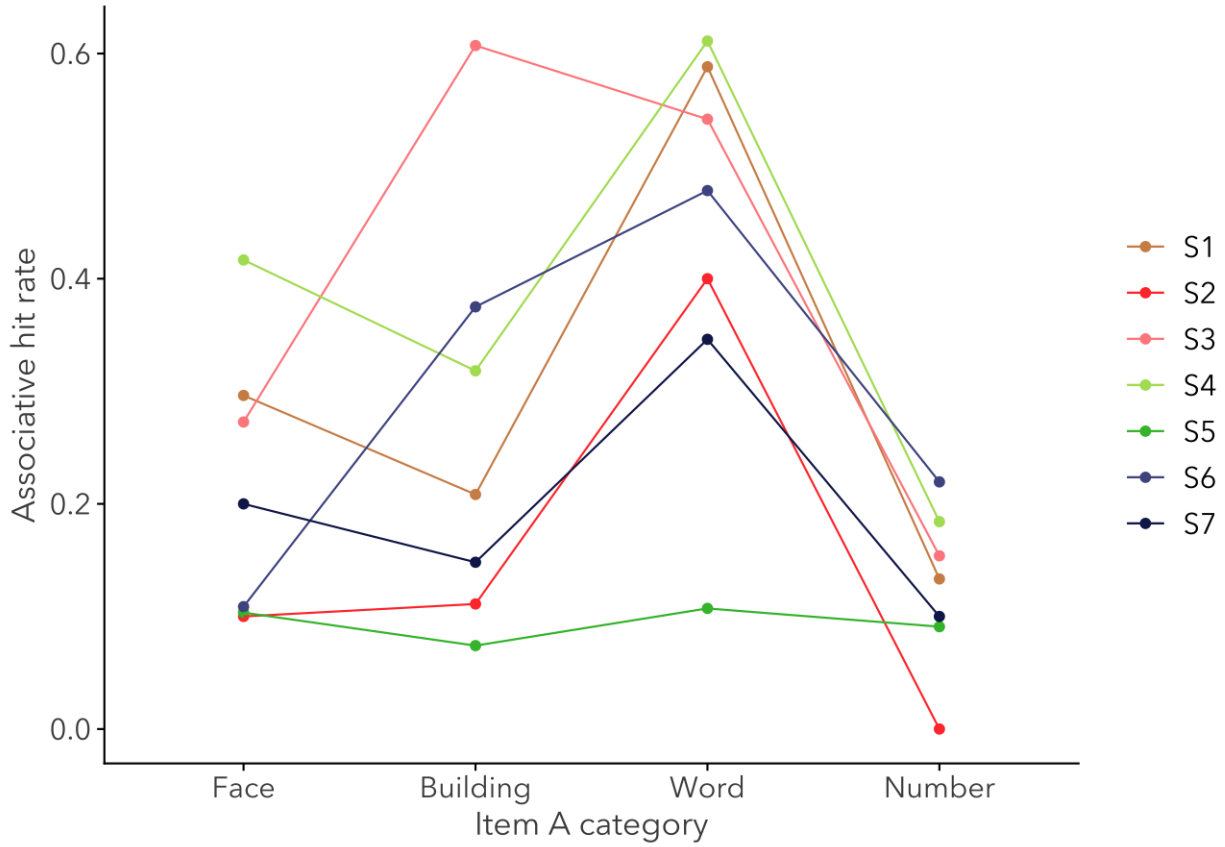
cue1_category	estimate	lower.HPD	upper.HPD
Face	0.209	0.080	0.340
Building	0.250	0.118	0.379
Word	0.343	0.212	0.476
Number	0.086	-0.031	0.225

Table 11: emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
cue1_category	Face - Building	0	-0.040	-0.162	0.074
cue1_category	Face - Word	0	-0.132	-0.248	-0.010
cue1_category	Face - Number	0	0.122	0.003	0.245
cue1_category	Building - Word	0	-0.093	-0.203	0.026
cue1_category	Building - Number	0	0.164	0.039	0.272
cue1_category	Word - Number	0	0.258	0.132	0.377

Table 12: emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
cue1_category	Building - Face	0	0.040	-0.074	0.162
cue1_category	Word - Face	0	0.132	0.010	0.248
cue1_category	Word - Building	0	0.093	-0.026	0.203
cue1_category	Number - Face	0	-0.122	-0.245	-0.003
cue1_category	Number - Building	0	-0.164	-0.272	-0.039
cue1_category	Number - Word	0	-0.258	-0.377	-0.132




```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: prop ~ 1 + cue1_category + (1 | subj)
##    Data: behavior.cat (Number of observations: 28)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 7)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.14      0.07     0.04     0.31 1.00      814     1515
```

```
##
## Regression Coefficients:
##                       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## Intercept                 0.22      0.07     0.08     0.36 1.00     1240
## cue1_categoryBuilding     0.05      0.06    -0.08     0.17 1.00     2230
## cue1_categoryWord         0.22      0.06     0.09     0.34 1.00     2278
## cue1_categoryNumber      -0.09      0.06    -0.21     0.03 1.00     2299
##                       Tail_ESS
## Intercept                 1910
## cue1_categoryBuilding     2429
## cue1_categoryWord         2299
## cue1_categoryNumber       2461
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.11      0.02     0.08     0.16 1.00     1547     2359
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
## prop ~ 1 + cue1_category + (1 | subj)
```

```
## prop ~ 1 + (1 | subj)
```

```
##            elpd_diff se_diff elpd_loo se_elpd_loo p_loo se_p_loo looic se_looic
## mod          0.0       0.0    16.2      4.6         9.3   2.6    -32.4   9.2
## mod.simple  -8.9       4.5     7.3      2.9         4.7   0.9    -14.6   5.7
```

```
## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'mod' is the best model (ELPD = 16.22),
## followed by 'mod.simple' (diff-ELPD = -8.91 +- 4.53, p = 0.049)
```

Table 13: emmeans

cue1_category	estimate	lower.HPD	upper.HPD
Face	0.217	0.085	0.359
Building	0.263	0.131	0.405
Word	0.440	0.308	0.587
Number	0.126	-0.002	0.268

Table 14: emmeans

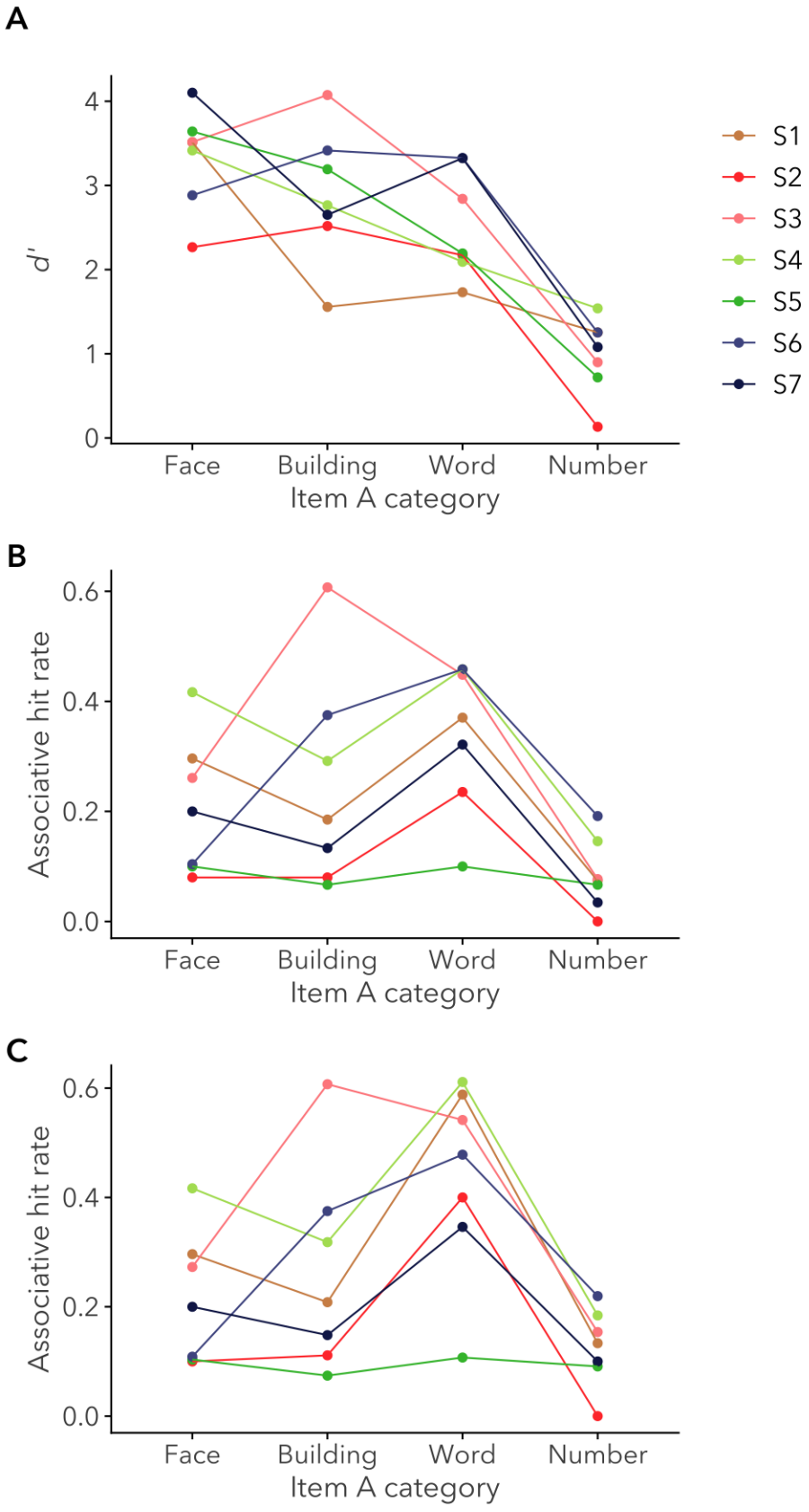
term	contrast	null.value	estimate	lower.HPD	upper.HPD
cue1_category	Face - Building	0	-0.047	-0.178	0.072

cue1_category	Face - Word	0	-0.222	-0.348	-0.100
cue1_category	Face - Number	0	0.091	-0.034	0.205
cue1_category	Building - Word	0	-0.174	-0.295	-0.052
cue1_category	Building - Number	0	0.138	0.025	0.259
cue1_category	Word - Number	0	0.313	0.198	0.445

Table 15: emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
cue1_category	Building - Face	0	0.047	-0.072	0.178
cue1_category	Word - Face	0	0.222	0.100	0.348
cue1_category	Word - Building	0	0.174	0.052	0.295
cue1_category	Number - Face	0	-0.091	-0.205	0.034
cue1_category	Number - Building	0	-0.138	-0.259	-0.025
cue1_category	Number - Word	0	-0.313	-0.445	-0.198

Fig S4


A
d′
4
3
2
1
0
Face
Building
Word
Number
Item A category
S1
S2
S3
S4
S5
S6
S7
B
Associative hit rate
0.6
0.4
0.2
0.0
Face
Building
Word
Number
Item A category
C
Associative hit rate
0.6
0.4
0.2
0.0
Face
Building
Word
Number
Item A category

```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: prop ~ 1 + cue2_category + (1 | subj)
##    Data: behavior.cat (Number of observations: 28)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 7)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.13      0.06     0.06     0.27 1.00      853     1530
##
## Regression Coefficients:
##                       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## Intercept                 0.26      0.06     0.13     0.38 1.00      948
## cue2_categoryBuilding    -0.06      0.04    -0.15     0.02 1.00     1755
## cue2_categoryWord        -0.06      0.04    -0.14     0.03 1.00     1927
## cue2_categoryNumber      -0.04      0.04    -0.13     0.04 1.00     2027
##                       Tail_ESS
## Intercept                 1429
## cue2_categoryBuilding     2260
## cue2_categoryWord         2139
## cue2_categoryNumber       2263
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.08      0.01     0.06     0.11 1.00     1587     1661
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
## prop ~ 1 + cue2_category + (1 | subj)
```

```
## prop ~ 1 + (1 | subj)
```

```
##            elpd_diff se_diff elpd_loo se_elpd_loo p_loo se_p_loo looic se_looic
## mod.simple   0.0       0.0    28.2      4.3         7.0   1.9    -56.5   8.6
## mod         -2.1       2.0    26.1      3.6         9.7   2.0    -52.3   7.2
```

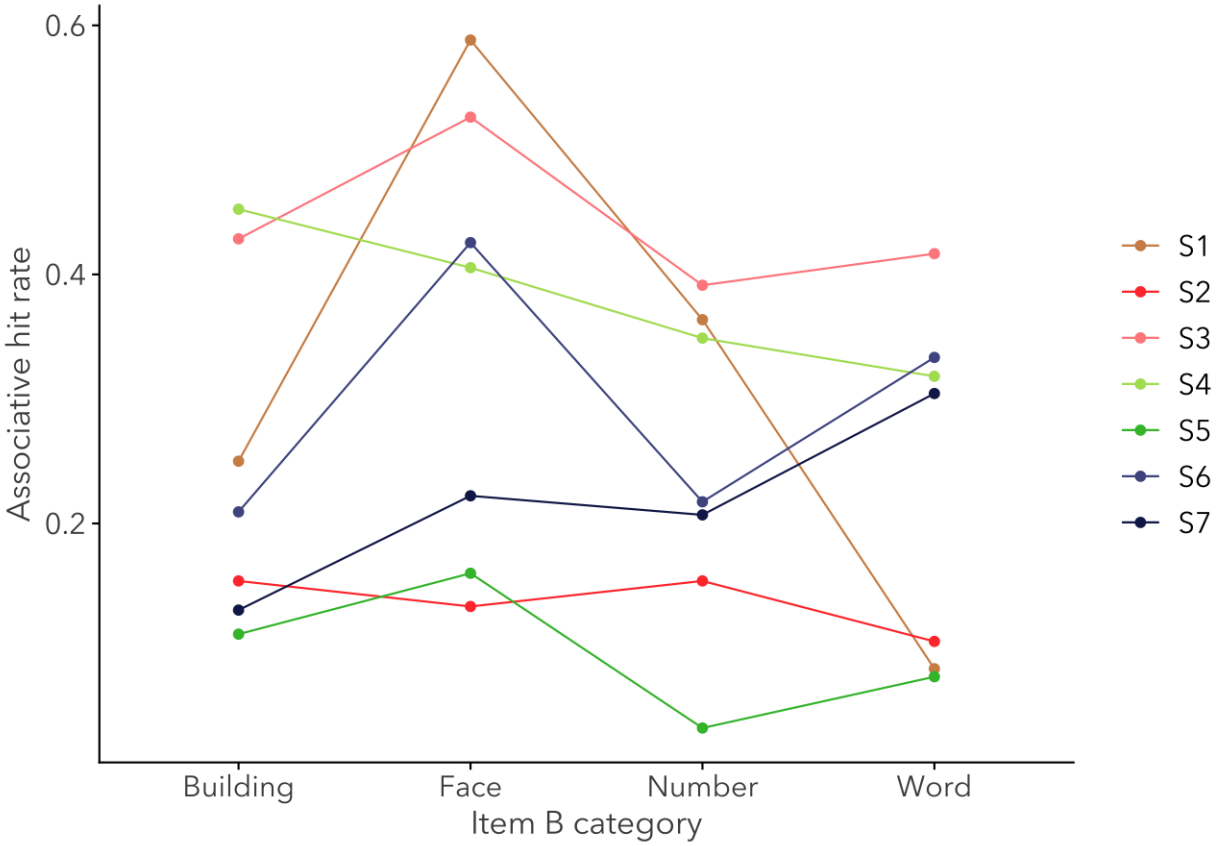
```
## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'mod.simple' is the best model (ELPD =
## 28.25), followed by 'mod' (diff-ELPD = -2.10 +- 1.98, p = 0.289)
```

Table 16: emmeans

cue2_category	estimate	lower.HPD	upper.HPD
Face	0.262	0.134	0.380
Building	0.195	0.077	0.323
Word	0.205	0.082	0.328
Number	0.218	0.095	0.338

Table 17: emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
cue2_category	Face - Building	0	0.065	-0.022	0.150
cue2_category	Face - Word	0	0.056	-0.030	0.142
cue2_category	Face - Number	0	0.043	-0.039	0.127
cue2_category	Building - Word	0	-0.009	-0.095	0.072
cue2_category	Building - Number	0	-0.023	-0.100	0.068
cue2_category	Word - Number	0	-0.014	-0.088	0.075




```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: prop ~ 1 + cue2_category + (1 | subj)
##    Data: behavior.cat (Number of observations: 28)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 7)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.15      0.07     0.07     0.32 1.00     1026     1895
##
## Regression Coefficients:
##                     Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept               0.25      0.07     0.10     0.39 1.00     1234     1712
## cue2_categoryFace       0.10      0.05    -0.00     0.21 1.00     2337     2529
## cue2_categoryNumber    -0.00      0.05    -0.11     0.11 1.00     2166     2424
## cue2_categoryWord      -0.01      0.05    -0.12     0.09 1.00     2207     2477
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.10      0.02     0.07     0.14 1.00     1652     1925
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
## prop ~ 1 + cue2_category + (1 | subj)
```

```
## prop ~ 1 + (1 | subj)
```

```
##            elpd_diff se_diff elpd_loo se_elpd_loo p_loo se_p_loo looic se_looic
## mod          0.0       0.0    18.9      5.2        10.4   3.1    -37.8  10.5
## mod.simple  -0.9       2.8    18.0      6.4         8.1   3.3    -36.1  12.9
```

```
## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'mod' is the best model (ELPD = 18.92),
## followed by 'mod.simple' (diff-ELPD = -0.89 +- 2.78, p = 0.749)
```

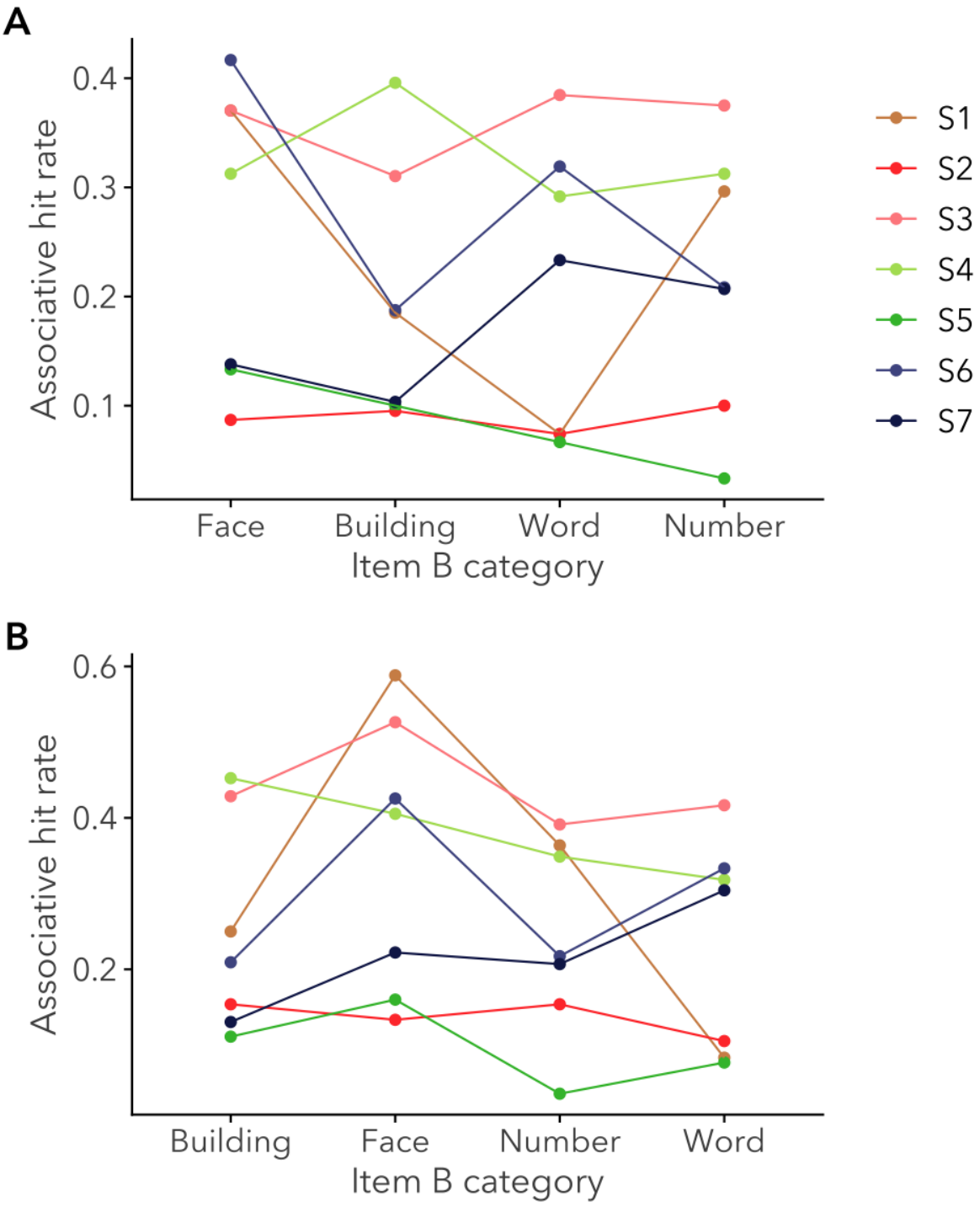
Table 18: emmeans

cue2_category	estimate	lower.HPD	upper.HPD
Building	0.246	0.107	0.389
Face	0.352	0.209	0.492
Number	0.244	0.096	0.380
Word	0.233	0.075	0.368

Table 19: emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
cue2_category	Building - Face	0	-0.104	-0.207	0.004
cue2_category	Building - Number	0	0.002	-0.107	0.109
cue2_category	Building - Word	0	0.011	-0.090	0.121
cue2_category	Face - Number	0	0.106	0.005	0.221
cue2_category	Face - Word	0	0.117	0.013	0.226
cue2_category	Number - Word	0	0.010	-0.096	0.124

Fig S5


A
Associative hit rate
0.4
0.3
0.2
0.1
Face
Building
Word
Number
Item B category
S1
S2
S3
S4
S5
S6
S7
B
Associative hit rate
0.6
0.4
0.2
Building
Face
Number
Word
Item B category

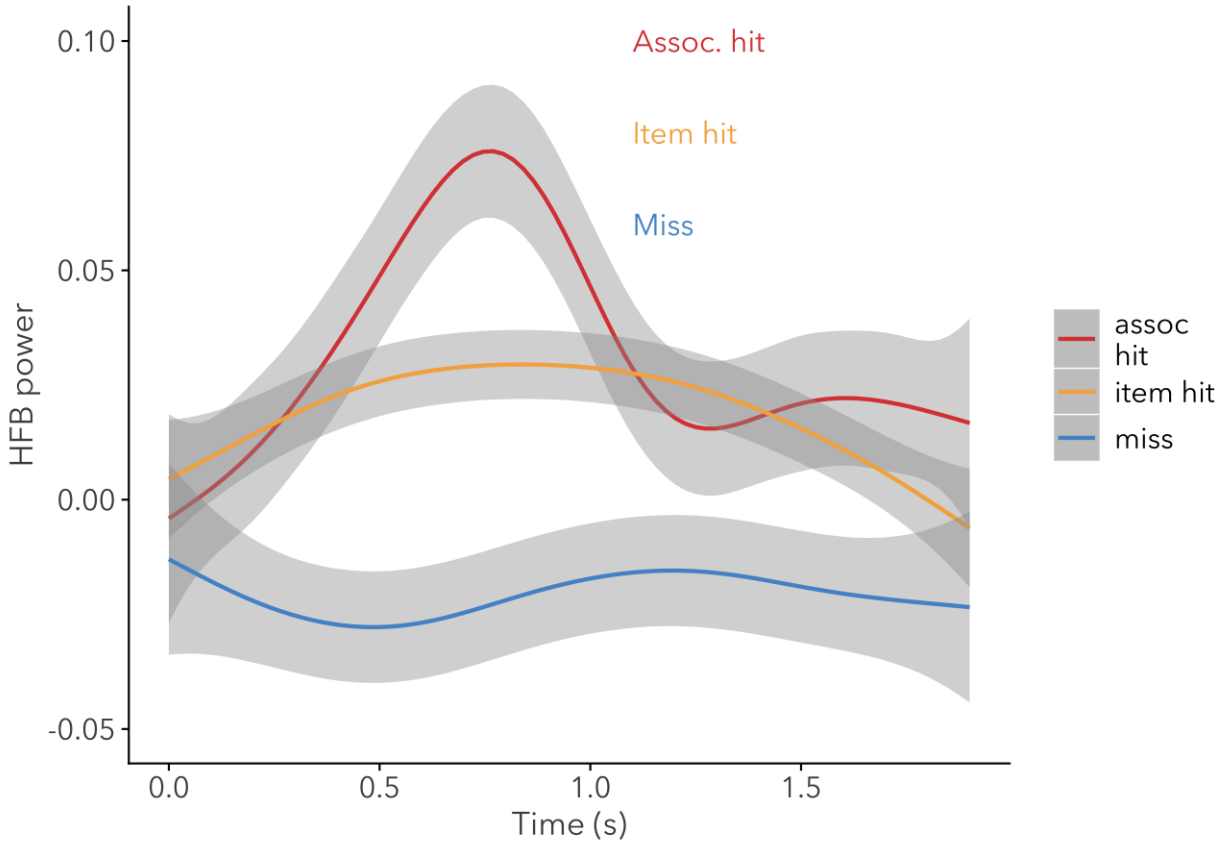
HFB

Study

For all univariate HFB analyses, HFB was first divided into 100-ms time bins separately for each electrode.

For study phase item A univariate analyses, the models included time-bin, item A category, and subsequent memory outcome as predictors and controls.

For study phase item B univariate analyses, the models included time-bin, both item A and item B categories, and subsequent memory outcome as predictors and controls.



Item A

```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: value ~ 1 + time + cue1_category + trial_type_combo + (1 | subj/chan)
##    Data: . (Number of observations: 50680)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 7)
```

```
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.05      0.03     0.02     0.12 1.00     1004     1747
##
## ~subj:chan (Number of levels: 30)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.03      0.01     0.02     0.04 1.00     1223     1779
##
## Regression Coefficients:
##                          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## Intercept                    0.04      0.02    -0.01     0.08 1.00     1571
## time                        -0.00      0.00    -0.01     0.00 1.00     5876
## cue1_categoryBuilding       -0.00      0.00    -0.01     0.00 1.00     4025
## cue1_categoryWord           -0.05      0.00    -0.06    -0.04 1.00     4079
## cue1_categoryNumber         -0.04      0.00    -0.05    -0.03 1.00     4177
## trial_type_comboitemhit     -0.01      0.00    -0.02    -0.00 1.00     4591
## trial_type_combomiss        -0.03      0.01    -0.04    -0.02 1.00     3847
##                          Tail_ESS
## Intercept                    2160
## time                         2251
## cue1_categoryBuilding        2768
## cue1_categoryWord            3232
## cue1_categoryNumber          3241
## trial_type_comboitemhit      3049
## trial_type_combomiss         3010
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.35      0.00     0.35     0.36 1.00     5627     2685
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
## value ~ 1 + time + cue1_category + trial_type_combo + (1 | subj/chan)
```

```
## value ~ 1 + time + cue1_category + (1 | subj/chan)
```

```
##             elpd_diff se_diff  elpd_loo se_elpd_loo p_loo    se_p_loo looic
## study.mod.A      0.0       0.0 -19332.1    187.8        35.6      0.3  38664.1
## mod.simple     -18.4       6.4 -19350.5    187.6        33.2      0.3  38701.0
##             se_looic
## study.mod.A    375.7
## mod.simple     375.2
```

```
## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'study.mod.A' is the best model (ELPD =
## -19332.07), followed by 'mod.simple' (diff-ELPD = -18.41 +- 6.37, p = 0.004)
```

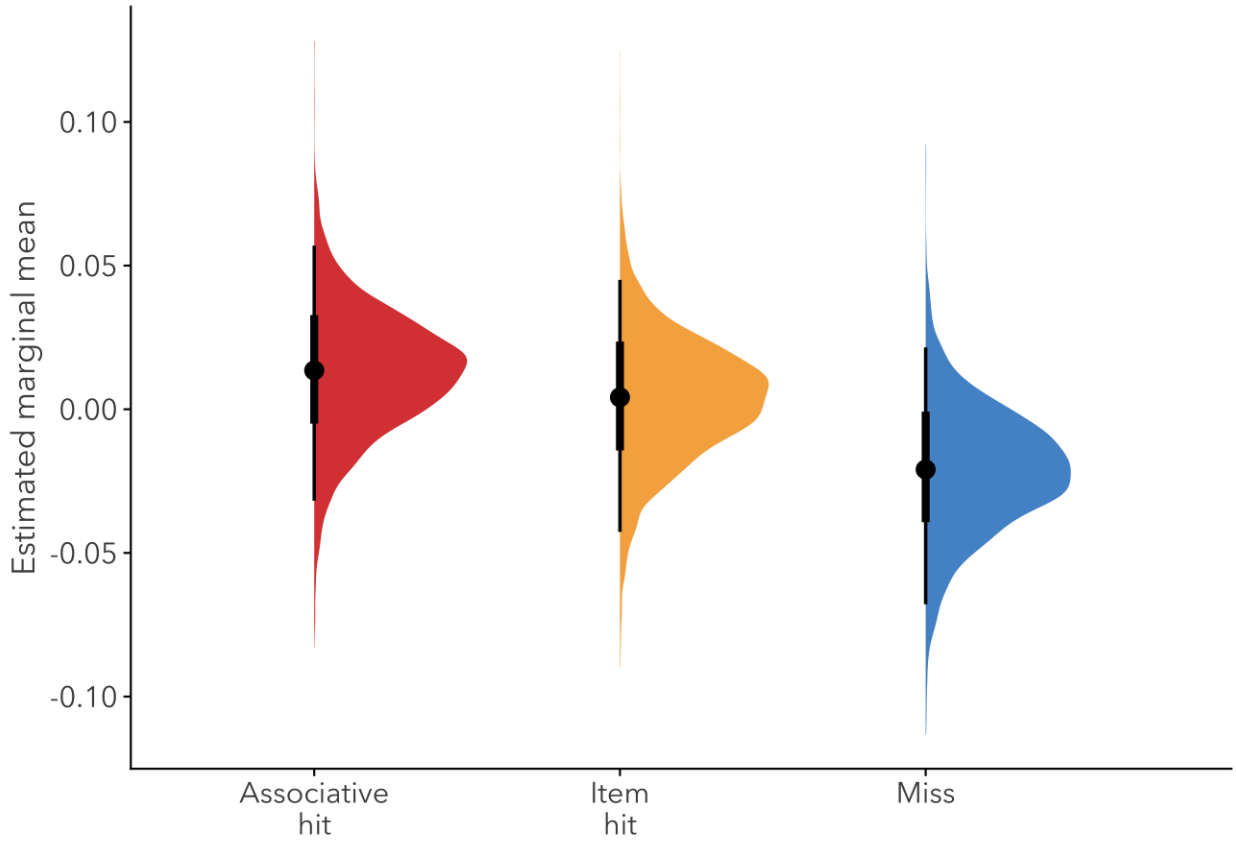
effect	component	term	estimate	std.error	conf.low	conf.high
fixed	cond	(Intercept)	0.0385262	0.0223479	-0.0078240	0.0820065
fixed	cond	time	-0.0037920	0.0028297	-0.0093457	0.0017265
fixed	cond	cue1_categoryBuilding	-0.0042970	0.0043816	-0.0129994	0.0042026
fixed	cond	cue1_categoryWord	-0.0461532	0.0045791	-0.0551821	-0.0374264
fixed	cond	cue1_categoryNumber	-0.0382682	0.0047893	-0.0477800	-0.0289610
fixed	cond	trial_type_comboitemhit	-0.0092298	0.0040087	-0.0170253	-0.0013787
fixed	cond	trial_type_combomiss	-0.0340527	0.0053533	-0.0444766	-0.0236299

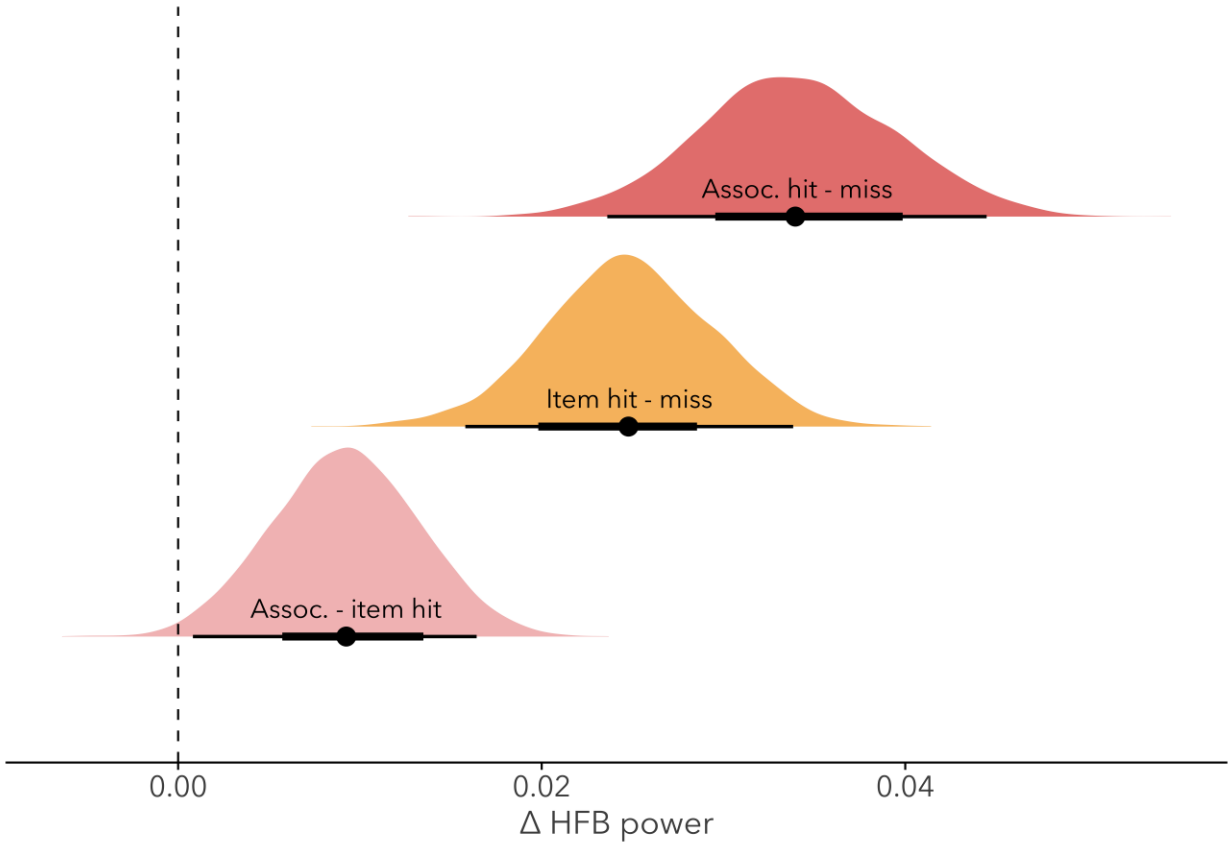
Table 21: study A emmeans

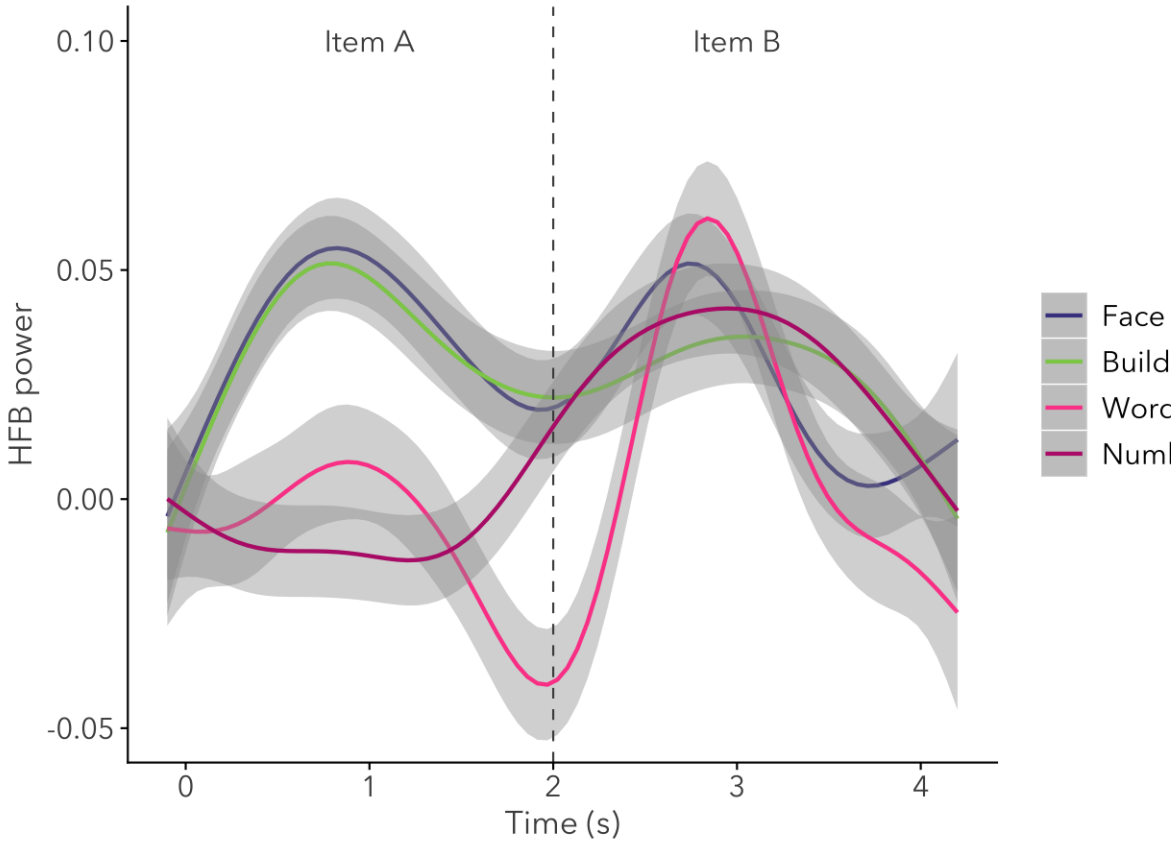
trial_type_combo	estimate	lower.HPD	upper.HPD
assoc hit	0.014	-0.035	0.053
item hit	0.004	-0.043	0.044
miss	-0.021	-0.066	0.023

Table 22: study A emmeans

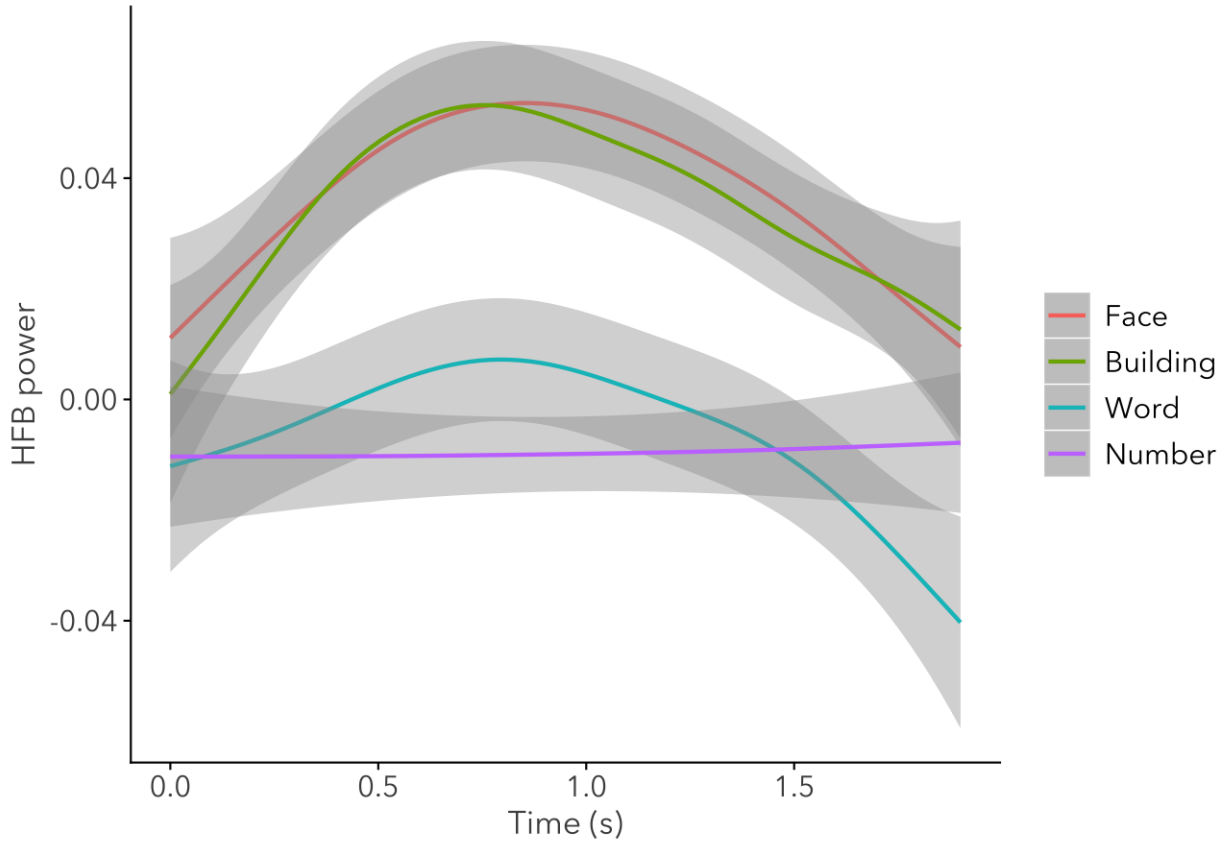
term	contrast	null.value	estimate	lower.HPD	upper.HPD
trial_type_combo	assoc hit - item hit	0	0.009	0.001	0.017
trial_type_combo	assoc hit - miss	0	0.034	0.024	0.044
trial_type_combo	item hit - miss	0	0.025	0.016	0.034


Estimated marginal mean
0.10
0.05
0.00
-0.05
-0.10
Associative hit
Item hit
Miss


Assoc. hit - miss
Item hit - miss
Assoc. - item hit
0.00
0.02
0.04
Δ HFB power



Stimulus category




```
## [1] "adapt_delta: 0.8"
```

```
## [1] "max_treedepth: 10"
```

```
## value ~ 1 + time + cue1_category + trial_type_combo + (1 | subj/chan)
```

```
## value ~ 1 + time + trial_type_combo + (1 | subj/chan)
```

```
##             elpd_diff se_diff  elpd_loo se_elpd_loo p_loo    se_p_loo looic
## study.mod.A       0.0       0.0 -19332.1    187.8        35.6      0.3  38664.1
## mod.simple      -71.1      12.3 -19403.2    186.9        32.1      0.3  38806.4
##             se_looic
## study.mod.A    375.7
## mod.simple     373.9
```

```
## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'study.mod.A' is the best model (ELPD =
## -19332.07), followed by 'mod.simple' (diff-ELPD = -71.14 +- 12.31, p < .001)
```

Table 23: study A emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD

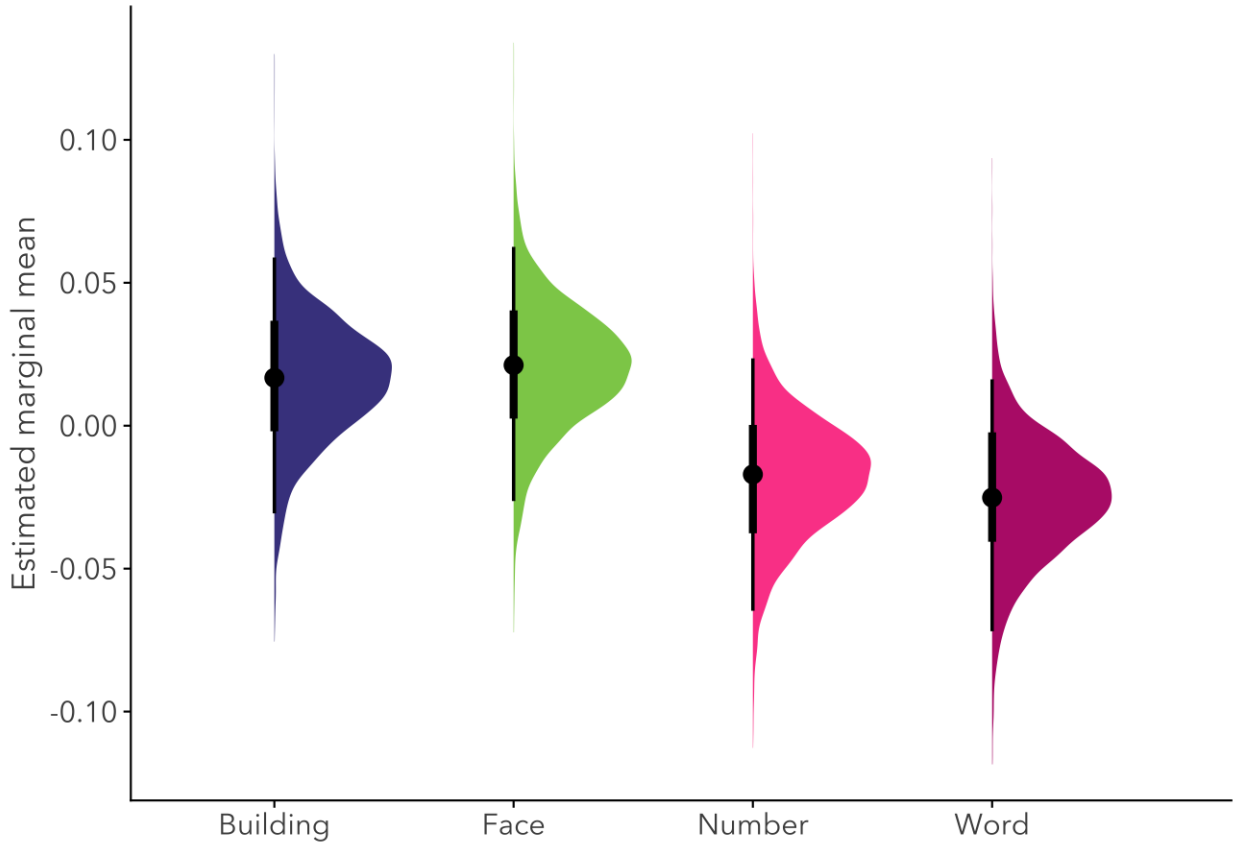
cue1_category	Building - Face	0	-0.004	-0.013	0.004
cue1_category	Word - Face	0	-0.046	-0.055	-0.037
cue1_category	Word - Building	0	-0.042	-0.051	-0.033
cue1_category	Number - Face	0	-0.038	-0.048	-0.029
cue1_category	Number - Building	0	-0.034	-0.043	-0.024
cue1_category	Number - Word	0	0.008	-0.001	0.018

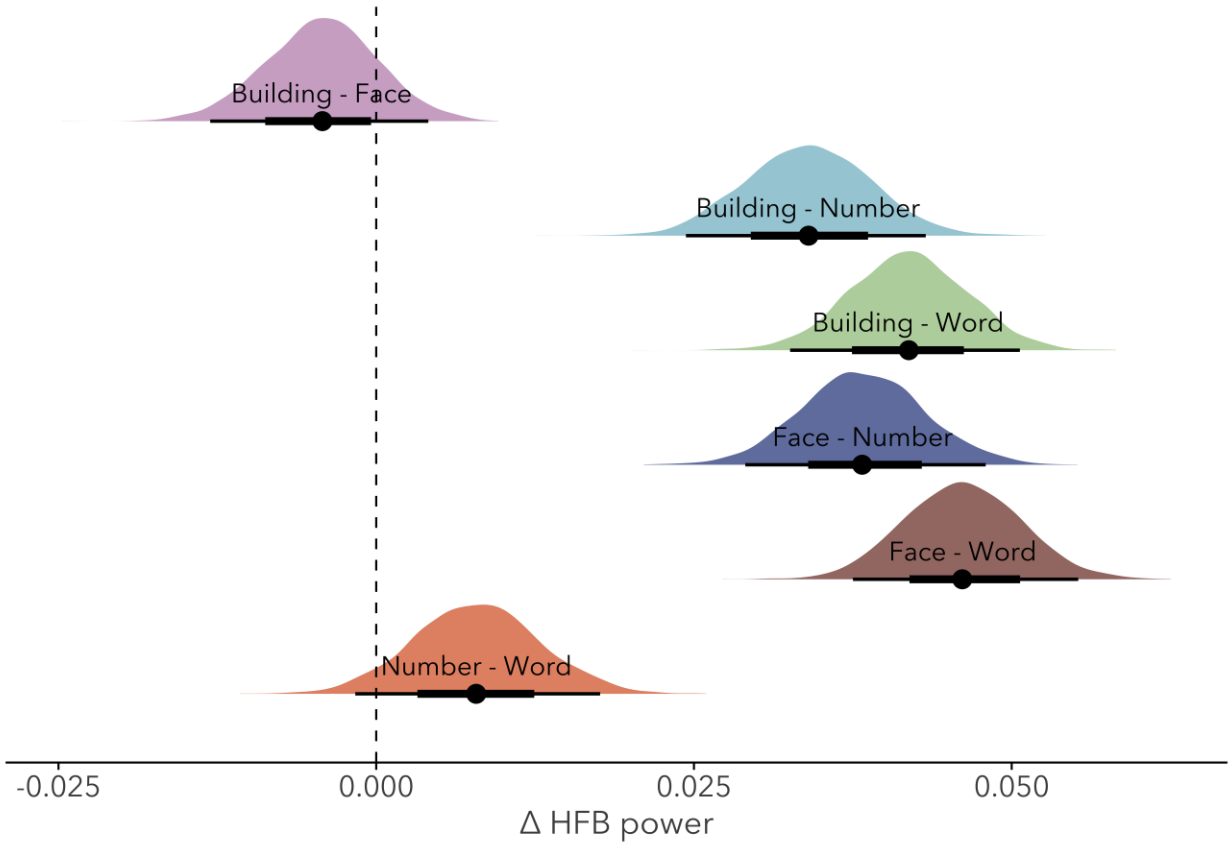
Table 24: study A emmeans

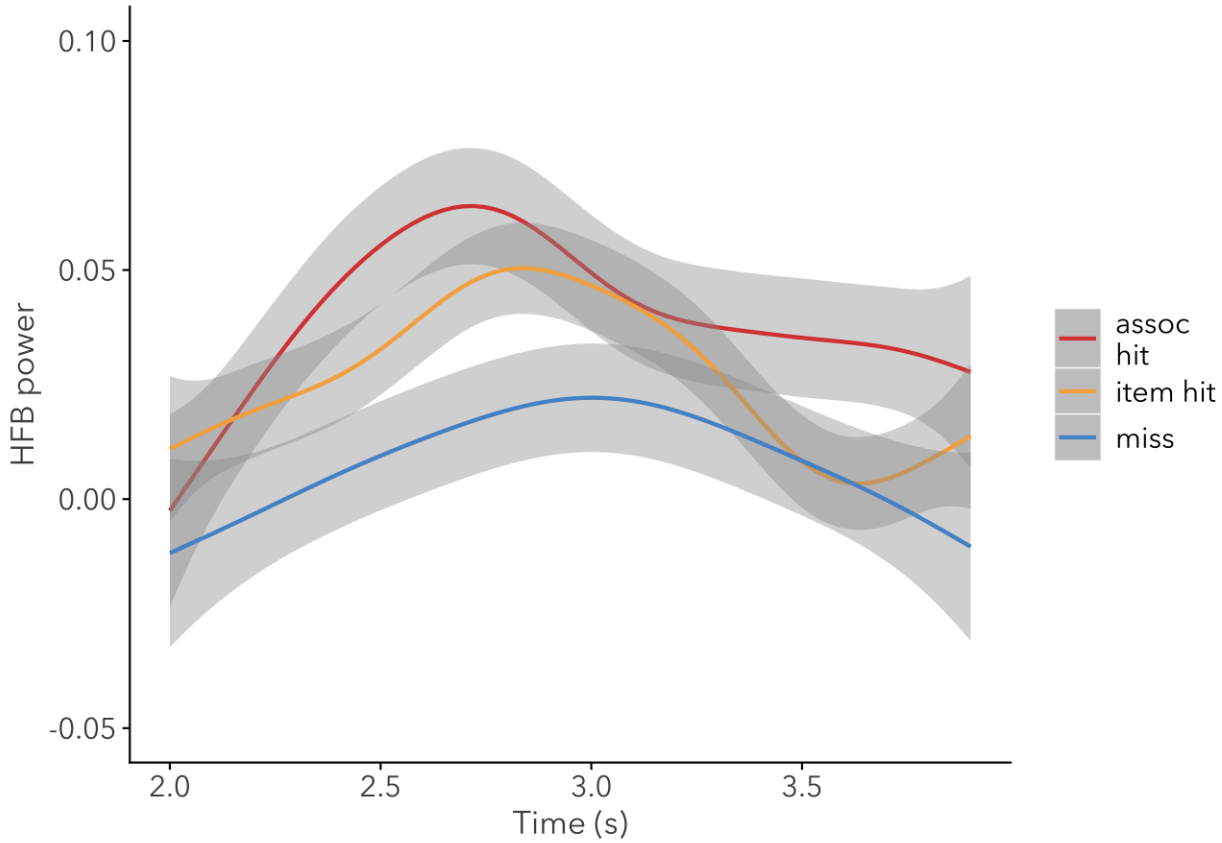
cue1_category	estimate	lower.HPD	upper.HPD
Face	0.021	-0.026	0.063
Building	0.017	-0.031	0.058
Word	-0.025	-0.070	0.018
Number	-0.017	-0.063	0.025

Table 25: study A emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
cue1_category	Face - Building	0	0.004	-0.004	0.013
cue1_category	Face - Word	0	0.046	0.037	0.055
cue1_category	Face - Number	0	0.038	0.029	0.048
cue1_category	Building - Word	0	0.042	0.033	0.051
cue1_category	Building - Number	0	0.034	0.024	0.043
cue1_category	Word - Number	0	-0.008	-0.018	0.001


Estimated marginal mean
0.10
0.05
0.00
-0.05
-0.10
Building
Face
Number
Word


Building - Face
Building - Number
Building - Word
Face - Number
Face - Word
Number - Word
-0.025
0.000
0.025
0.050
Δ HFB power



Item B

```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: value ~ 1 + time + cue1_category + cue2_category + trial_type_combo + (1 | subj/chan)
##    Data: . (Number of observations: 50680)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 7)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.06      0.03     0.02     0.12 1.00     1077     1765
##
## ~subj:chan (Number of levels: 30)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.03      0.01     0.02     0.04 1.00     1013     1613
##
## Regression Coefficients:
##                          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## Intercept                    0.03      0.03    -0.02     0.08 1.00     1572
## time                        -0.00      0.00    -0.01     0.00 1.00     6229
## cue1_categoryBuilding        0.00      0.00    -0.01     0.01 1.00     3782
```

```
## cue1_categoryWord          -0.02      0.00    -0.03    -0.01 1.00     3890
## cue1_categoryNumber         0.02      0.01     0.01     0.03 1.00     3664
## cue2_categoryFace           0.00      0.00    -0.01     0.01 1.00     3948
## cue2_categoryNumber         0.04      0.00     0.03     0.04 1.00     4215
## cue2_categoryWord          -0.01      0.00    -0.02    -0.00 1.00     3862
## trial_type_comboitemhit    -0.01      0.00    -0.02    -0.00 1.00     4188
## trial_type_combomiss       -0.04      0.01    -0.05    -0.02 1.00     3882
##                          Tail_ESS
## Intercept                    1983
## time                         2526
## cue1_categoryBuilding        3274
## cue1_categoryWord            3149
## cue1_categoryNumber          3213
## cue2_categoryFace            3416
## cue2_categoryNumber          3356
## cue2_categoryWord            3417
## trial_type_comboitemhit      3033
## trial_type_combomiss         2620
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.37      0.00     0.36     0.37 1.00     6203     2539
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).

## [1] "adapt_delta: 0.99"

## [1] "max_treedepth: 10"

## value ~ 1 + time + cue1_category + cue2_category + trial_type_combo + (1 | subj/chan)

## value ~ 1 + time + cue1_category + cue2_category + (1 | subj/chan)

##             elpd_diff se_diff  elpd_loo se_elpd_loo p_loo    se_p_loo looic
## study.mod.B      0.0       0.0 -21095.9    201.7        38.8      0.4  42191.9
## mod.simple     -18.1       6.3 -21114.0    201.3        36.4      0.4  42228.1
##             se_looic
## study.mod.B    403.4
## mod.simple     402.6

## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'study.mod.B' is the best model (ELPD =
## -21095.94), followed by 'mod.simple' (diff-ELPD = -18.09 +- 6.27, p = 0.004)
```

effect	component	term	estimate	std.error	conf.low	conf.high
fixed	cond	(Intercept)	0.0332494	0.0263830	-0.0200802	0.0846775
fixed	cond	time	-0.0033595	0.0028926	-0.0091182	0.0020255
fixed	cond	cue1_categoryBuilding	0.0002612	0.0048898	-0.0091091	0.0098699

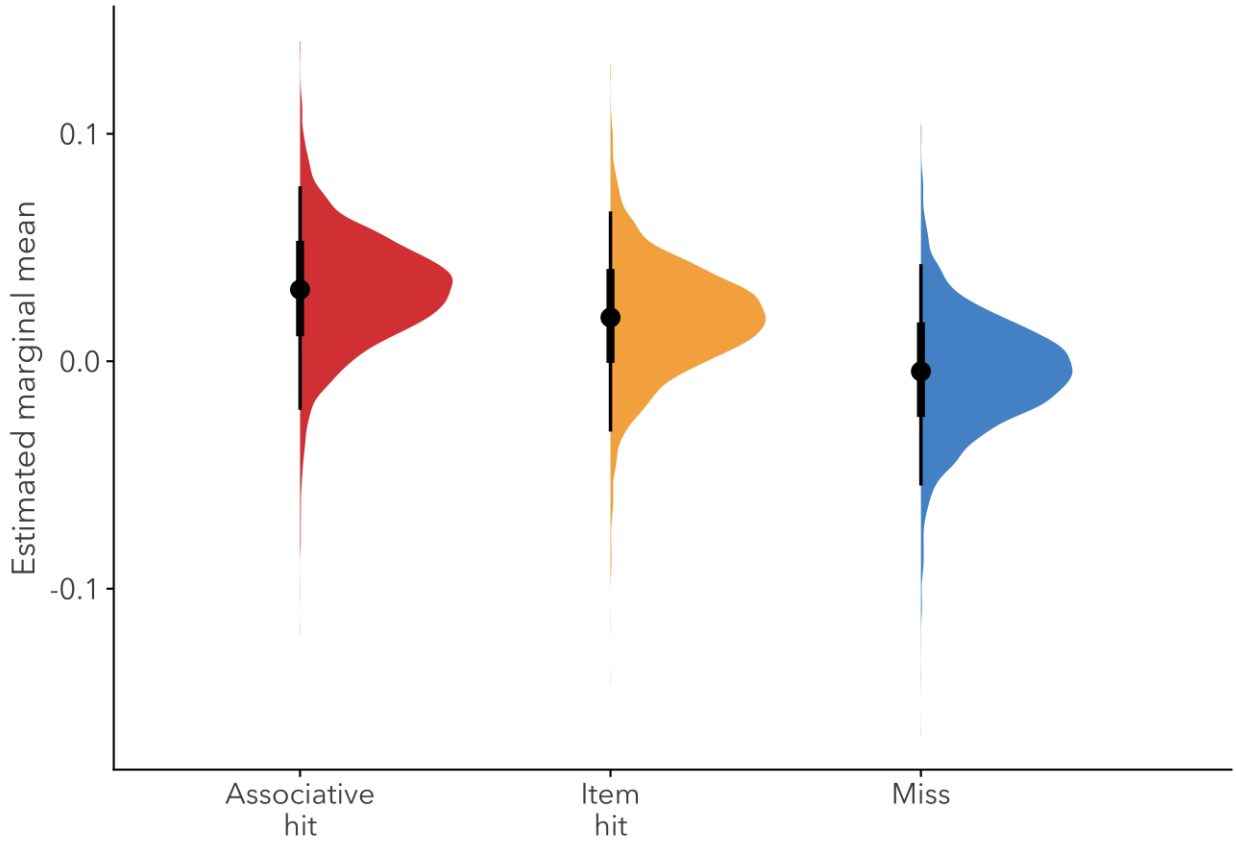
fixed	cond	cue1_categoryWord	-0.0211707	0.0048536	-0.0306922	-0.0119375
fixed	cond	cue1_categoryNumber	0.0238913	0.0051185	0.0138650	0.0337555
fixed	cond	cue2_categoryFace	0.0015335	0.0048245	-0.0078845	0.0111254
fixed	cond	cue2_categoryNumber	0.0356927	0.0048359	0.0261262	0.0448869
fixed	cond	cue2_categoryWord	-0.0098128	0.0048307	-0.0192906	-0.0003169
fixed	cond	trial_type_comboitemhit	-0.0123849	0.0042286	-0.0202838	-0.0041772
fixed	cond	trial_type_combomiss	-0.0360003	0.0055631	-0.0468805	-0.0246835

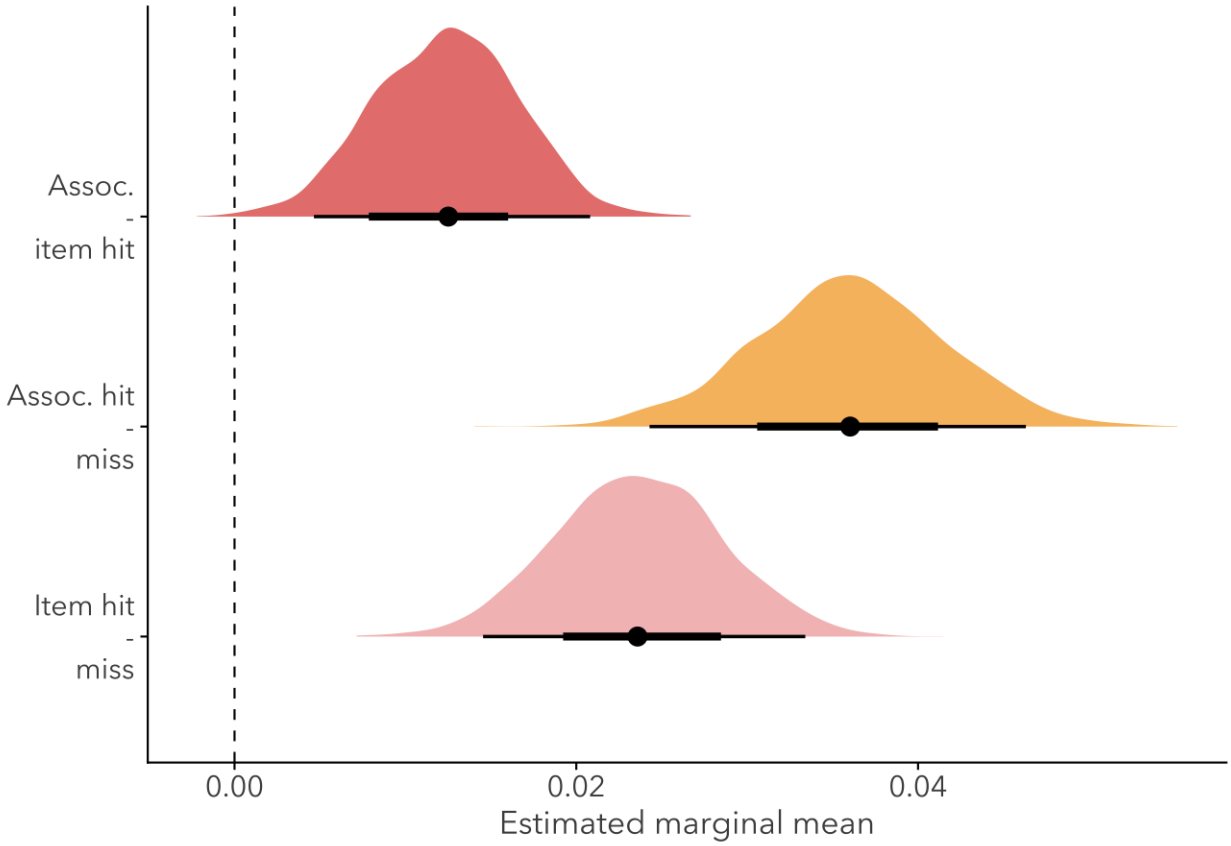
Table 27: study B emmeans

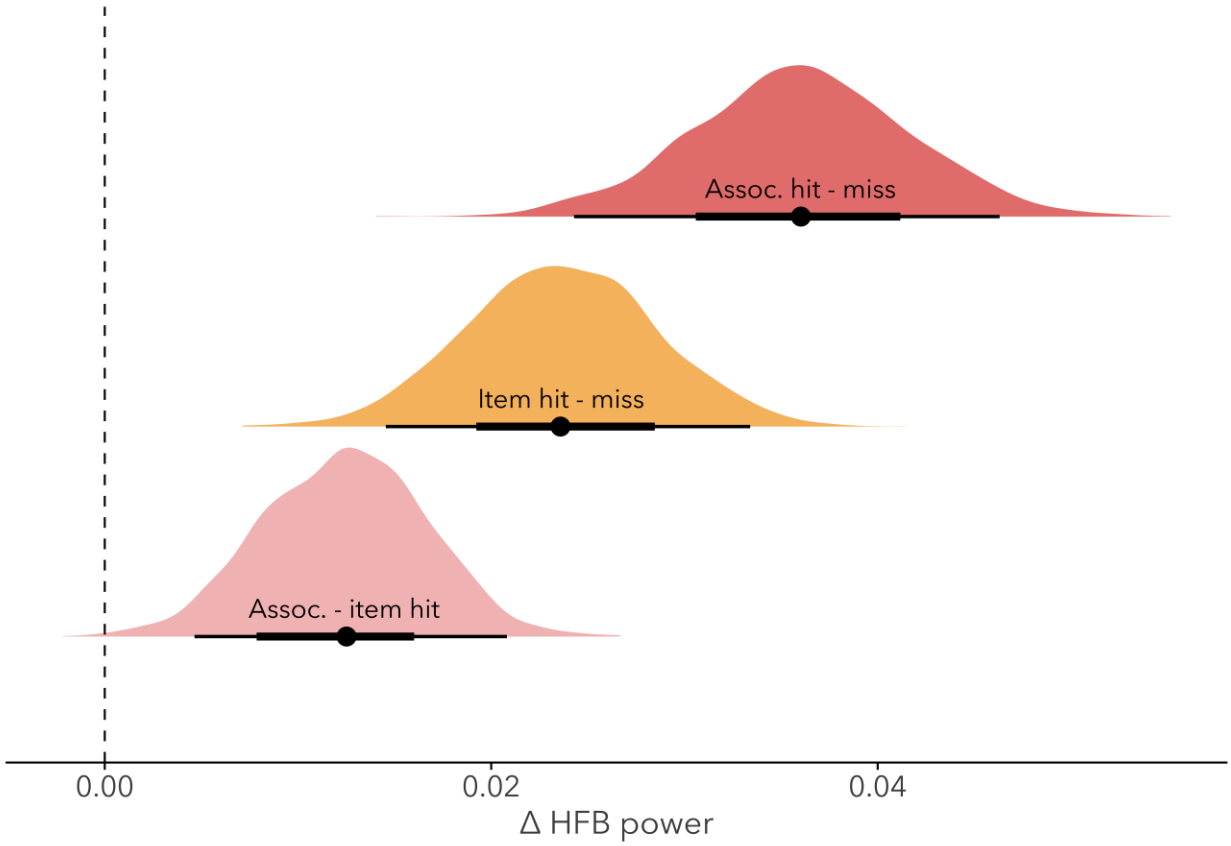
trial_type_combo	estimate	lower.HPD	upper.HPD
assoc hit	0.032	-0.021	0.077
item hit	0.019	-0.031	0.066
miss	-0.004	-0.056	0.042

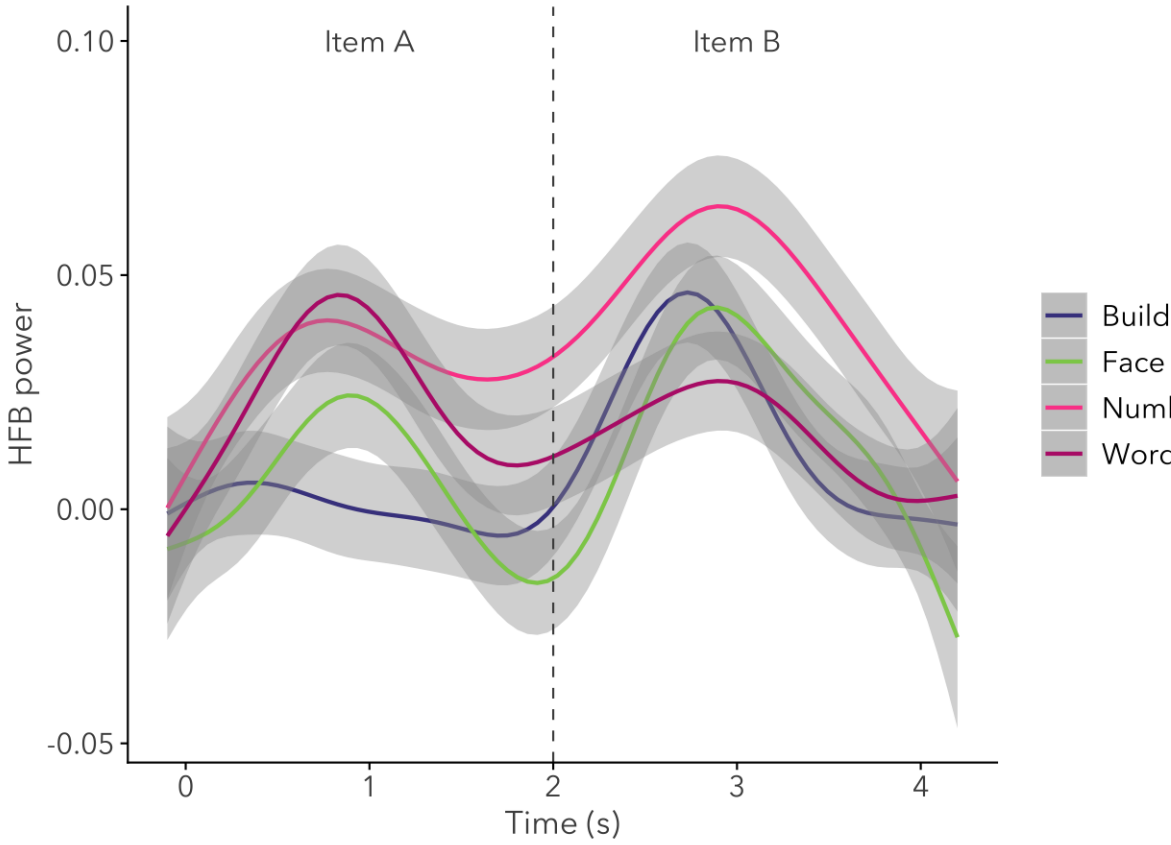
Table 28: study B emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
trial_type_combo	assoc hit - item hit	0	0.013	0.004	0.020
trial_type_combo	assoc hit - miss	0	0.036	0.024	0.046
trial_type_combo	item hit - miss	0	0.024	0.014	0.033

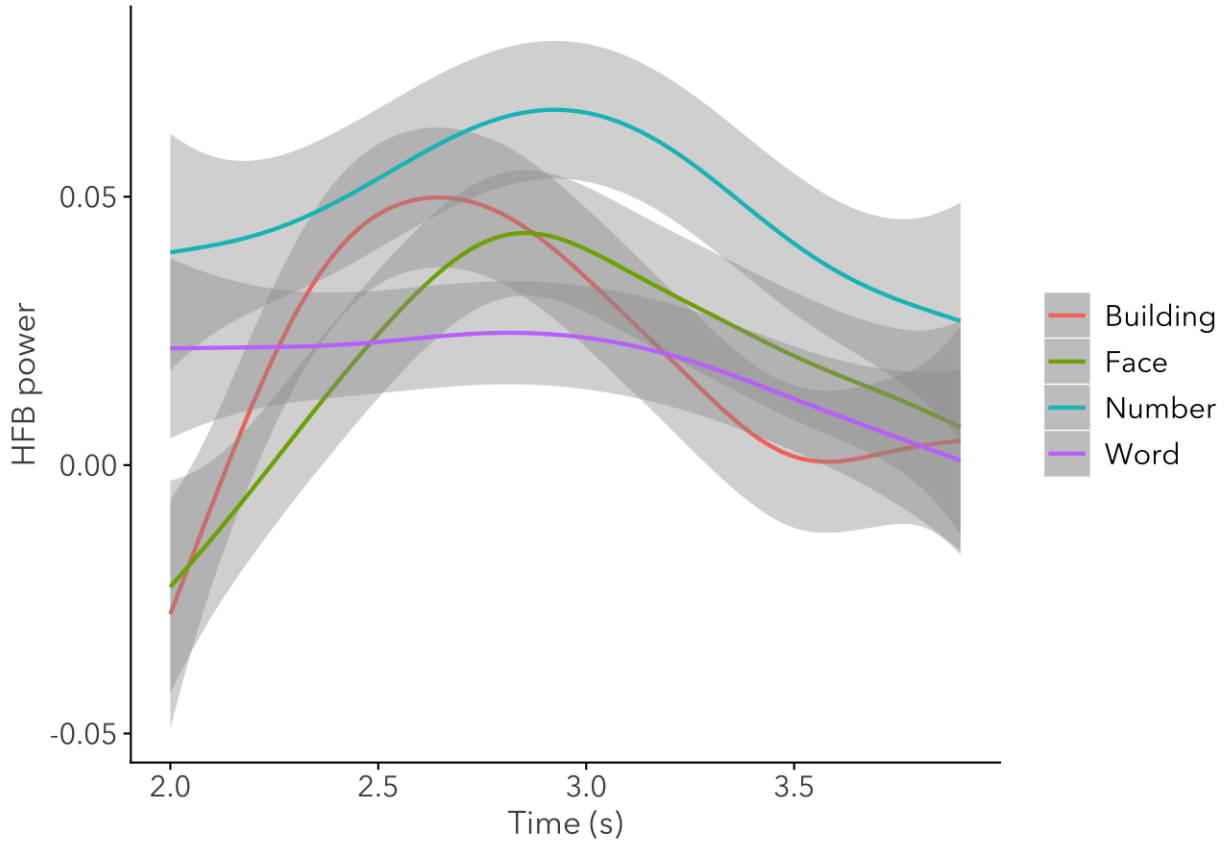



Assoc.
-
item hit
Assoc. hit
-
miss
Item hit
-
miss
0.00
0.02
0.04
Estimated marginal mean


Assoc. hit - miss
Item hit - miss
Assoc. - item hit
0.00
0.02
0.04
Δ HFB power



Stimulus category




```
## [1] "adapt_delta: 0.8"

## [1] "max_treedepth: 10"

## value ~ 1 + time + cue1_category + cue2_category + trial_type_combo + (1 | subj/chan)

## value ~ 1 + time + cue1_category + trial_type_combo + (1 | subj/chan)

##             elpd_diff se_diff  elpd_loo se_elpd_loo p_loo    se_p_loo looic
## study.mod.B      0.0       0.0 -21095.9    201.7        38.8      0.4  42191.9
## mod.simple     -44.0       9.9 -21139.9    203.4        35.4      0.3  42279.8
##             se_looic
## study.mod.B    403.4
## mod.simple     406.8

## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'study.mod.B' is the best model (ELPD =
## -21095.94), followed by 'mod.simple' (diff-ELPD = -43.95 +- 9.90, p < .001)
```

Table 29: study B emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD

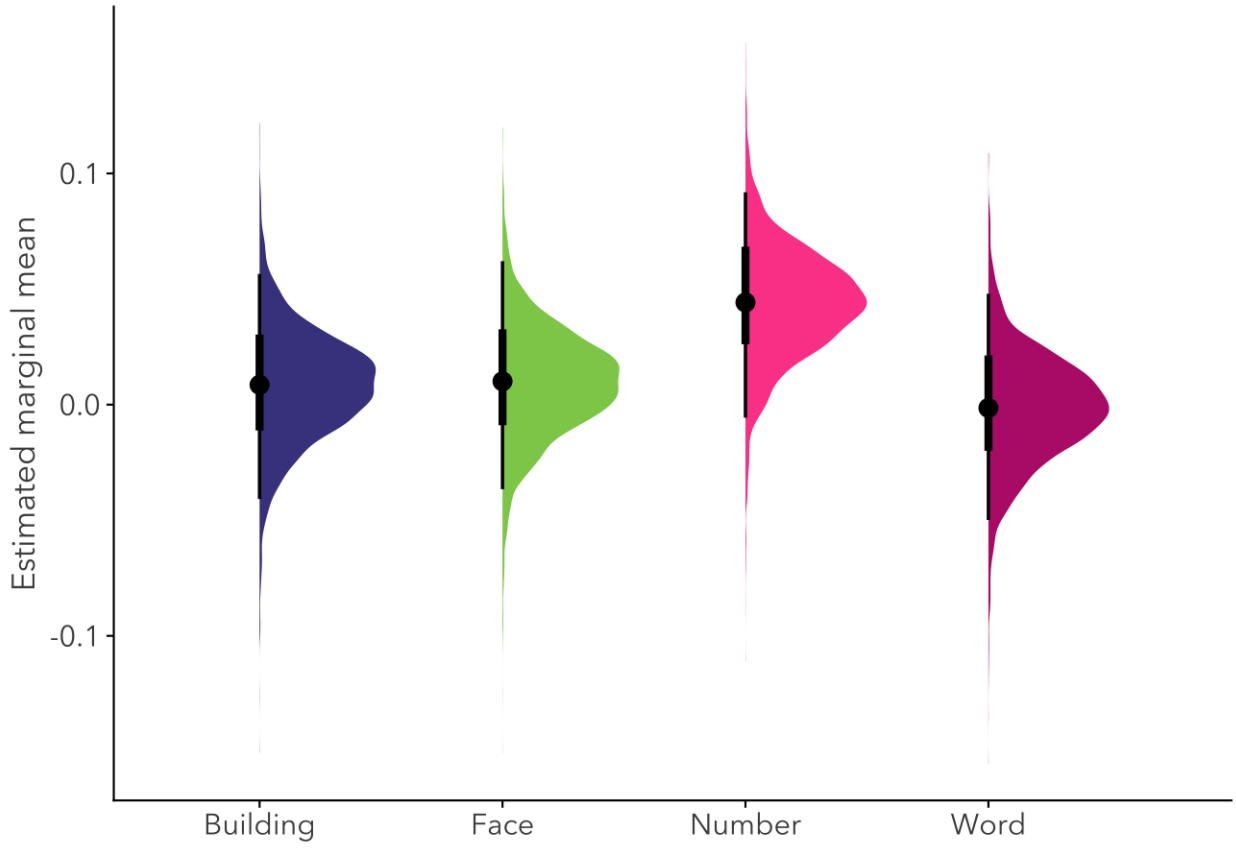
cue2_category	Face - Building	0	0.002	-0.008	0.011
cue2_category	Number - Building	0	0.036	0.027	0.045
cue2_category	Number - Face	0	0.034	0.024	0.044
cue2_category	Word - Building	0	-0.010	-0.019	0.000
cue2_category	Word - Face	0	-0.011	-0.022	-0.003
cue2_category	Word - Number	0	-0.045	-0.055	-0.036

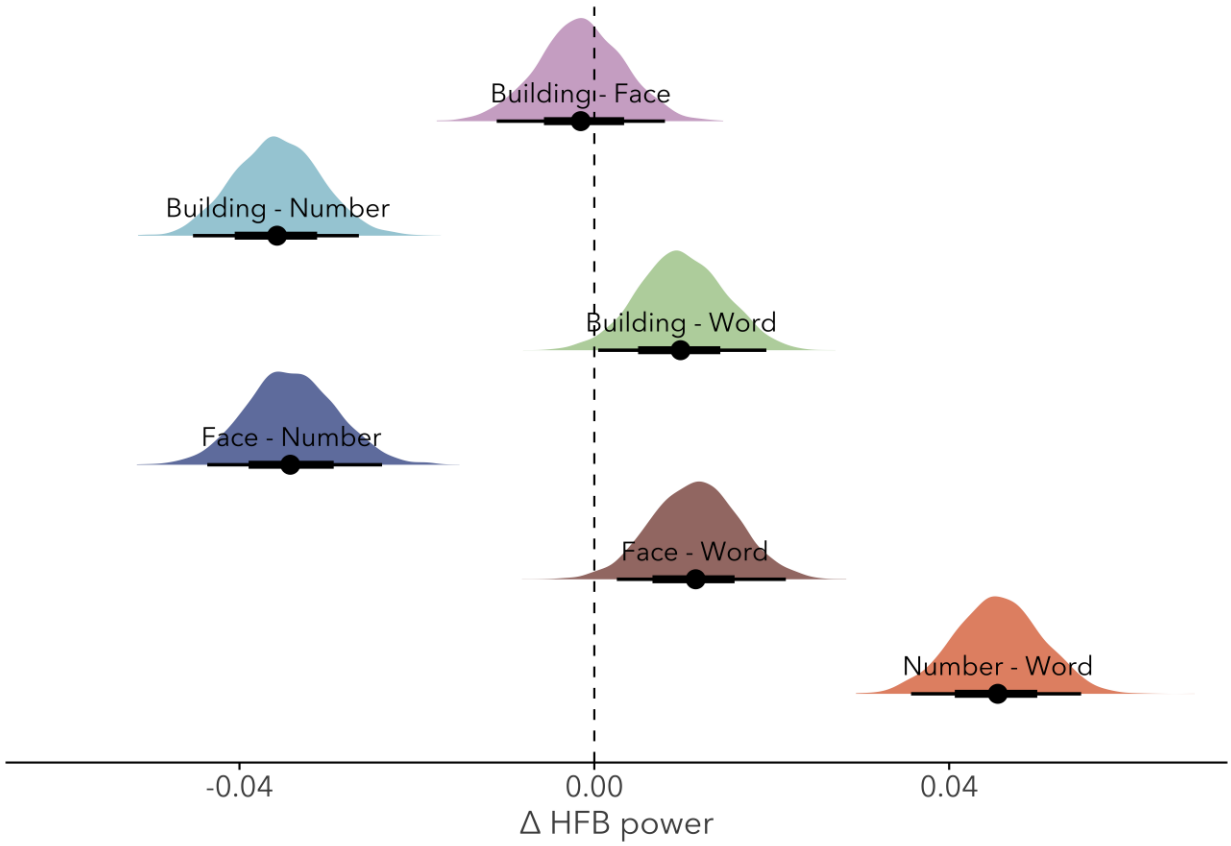
Table 30: study B emmeans

cue2_category	estimate	lower.HPD	upper.HPD
Building	0.009	-0.041	0.057
Face	0.010	-0.036	0.062
Number	0.044	-0.004	0.093
Word	-0.001	-0.049	0.048

Table 31: study B emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
cue2_category	Building - Face	0	-0.002	-0.011	0.008
cue2_category	Building - Number	0	-0.036	-0.045	-0.027
cue2_category	Building - Word	0	0.010	0.000	0.019
cue2_category	Face - Number	0	-0.034	-0.044	-0.024
cue2_category	Face - Word	0	0.011	0.003	0.022
cue2_category	Number - Word	0	0.045	0.036	0.055

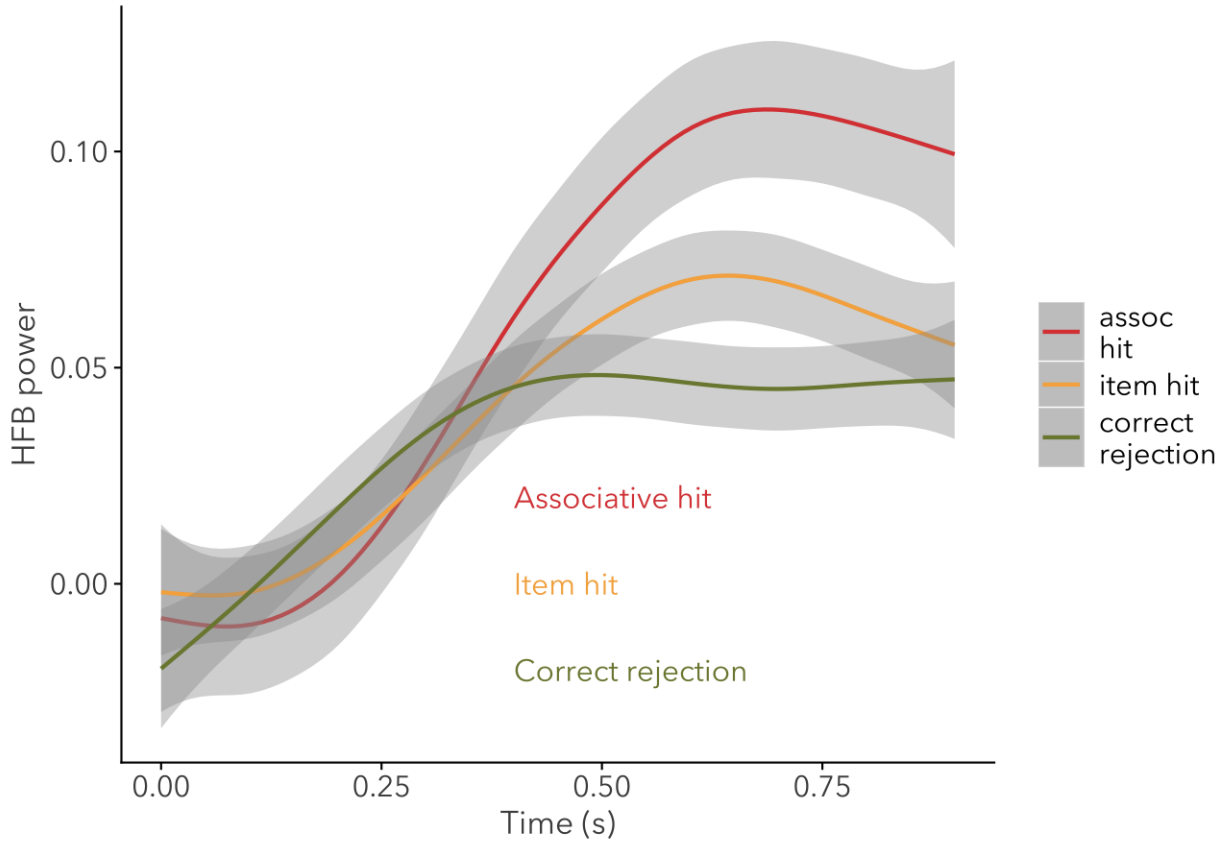

Estimated marginal mean
0.1
0.0
-0.1
Building
Face
Number
Word



Test cue

For test phase univariate analyses on the effects of item categories, *the lure trials were excluded*, and the model included time-bin, item A and item B categories, and subsequent memory outcome as predictors and controls.

For test phase univariate analyses on the effects of memory outcome, *all trials* were included, and time-bin and item A category were included as control variables.




```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: value ~ 1 + time + cue1_category + trial_type_combo + (1 | subj/chan)
##    Data: . (Number of observations: 48610)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 7)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.06      0.03     0.03     0.14 1.01      910     1089
##
## ~subj:chan (Number of levels: 30)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.03      0.01     0.02     0.04 1.00     1088     1551
##
## Regression Coefficients:
##                                   Estimate Est.Error l-95% CI u-95% CI Rhat
## Intercept                             0.01      0.03    -0.05     0.06 1.00
## time                                  0.08      0.01     0.07     0.09 1.00
## cue1_categoryBuilding                 0.02      0.00     0.01     0.03 1.00
```

```
## cue1_categoryWord                     -0.01      0.00    -0.01     0.00 1.00
## cue1_categoryNumber                   -0.04      0.00    -0.05    -0.03 1.00
## trial_type_comboitemhit               -0.01      0.00    -0.02    -0.00 1.00
## trial_type_combocorrect\nrejection    -0.02      0.00    -0.03    -0.01 1.00
## trial_type_combomiss                  -0.02      0.01    -0.04    -0.01 1.00
## trial_type_combofa                    -0.00      0.01    -0.02     0.02 1.00
##                                    Bulk_ESS Tail_ESS
## Intercept                              1159     1651
## time                                   4399     2637
## cue1_categoryBuilding                  3173     3121
## cue1_categoryWord                      3854     3131
## cue1_categoryNumber                    3905     3444
## trial_type_comboitemhit                3387     2645
## trial_type_combocorrect\nrejection     3351     3022
## trial_type_combomiss                   3238     2909
## trial_type_combofa                     3513     3051
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.33      0.00     0.33     0.34 1.00     6186     2901
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).

## [1] "adapt_delta: 0.99"

## [1] "max_treedepth: 10"

## value ~ 1 + time + cue1_category + trial_type_combo + (1 | subj/chan)

## value ~ 1 + time + cue1_category + (1 | subj/chan)

##              elpd_diff se_diff  elpd_loo se_elpd_loo p_loo    se_p_loo looic
## test.mod.cue      0.0       0.0 -15788.0    167.0        37.5      0.3  31575.9
## mod.simple       -8.5       5.0 -15796.5    167.0        33.7      0.2  31592.9
##              se_looic
## test.mod.cue    334.0
## mod.simple      334.0

## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'test.mod.cue' is the best model (ELPD =
## -15787.97), followed by 'mod.simple' (diff-ELPD = -8.48 +- 4.98, p = 0.089)
```

effect	component	term	estimate	std.error	conf.low	conf.high
fixed	cond	(Intercept)	0.0088074	0.0261859	-0.0455838	0.0584690
fixed	cond	time	0.0800591	0.0054177	0.0694950	0.0907850
fixed	cond	cue1_categoryBuilding	0.0211562	0.0040991	0.0132016	0.0293108
fixed	cond	cue1_categoryWord	-0.0050169	0.0043658	-0.0134885	0.0033507
fixed	cond	cue1_categoryNumber	-0.0377589	0.0045582	-0.0468364	-0.0290731

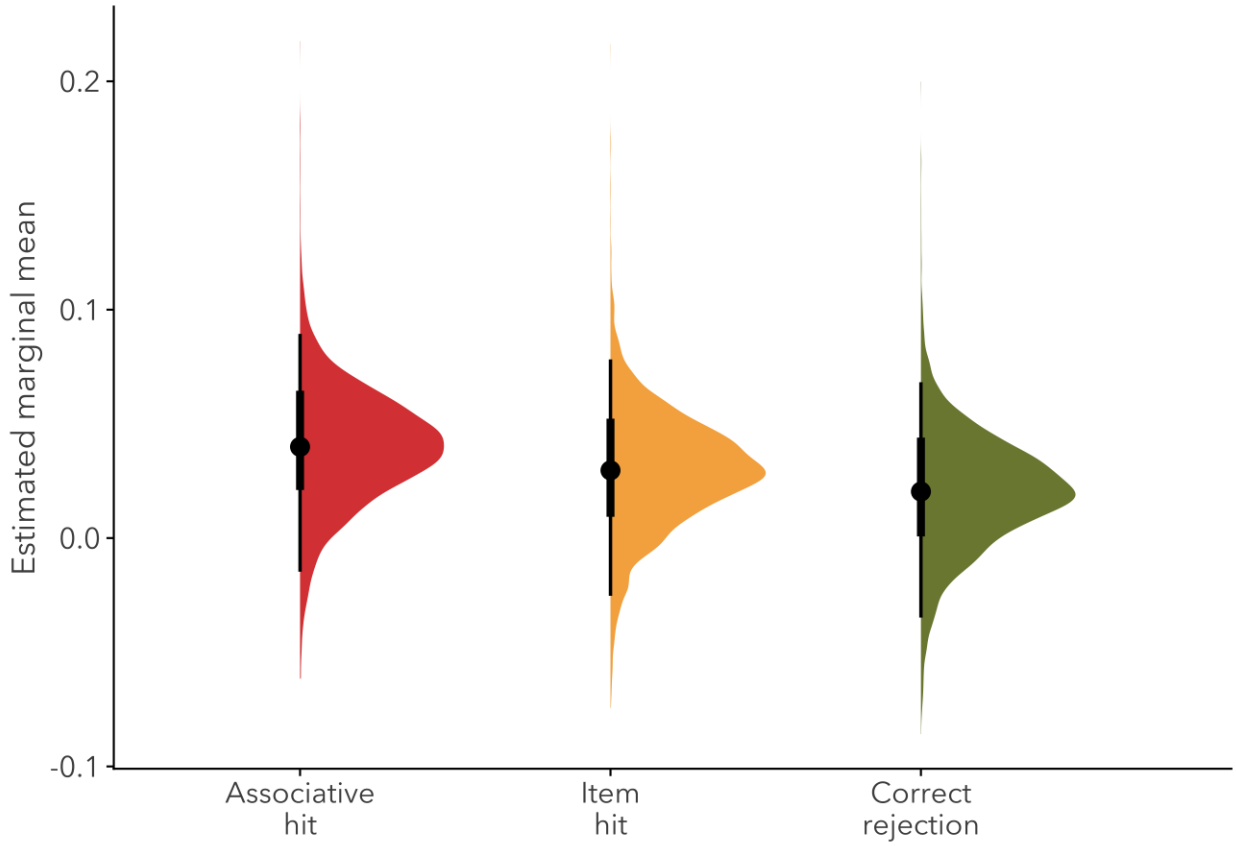
fixed	cond	trial_type_comboitemhit	-0.0101404	0.0049287	-0.0199487	-0.0004625
fixed	cond	trial_type_combocorrect rejection	-0.0194754	0.0047894	-0.0290169	-0.0101184
fixed	cond	trial_type_combomiss	-0.0244909	0.0065449	-0.0376094	-0.0118010
fixed	cond	trial_type_combofa	-0.0043356	0.0100596	-0.0238877	0.0150877

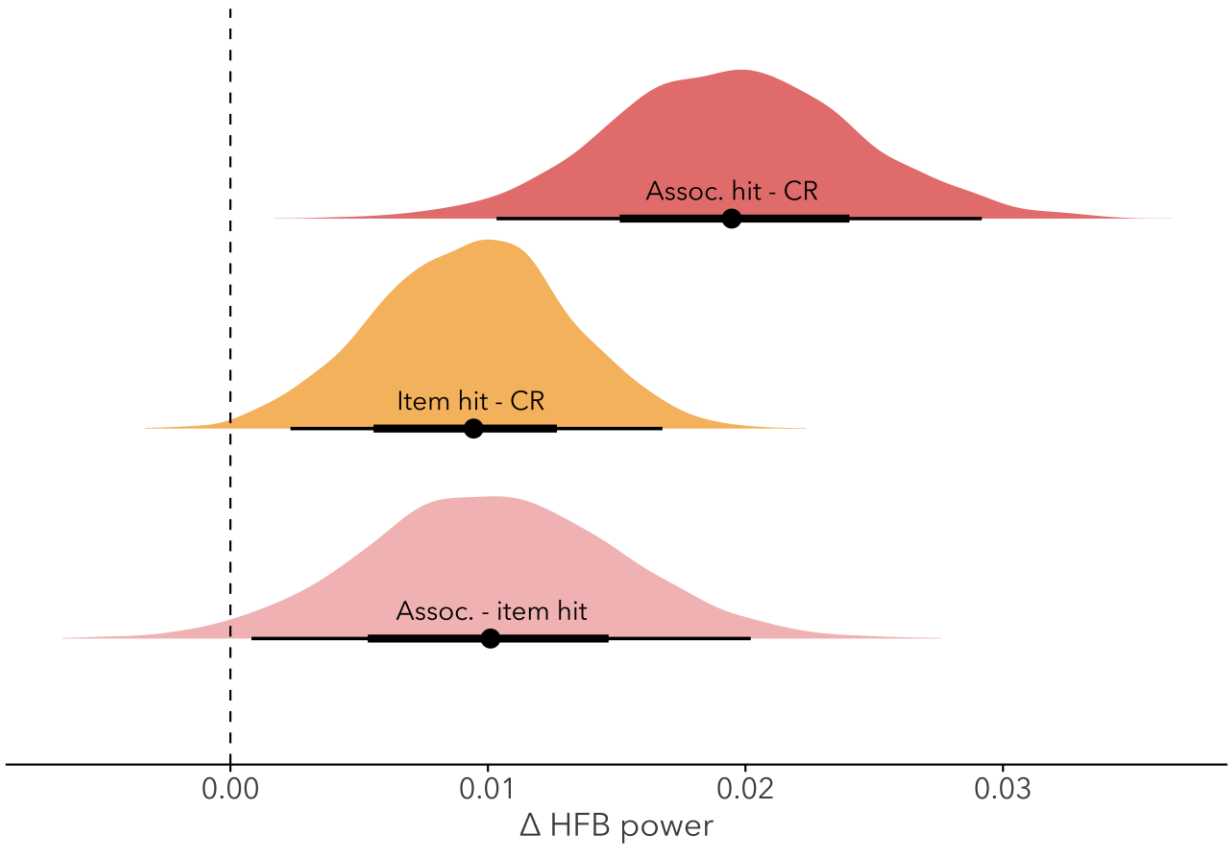
Table 33: test cue emmeans

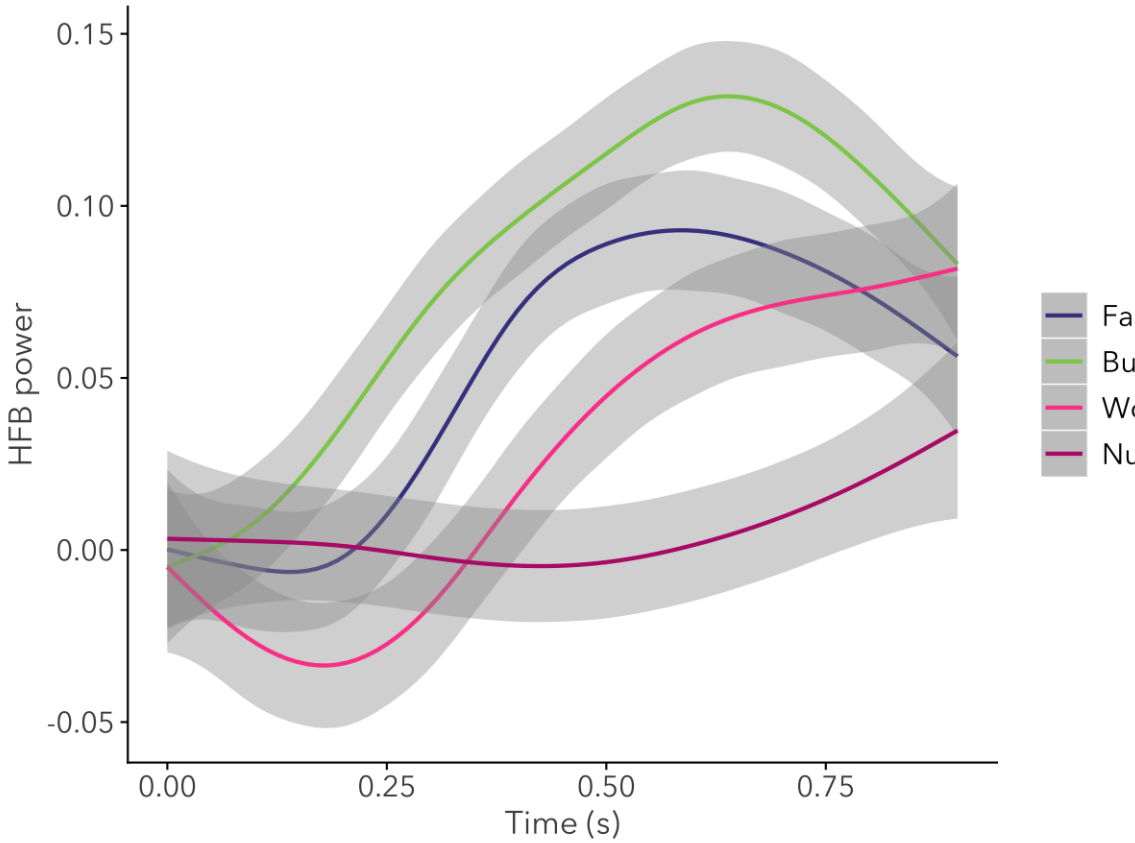
trial_type_combo	estimate	lower.HPD	upper.HPD
assoc hit	0.040	-0.015	0.089
item hit	0.030	-0.024	0.079
correct rejection	0.020	-0.035	0.068
miss	0.015	-0.040	0.063
fa	0.036	-0.020	0.089

Table 34: test cue emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
trial_type_combo	assoc hit - item hit	0	0.010	0.000	0.020
trial_type_combo	assoc hit - correct rejection	0	0.019	0.011	0.030
trial_type_combo	assoc hit - miss	0	0.025	0.012	0.038
trial_type_combo	assoc hit - fa	0	0.004	-0.016	0.023
trial_type_combo	item hit - correct rejection	0	0.009	0.002	0.017
trial_type_combo	item hit - miss	0	0.014	0.003	0.025
trial_type_combo	item hit - fa	0	-0.006	-0.025	0.012
trial_type_combo	correct rejection - miss	0	0.005	-0.006	0.015
trial_type_combo	correct rejection - fa	0	-0.015	-0.033	0.003
trial_type_combo	miss - fa	0	-0.020	-0.040	-0.001


Estimated marginal mean
0.2
0.1
0.0
-0.1
Associative hit
Item hit
Correct rejection


Assoc. hit - CR
Item hit - CR
Assoc. - item hit
0.00
0.01
0.02
0.03
Δ HFB power



Item category effects

```
## [1] "adapt_delta: 0.999"
```

```
## [1] "max_treedepth: 10"
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: value ~ 1 + time + cue1_category + cue2_category + trial_type_combo + (1 | subj/chan)
##    Data: . (Number of observations: 29670)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 7)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.07      0.03     0.02     0.14 1.01     1023     1420
##
## ~subj:chan (Number of levels: 30)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.04      0.01     0.03     0.05 1.00     1182     1889
##
## Regression Coefficients:
##                          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## Intercept                    0.01      0.03    -0.05     0.07 1.00     1380
## time                         0.09      0.01     0.08     0.10 1.00     4875
## cue1_categoryBuilding        0.01      0.01    -0.00     0.02 1.00     3477
```

```
## cue1_categoryWord          -0.04      0.01    -0.05    -0.03 1.00     3560
## cue1_categoryNumber        -0.05      0.01    -0.06    -0.04 1.00     3780
## cue2_categoryFace           0.03      0.01     0.02     0.04 1.00     3570
## cue2_categoryNumber         0.02      0.01     0.00     0.03 1.00     3750
## cue2_categoryWord          -0.02      0.01    -0.03    -0.00 1.00     3704
## trial_type_comboitemhit    -0.01      0.00    -0.02    -0.00 1.00     4301
## trial_type_combomiss       -0.03      0.01    -0.04    -0.02 1.00     4247
##                          Tail_ESS
## Intercept                    1746
## time                         3134
## cue1_categoryBuilding        3214
## cue1_categoryWord            3098
## cue1_categoryNumber          3160
## cue2_categoryFace            3449
## cue2_categoryNumber          3190
## cue2_categoryWord            3056
## trial_type_comboitemhit      3291
## trial_type_combomiss         3270
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.33      0.00     0.33     0.34 1.00     5866     2610
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

Stimulus cat A

```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
## value ~ 1 + time + cue1_category + cue2_category + trial_type_combo + (1 | subj/chan)
```

```
## value ~ 1 + time + cue2_category + trial_type_combo + (1 | subj/chan)
```

```
##              elpd_diff se_diff elpd_loo se_elpd_loo p_loo   se_p_loo looic
## test.mod.cue     0.0       0.0 -9529.7    128.9        38.5     0.4  19059.5
## mod.simple     -57.2      10.9 -9586.9    129.1        35.2     0.3  19173.8
##              se_looic
## test.mod.cue   257.7
## mod.simple     258.2
```

```
## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'test.mod.cue' is the best model (ELPD =
## -9529.73), followed by 'mod.simple' (diff-ELPD = -57.20 +- 10.87, p < .001)
```

Table 35: test cue emmeans

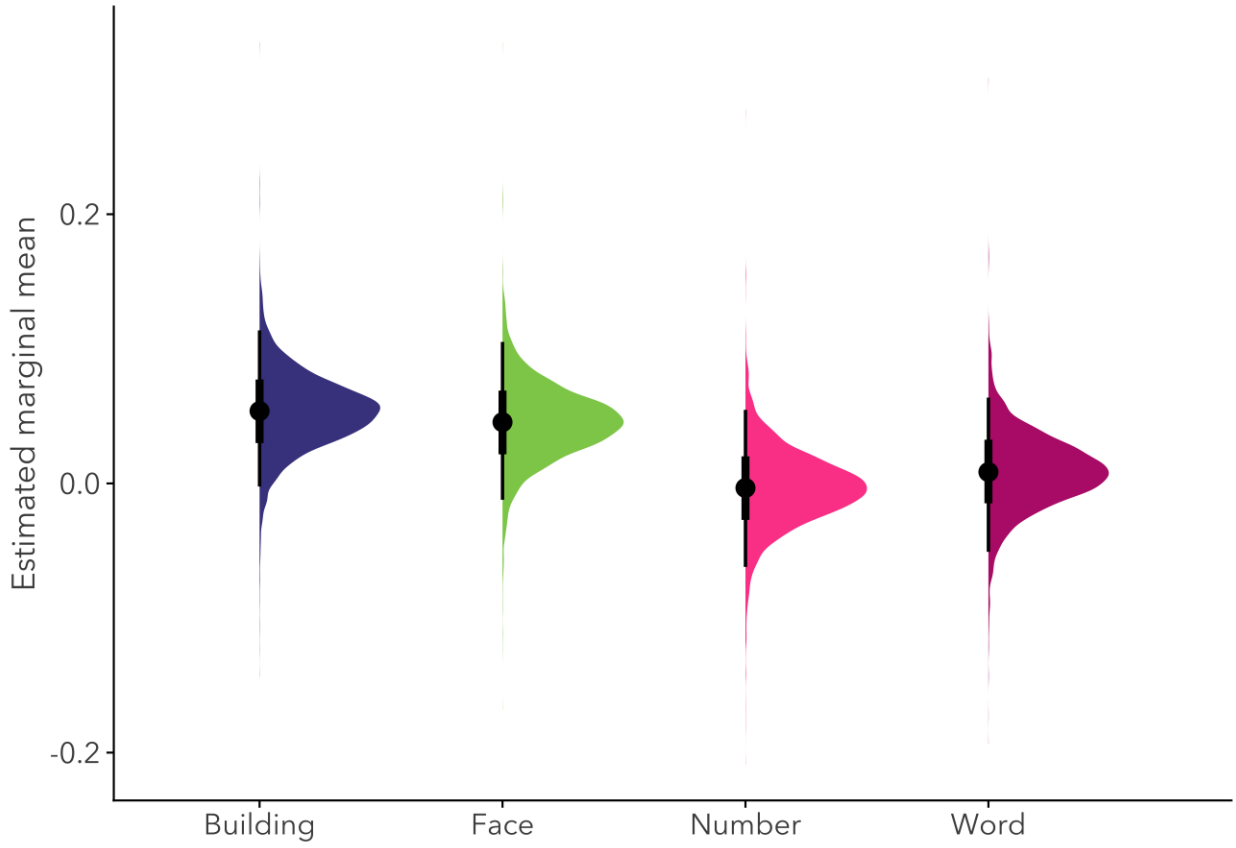
term	contrast	null.value	estimate	lower.HPD	upper.HPD
cue1_category	Building - Face	0	0.008	-0.003	0.020
cue1_category	Word - Face	0	-0.037	-0.048	-0.026
cue1_category	Word - Building	0	-0.045	-0.056	-0.033
cue1_category	Number - Face	0	-0.049	-0.060	-0.036
cue1_category	Number - Building	0	-0.057	-0.069	-0.045
cue1_category	Number - Word	0	-0.012	-0.024	0.000

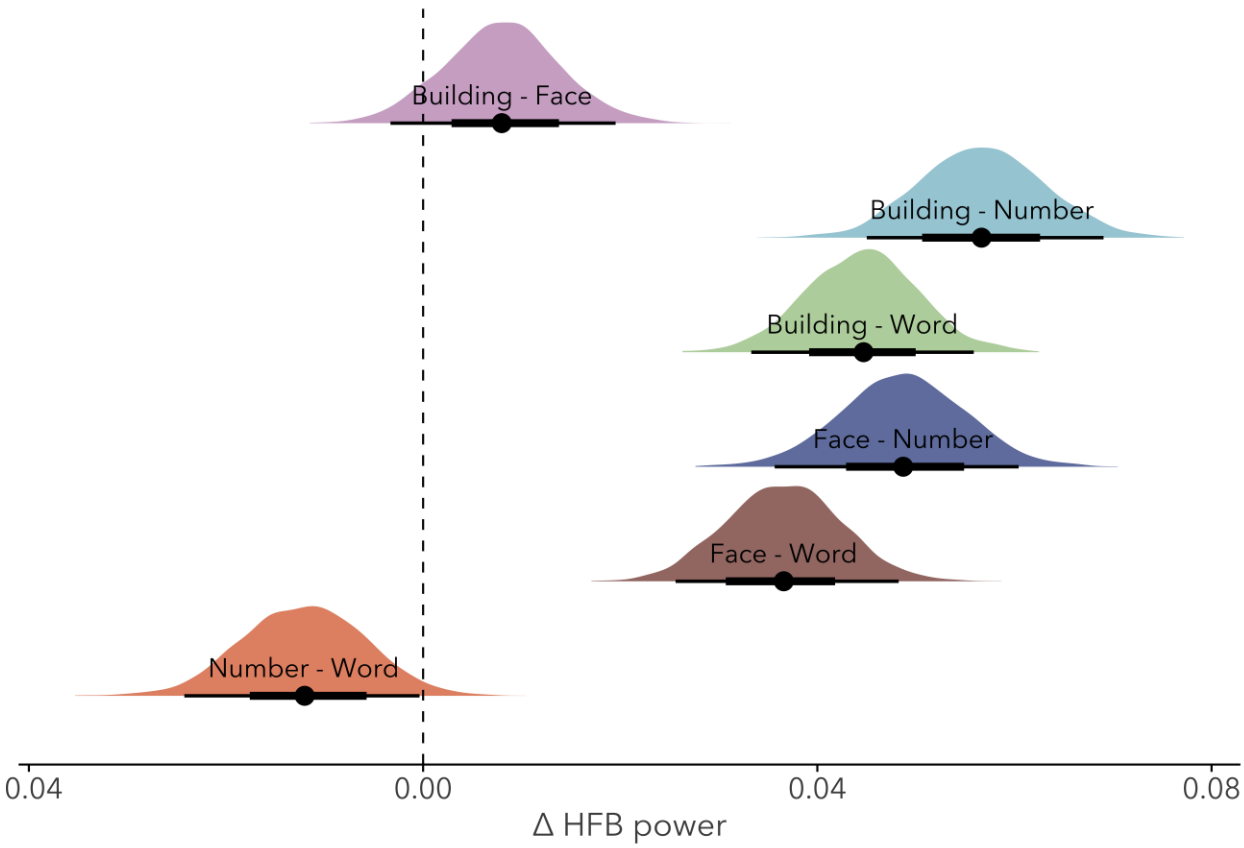
Table 36: test cue emmeans

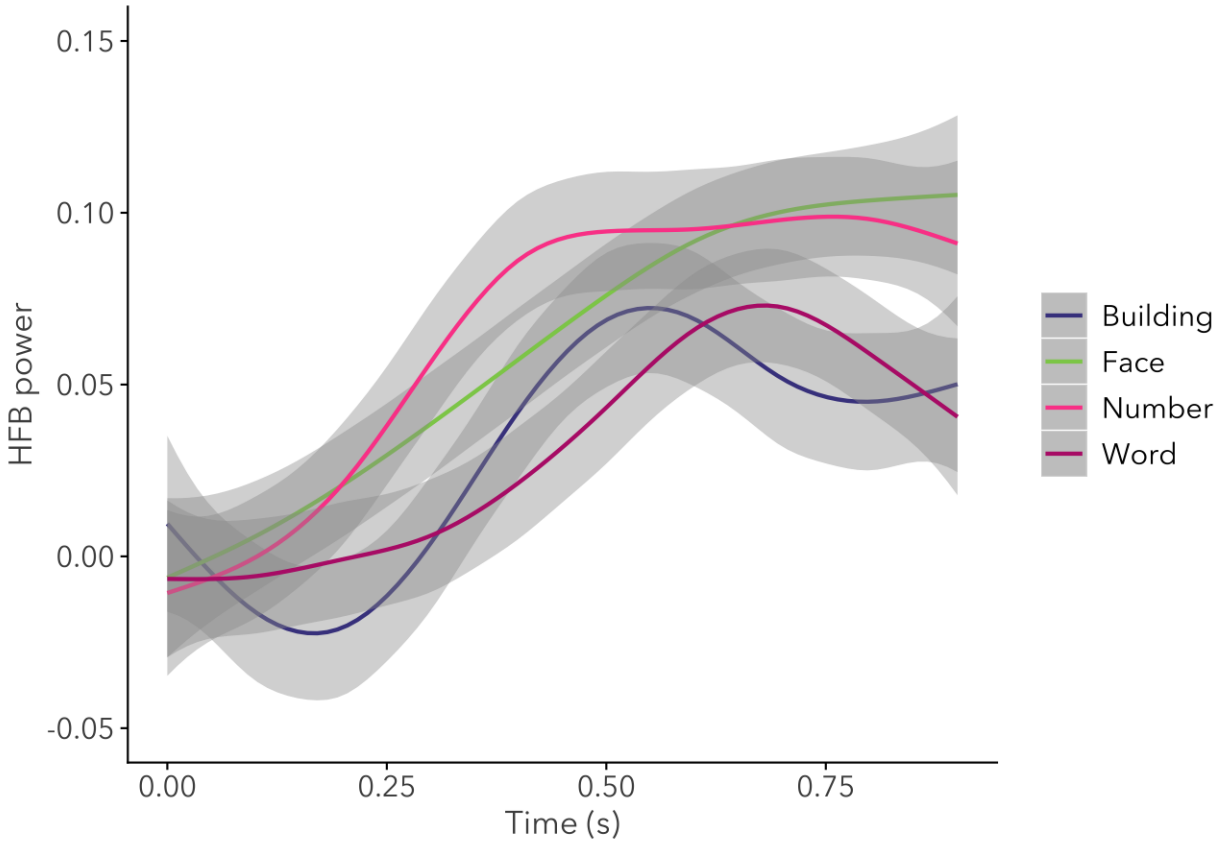
cue1_category	estimate	lower.HPD	upper.HPD
Face	0.045	-0.013	0.103
Building	0.054	-0.001	0.114
Word	0.008	-0.051	0.064
Number	-0.003	-0.062	0.055

Table 37: test cue emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
cue1_category	Face - Building	0	-0.008	-0.020	0.003
cue1_category	Face - Word	0	0.037	0.026	0.048
cue1_category	Face - Number	0	0.049	0.036	0.060
cue1_category	Building - Word	0	0.045	0.033	0.056
cue1_category	Building - Number	0	0.057	0.045	0.069
cue1_category	Word - Number	0	0.012	0.000	0.024


Estimated marginal mean
0.2
0.0
-0.2
Building
Face
Number
Word


Building - Face
Building - Number
Building - Word
Face - Number
Face - Word
Number - Word
0.04
0.00
0.04
0.08
Δ HFB power



Stimulus cat B

```
## [1] "adapt_delta: 0.8"
```

```
## [1] "max_treedepth: 10"
```

```
## value ~ 1 + time + cue1_category + cue2_category + trial_type_combo + (1 | subj/chan)
```

```
## value ~ 1 + time + cue1_category + trial_type_combo + (1 | subj/chan)
```

```
##              elpd_diff se_diff elpd_loo se_elpd_loo p_loo   se_p_loo looic
## test.mod.cue     0.0       0.0 -9529.7    128.9        38.5     0.4  19059.5
## mod.simple     -31.2       8.3 -9560.9    129.0        35.0     0.3  19121.8
##              se_looic
## test.mod.cue   257.7
## mod.simple     257.9
```

```
## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'test.mod.cue' is the best model (ELPD =
## -9529.73), followed by 'mod.simple' (diff-ELPD = -31.16 +- 8.30, p < .001)
```

Table 38: test cue emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD

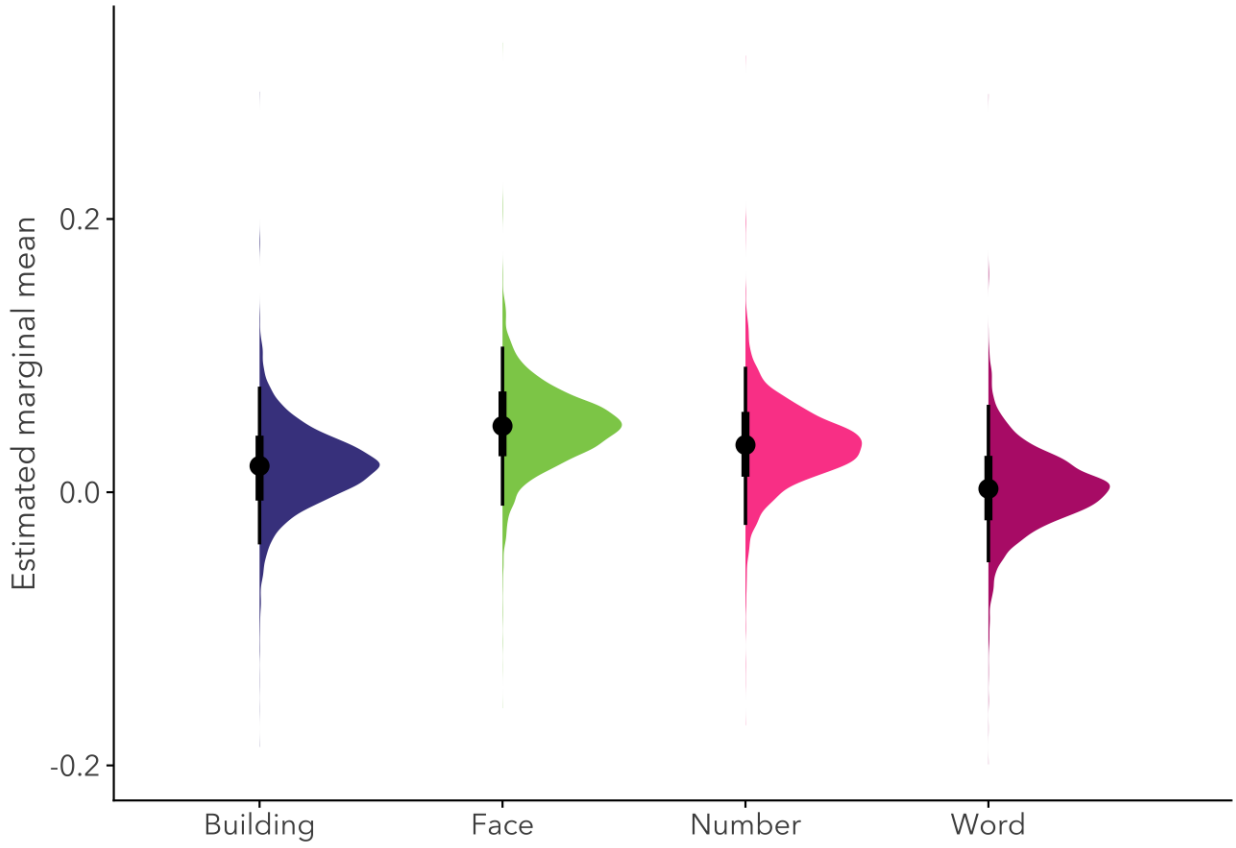
cue2_category	Face - Building	0	0.029	0.018	0.041
cue2_category	Number - Building	0	0.016	0.004	0.027
cue2_category	Number - Face	0	-0.014	-0.025	-0.002
cue2_category	Word - Building	0	-0.016	-0.028	-0.005
cue2_category	Word - Face	0	-0.046	-0.056	-0.034
cue2_category	Word - Number	0	-0.032	-0.044	-0.021

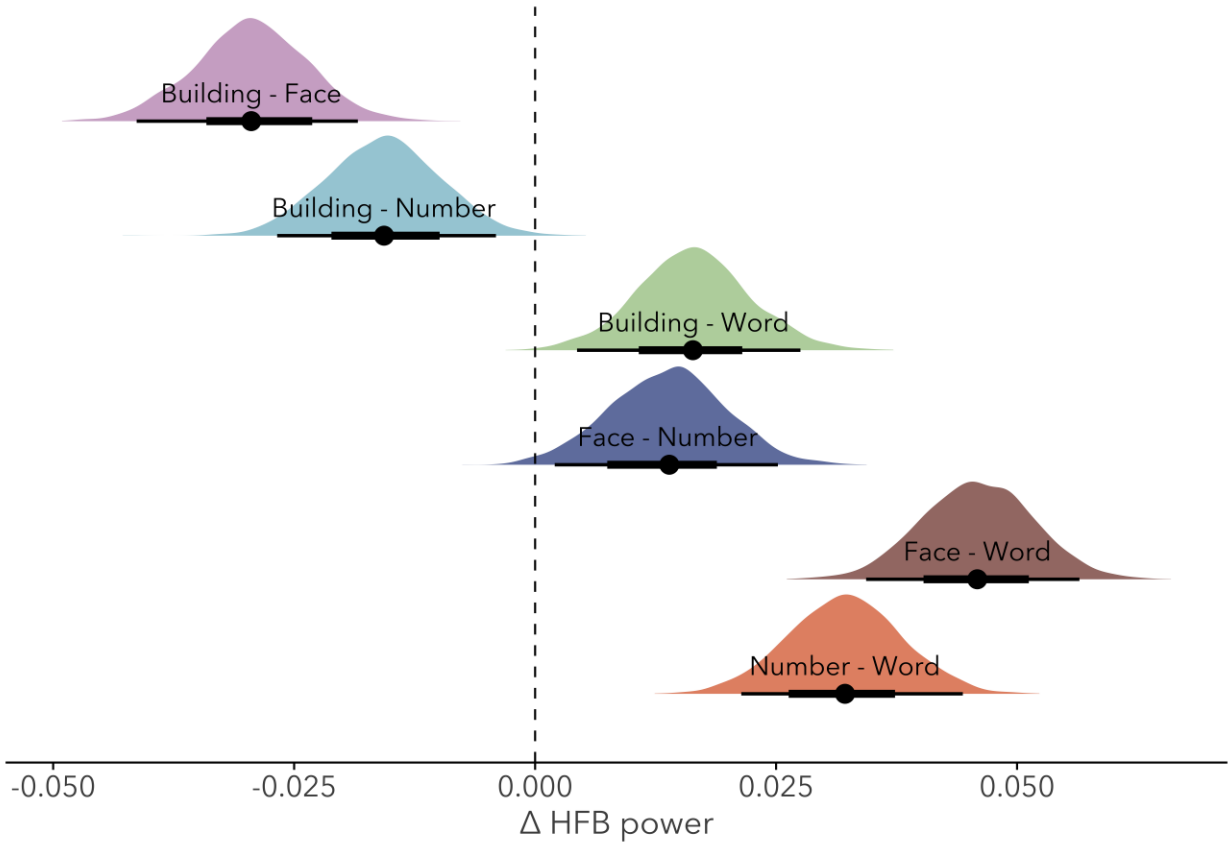
Table 39: test cue emmeans

cue2_category	estimate	lower.HPD	upper.HPD
Building	0.019	-0.038	0.077
Face	0.048	-0.010	0.106
Number	0.035	-0.024	0.091
Word	0.003	-0.053	0.062

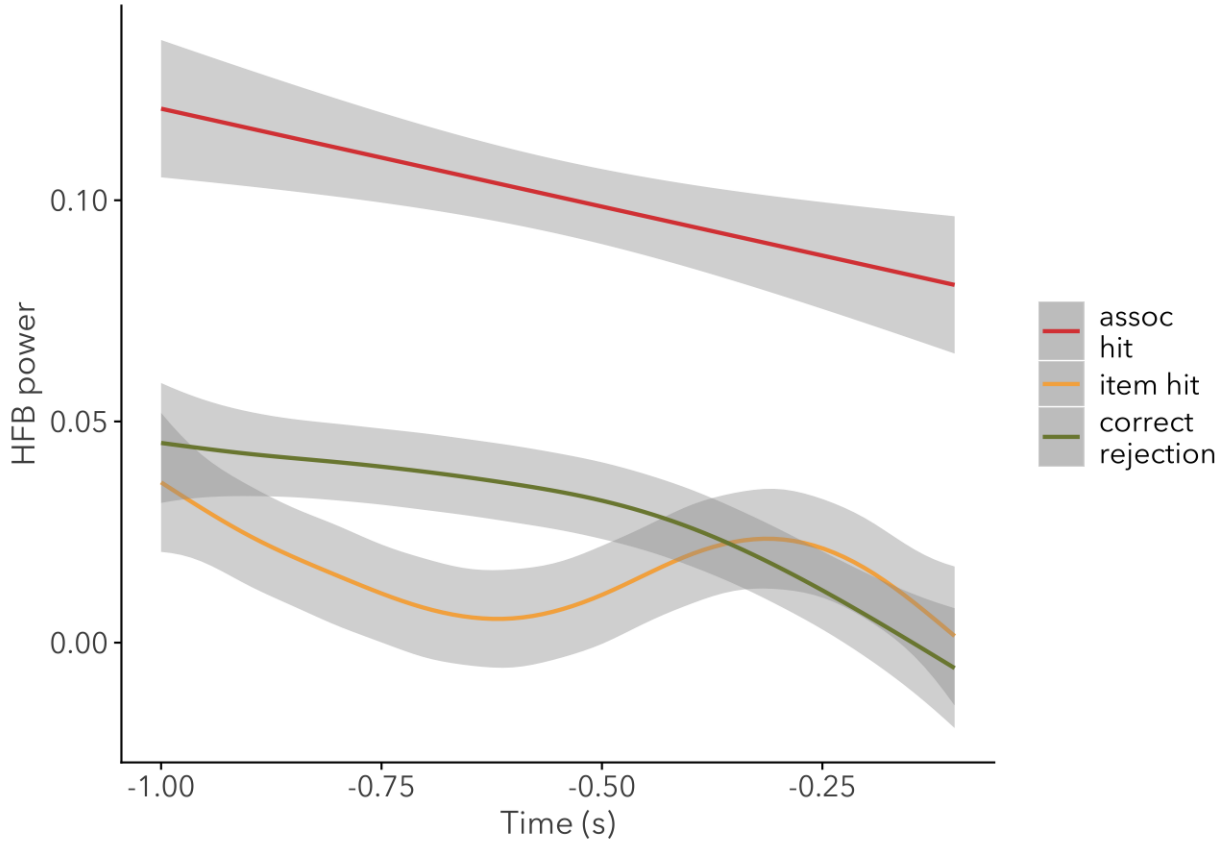
Table 40: test cue emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
cue2_category	Building - Face	0	-0.029	-0.041	-0.018
cue2_category	Building - Number	0	-0.016	-0.027	-0.004
cue2_category	Building - Word	0	0.016	0.005	0.028
cue2_category	Face - Number	0	0.014	0.002	0.025
cue2_category	Face - Word	0	0.046	0.034	0.056
cue2_category	Number - Word	0	0.032	0.021	0.044


Estimated marginal mean
0.2
0.0
-0.2
Building
Face
Number
Word


Building - Face
Building - Number
Building - Word
Face - Number
Face - Word
Number - Word
-0.050
-0.025
0.000
0.025
0.050
Δ HFB power

Test response




```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: value ~ 1 + time + cue1_category + trial_type_combo + (1 | subj/chan)
##    Data: . (Number of observations: 48610)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 7)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.08      0.03     0.04     0.17 1.00     1128     2018
##
## ~subj:chan (Number of levels: 30)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.03      0.01     0.02     0.05 1.00     1027     1837
##
## Regression Coefficients:
##                                  Estimate Est.Error l-95% CI u-95% CI Rhat
## Intercept                            0.05      0.03    -0.03     0.12 1.00
```

```
## time                                    -0.04      0.01    -0.05    -0.03 1.00
## cue1_categoryBuilding                    0.02      0.00     0.01     0.03 1.00
## cue1_categoryWord                        0.02      0.00     0.01     0.03 1.00
## cue1_categoryNumber                     -0.01      0.01    -0.02     0.00 1.00
## trial_type_comboitemhit                 -0.07      0.01    -0.08    -0.06 1.00
## trial_type_combocorrect\nrejection      -0.06      0.01    -0.07    -0.05 1.00
## trial_type_combomiss                    -0.08      0.01    -0.10    -0.07 1.00
## trial_type_combofa                      -0.04      0.01    -0.06    -0.02 1.00
##                                      Bulk_ESS Tail_ESS
## Intercept                                1640     1925
## time                                     4868     2904
## cue1_categoryBuilding                    4026     3132
## cue1_categoryWord                        4173     3157
## cue1_categoryNumber                      4338     3303
## trial_type_comboitemhit                  3969     3042
## trial_type_combocorrect\nrejection       3834     2882
## trial_type_combomiss                     4080     3447
## trial_type_combofa                       4597     3276
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.36      0.00     0.36     0.37 1.00     6057     3186
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).

## [1] "adapt_delta: 0.99"

## [1] "max_treedepth: 10"

## value ~ 1 + time + cue1_category + trial_type_combo + (1 | subj/chan)

## value ~ 1 + time + cue1_category + (1 | subj/chan)

##               elpd_diff se_diff  elpd_loo se_elpd_loo p_loo    se_p_loo
## test.mod.resp      0.0       0.0 -19910.8    163.6        38.1      0.3
## mod.simple      -102.8      14.6 -20013.6    163.3        33.8      0.2
##               looic    se_looic
## test.mod.resp  39821.6    327.2
## mod.simple     40027.2    326.5

## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'test.mod.resp' is the best model (ELPD =
## -19910.82), followed by 'mod.simple' (diff-ELPD = -102.76 +- 14.61, p < .001)
```

effect	component	term	estimate	std.error	conf.low	conf.high
fixed	cond	(Intercept)	0.0454877	0.0349505	-0.0265122	0.1160574
fixed	cond	time	-0.0424943	0.0058120	-0.0536533	-0.0310803
fixed	cond	cue1_categoryBuilding	0.0196673	0.0045921	0.0108953	0.0287988

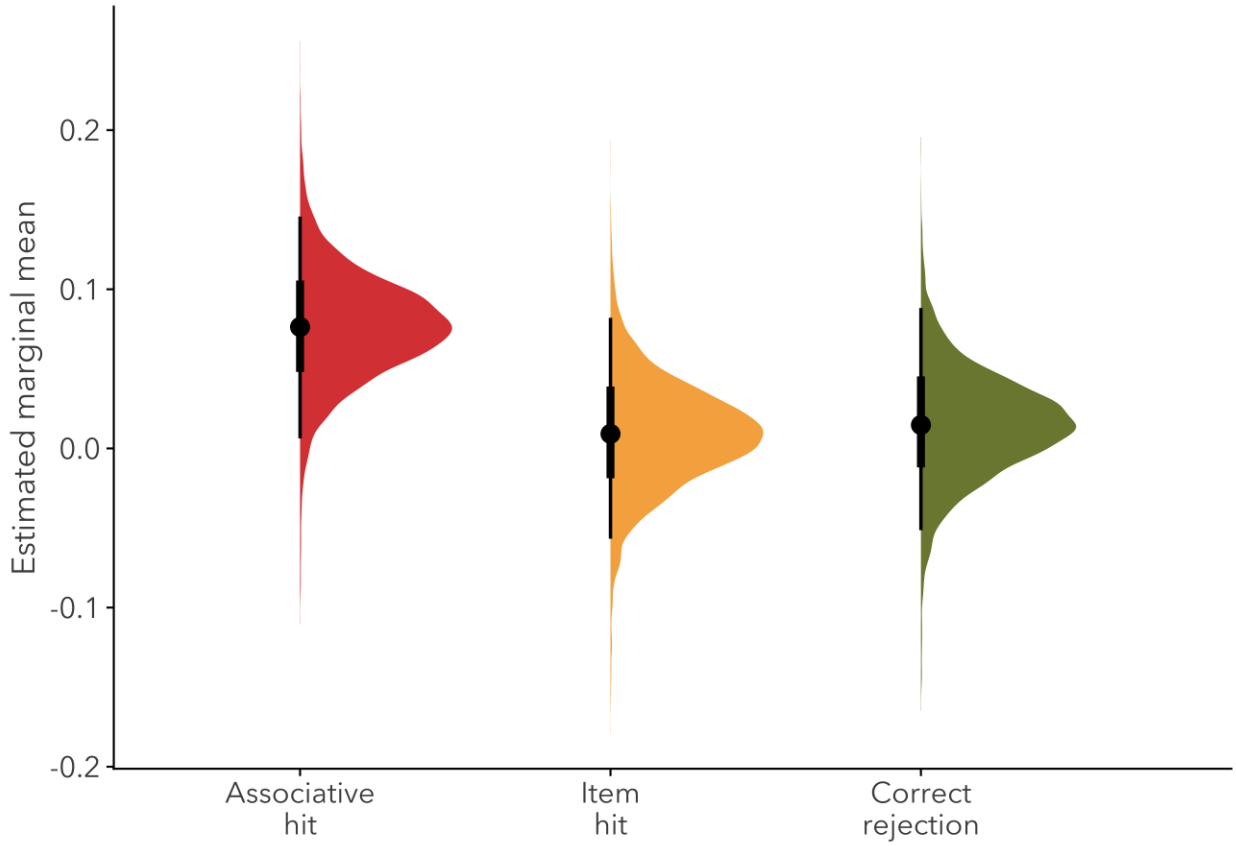
fixed	cond	cue1_categoryWord	0.0195889	0.0047932	0.0102508	0.0290305
fixed	cond	cue1_categoryNumber	-0.0101982	0.0050837	-0.0197851	0.0000795
fixed	cond	trial_type_comboitemhit	-0.0675105	0.0051813	-0.0774666	-0.0571782
fixed	cond	trial_type_combocorrect rejection	-0.0615582	0.0050784	-0.0716035	-0.0512409
fixed	cond	trial_type_combomiss	-0.0818937	0.0069227	-0.0953013	-0.0683660
fixed	cond	trial_type_combofa	-0.0416587	0.0108409	-0.0628473	-0.0206613

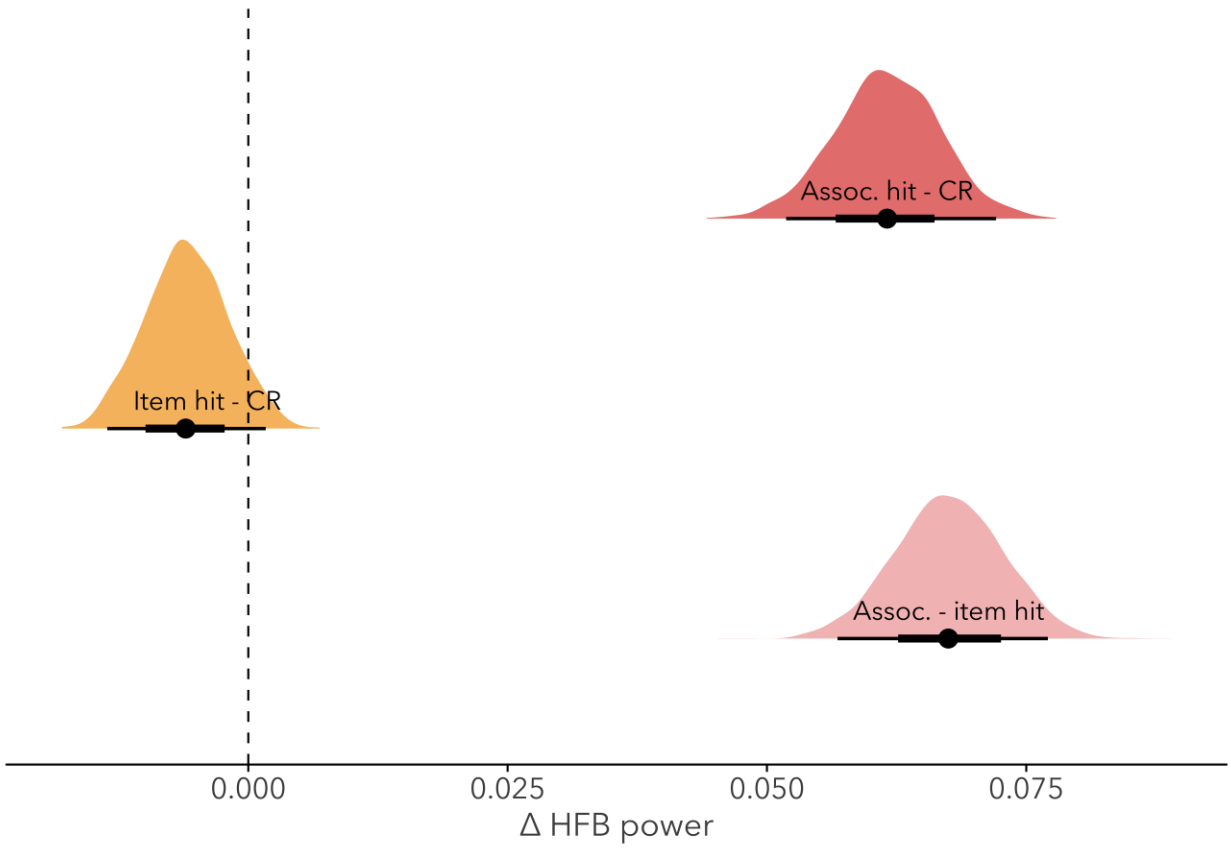
Table 42: test resp emmeans

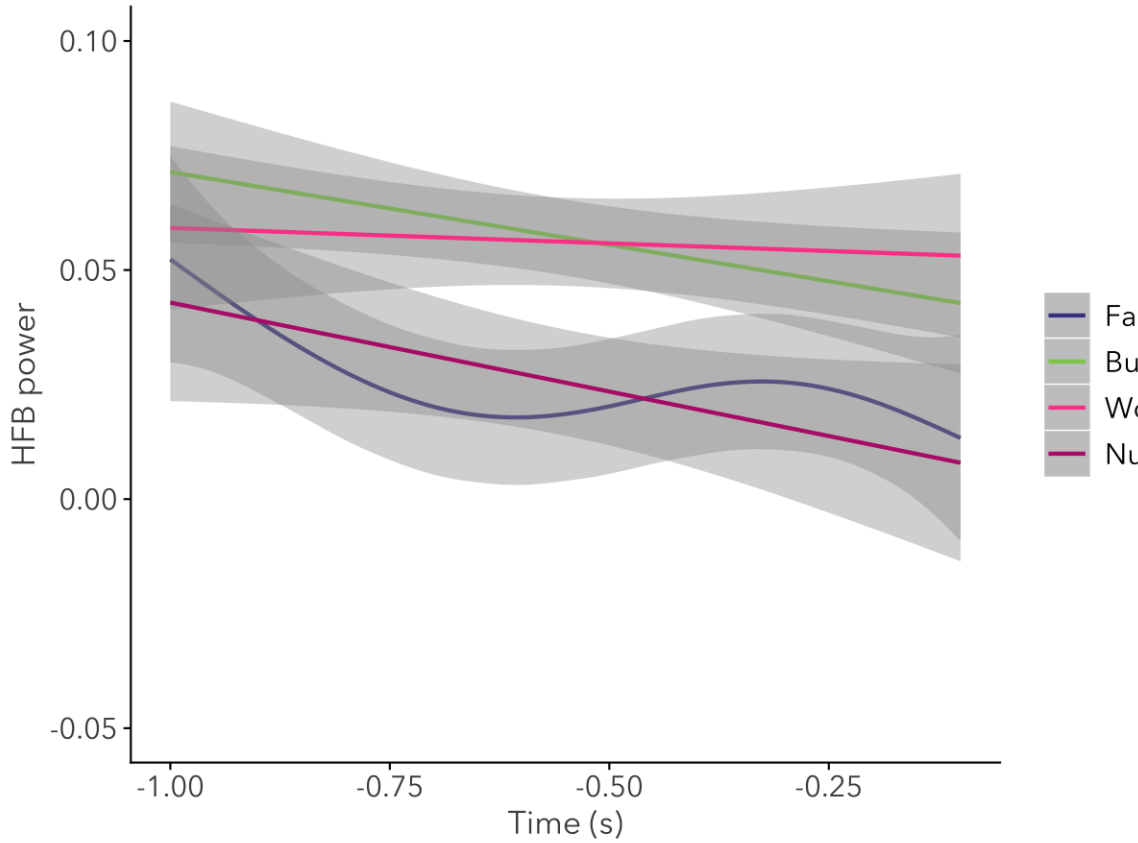
trial_type_combo	estimate	lower.HPD	upper.HPD
assoc hit	0.076	0.006	0.145
item hit	0.009	-0.055	0.084
correct rejection	0.015	-0.051	0.088
miss	-0.006	-0.075	0.063
fa	0.035	-0.035	0.107

Table 43: test resp emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
trial_type_combo	assoc hit - item hit	0	0.067	0.057	0.078
trial_type_combo	assoc hit - correct rejection	0	0.062	0.052	0.072
trial_type_combo	assoc hit - miss	0	0.082	0.068	0.095
trial_type_combo	assoc hit - fa	0	0.042	0.020	0.062
trial_type_combo	item hit - correct rejection	0	-0.006	-0.014	0.002
trial_type_combo	item hit - miss	0	0.014	0.003	0.027
trial_type_combo	item hit - fa	0	-0.026	-0.046	-0.005
trial_type_combo	correct rejection - miss	0	0.020	0.007	0.031
trial_type_combo	correct rejection - fa	0	-0.020	-0.040	0.001
trial_type_combo	miss - fa	0	-0.040	-0.062	-0.019


Estimated marginal mean
0.2
0.1
0.0
-0.1
-0.2
Associative hit
Item hit
Correct rejection


Assoc. hit - CR
Item hit - CR
Assoc. - item hit
0.000
0.025
0.050
0.075
Δ HFB power



Item category effects

```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: value ~ 1 + time + cue1_category + cue2_category + trial_type_combo + (1 | subj/chan)
##    Data: . (Number of observations: 29670)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 7)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.10      0.04     0.04     0.20 1.00     1172     1938
##
## ~subj:chan (Number of levels: 30)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.04      0.01     0.03     0.06 1.00     1326     1976
##
## Regression Coefficients:
##                          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## Intercept                    0.05      0.04    -0.04     0.12 1.00     1400
## time                        -0.04      0.01    -0.05    -0.02 1.00     4757
## cue1_categoryBuilding        0.01      0.01    -0.01     0.02 1.00     3779
```

```
## cue1_categoryWord           0.00      0.01    -0.01     0.02 1.00     4222
## cue1_categoryNumber        -0.02      0.01    -0.03    -0.00 1.00     3857
## cue2_categoryFace           0.03      0.01     0.01     0.04 1.00     3858
## cue2_categoryNumber         0.02      0.01     0.01     0.03 1.00     3885
## cue2_categoryWord          -0.00      0.01    -0.02     0.01 1.00     3851
## trial_type_comboitemhit    -0.07      0.01    -0.08    -0.06 1.00     4412
## trial_type_combomiss       -0.09      0.01    -0.10    -0.07 1.00     4648
##                          Tail_ESS
## Intercept                    1819
## time                         2948
## cue1_categoryBuilding        3210
## cue1_categoryWord            3102
## cue1_categoryNumber          2984
## cue2_categoryFace            3242
## cue2_categoryNumber          3031
## cue2_categoryWord            3137
## trial_type_comboitemhit      2941
## trial_type_combomiss         3582
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.36      0.00     0.36     0.37 1.00     6538     2982
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

Stimulus cat A

```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
## value ~ 1 + time + cue1_category + cue2_category + trial_type_combo + (1 | subj/chan)
```

```
## value ~ 1 + time + cue2_category + trial_type_combo + (1 | subj/chan)
```

```
##               elpd_diff se_diff  elpd_loo se_elpd_loo p_loo    se_p_loo
## test.mod.resp      0.0       0.0 -11971.9    126.4        38.8      0.4
## mod.simple        -3.6       3.7 -11975.5    126.4        35.6      0.3
##               looic    se_looic
## test.mod.resp  23943.8    252.7
## mod.simple     23951.1    252.7
```

```
## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'test.mod.resp' is the best model (ELPD =
## -11971.92), followed by 'mod.simple' (diff-ELPD = -3.63 +- 3.65, p = 0.320)
```

Table 44: test resp emmeans

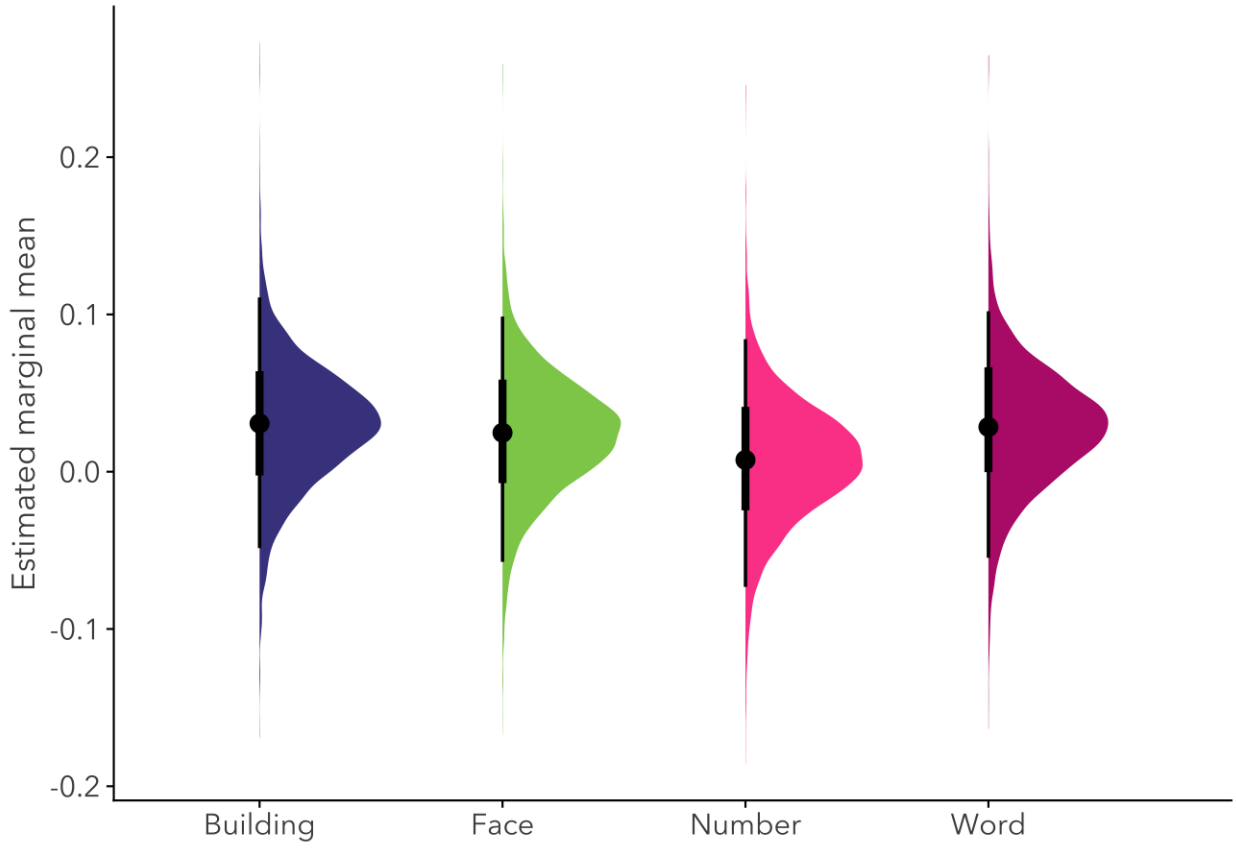
term	contrast	null.value	estimate	lower.HPD	upper.HPD
cue1_category	Building - Face	0	0.006	-0.006	0.018
cue1_category	Word - Face	0	0.004	-0.009	0.016
cue1_category	Word - Building	0	-0.003	-0.014	0.011
cue1_category	Number - Face	0	-0.017	-0.030	-0.003
cue1_category	Number - Building	0	-0.023	-0.036	-0.009
cue1_category	Number - Word	0	-0.021	-0.035	-0.008

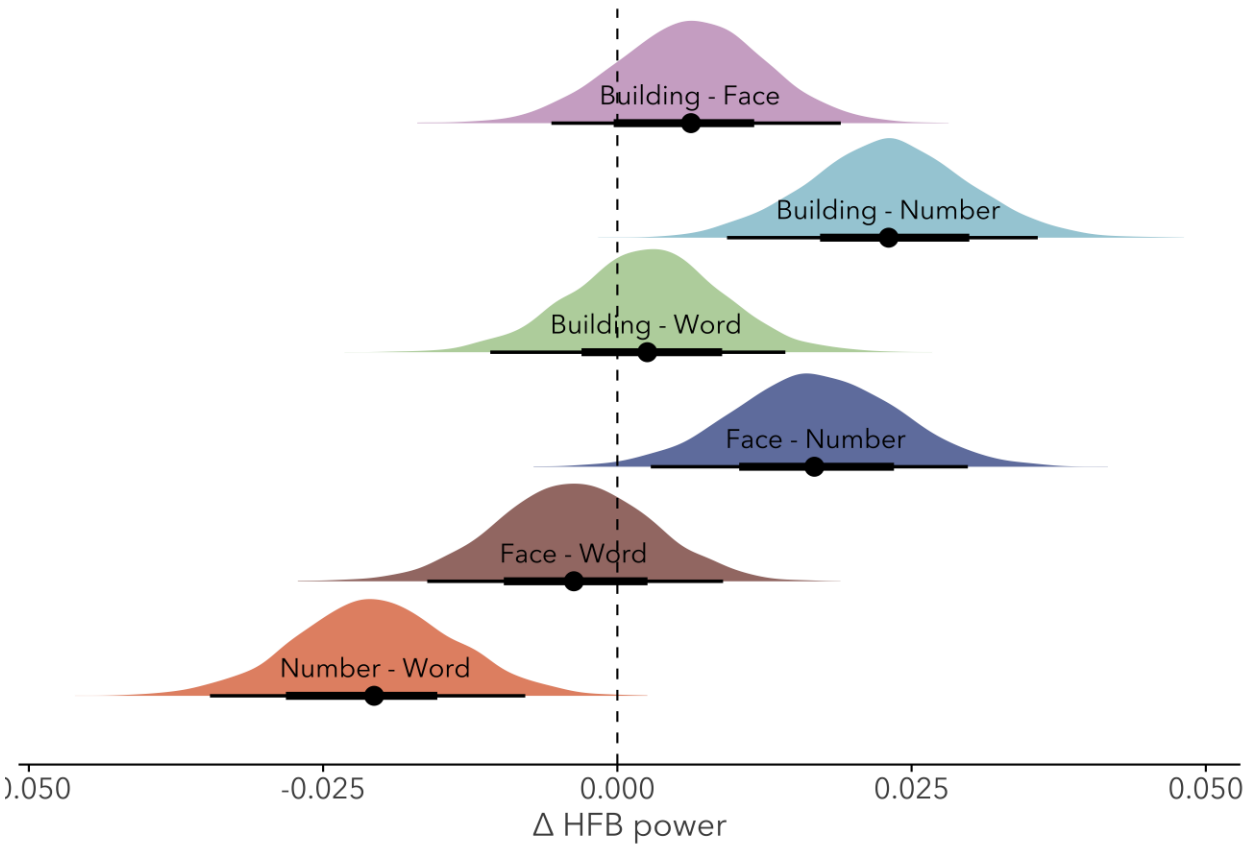
Table 45: test resp emmeans

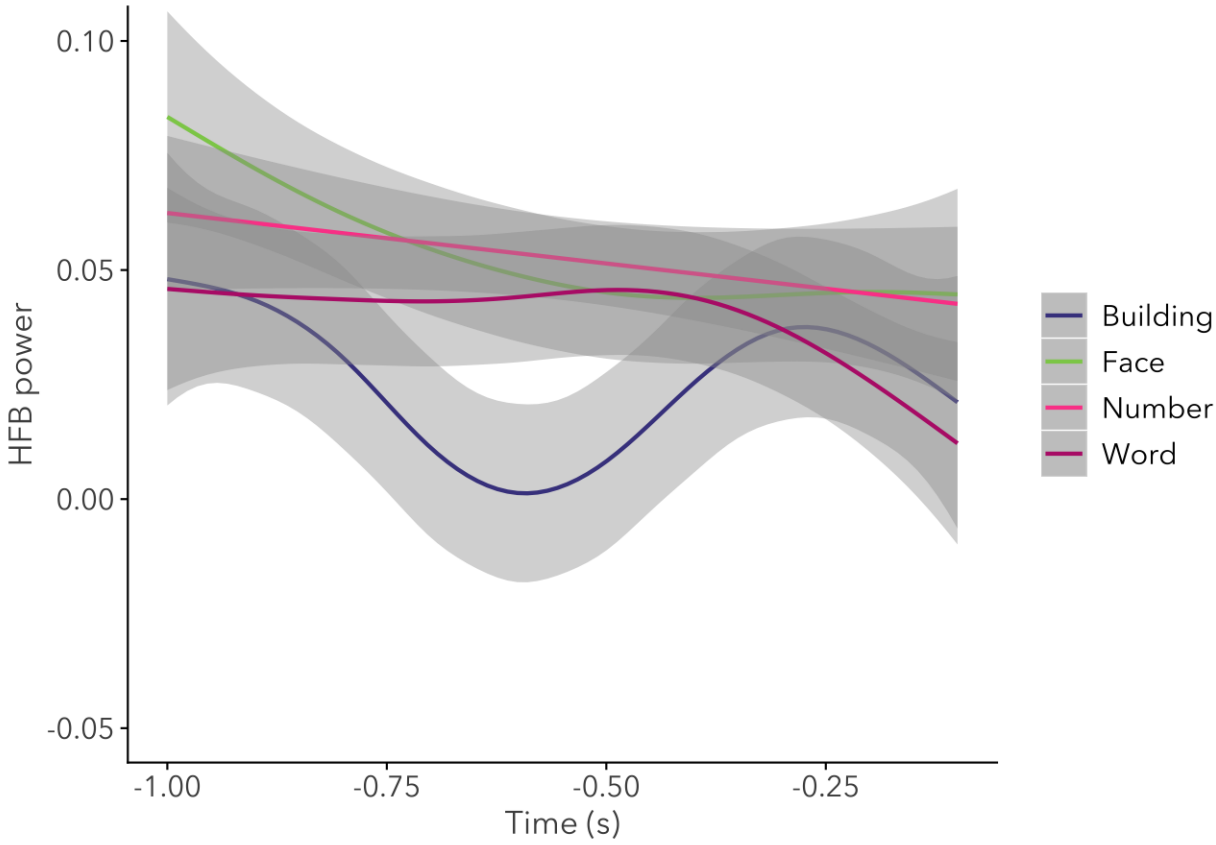
cue1_category	estimate	lower.HPD	upper.HPD
Face	0.025	-0.057	0.099
Building	0.031	-0.056	0.103
Word	0.028	-0.056	0.101
Number	0.008	-0.073	0.084

Table 46: test resp emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
cue1_category	Face - Building	0	-0.006	-0.018	0.006
cue1_category	Face - Word	0	-0.004	-0.016	0.009
cue1_category	Face - Number	0	0.017	0.003	0.030
cue1_category	Building - Word	0	0.003	-0.011	0.014
cue1_category	Building - Number	0	0.023	0.009	0.036
cue1_category	Word - Number	0	0.021	0.008	0.035


Estimated marginal mean
0.2
0.1
0.0
-0.1
-0.2
Building
Face
Number
Word


Building - Face
Building - Number
Building - Word
Face - Number
Face - Word
Number - Word
.050
-0.025
0.000
0.025
0.050
Δ HFB power



Stimulus cat B

```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
## value ~ 1 + time + cue1_category + cue2_category + trial_type_combo + (1 | subj/chan)
```

```
## value ~ 1 + time + cue1_category + trial_type_combo + (1 | subj/chan)
```

```
##               elpd_diff se_diff  elpd_loo se_elpd_loo p_loo    se_p_loo
## test.mod.resp      0.0       0.0 -11971.9    126.4        38.8      0.4
## mod.simple       -13.0       5.7 -11984.9    126.3        35.4      0.3
##               looic    se_looic
## test.mod.resp  23943.8    252.7
## mod.simple     23969.7    252.7
```

```
## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'test.mod.resp' is the best model (ELPD =
## -11971.92), followed by 'mod.simple' (diff-ELPD = -12.95 +- 5.67, p = 0.022)
```

Table 47: test resp emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD

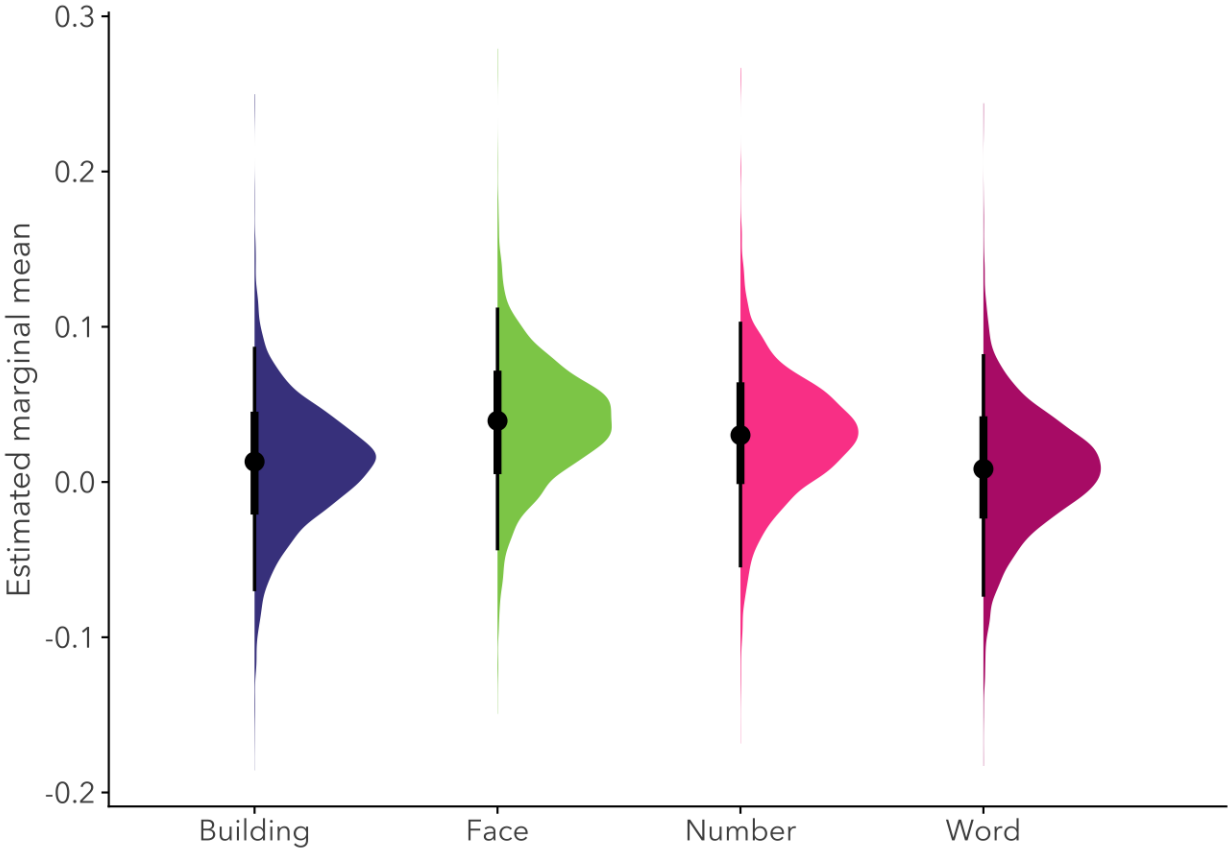
cue2_category	Face - Building	0	0.027	0.015	0.040
cue2_category	Number - Building	0	0.018	0.005	0.030
cue2_category	Number - Face	0	-0.009	-0.021	0.004
cue2_category	Word - Building	0	-0.004	-0.017	0.008
cue2_category	Word - Face	0	-0.031	-0.044	-0.018
cue2_category	Word - Number	0	-0.022	-0.034	-0.009

Table 48: test resp emmeans

cue2_category	estimate	lower.HPD	upper.HPD
Building	0.013	-0.070	0.088
Face	0.039	-0.044	0.112
Number	0.030	-0.055	0.103
Word	0.008	-0.071	0.085

Table 49: test resp emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
cue2_category	Building - Face	0	-0.027	-0.040	-0.015
cue2_category	Building - Number	0	-0.018	-0.030	-0.005
cue2_category	Building - Word	0	0.004	-0.008	0.017
cue2_category	Face - Number	0	0.009	-0.004	0.021
cue2_category	Face - Word	0	0.031	0.018	0.044
cue2_category	Number - Word	0	0.022	0.009	0.034


0.3
0.2
0.1
0.0
-0.1
-0.2
Estimated marginal mean
Building
Face
Number
Word

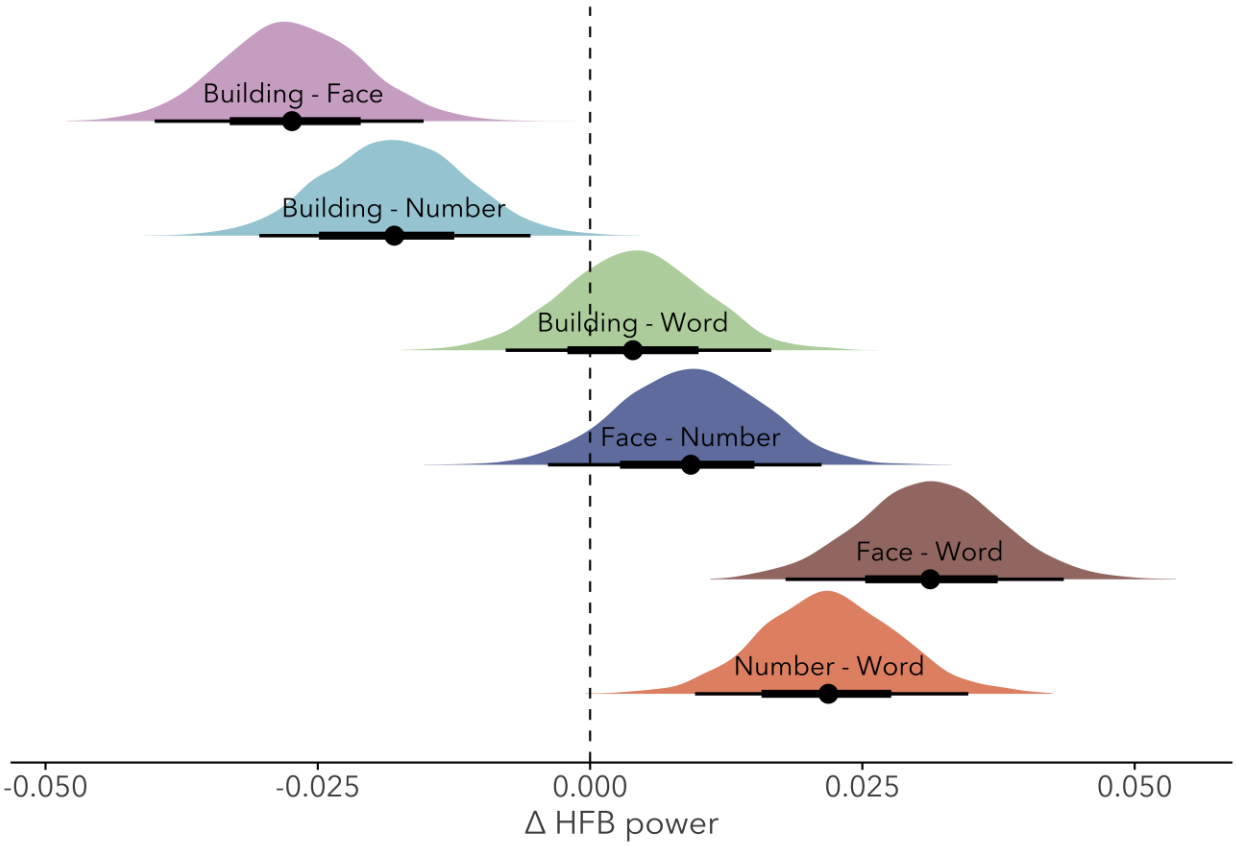
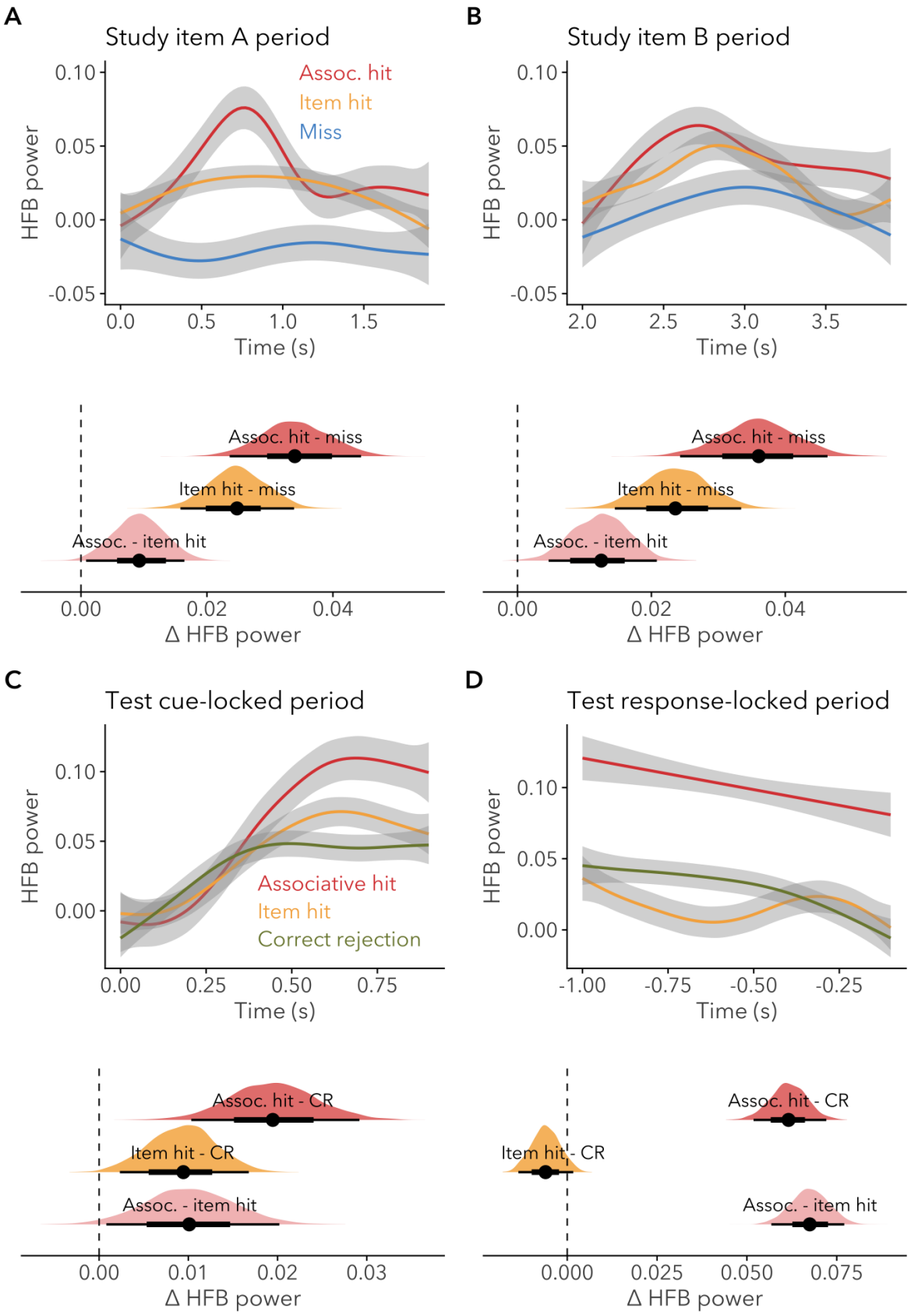

Building - Face
Building - Number
Building - Word
Face - Number
Face - Word
Number - Word
-0.050
-0.025
0.000
0.025
0.050
Δ HFB power

Fig 2


A
Study item A period
Assoc. hit
Item hit
Miss
HFB power
Time (s)
Assoc. hit - miss
Item hit - miss
Assoc. - item hit
Δ HFB power
B
Study item B period
C
Test cue-locked period
Associative hit
Item hit
Correct rejection
Assoc. hit - CR
Item hit - CR
D
Test response-locked period

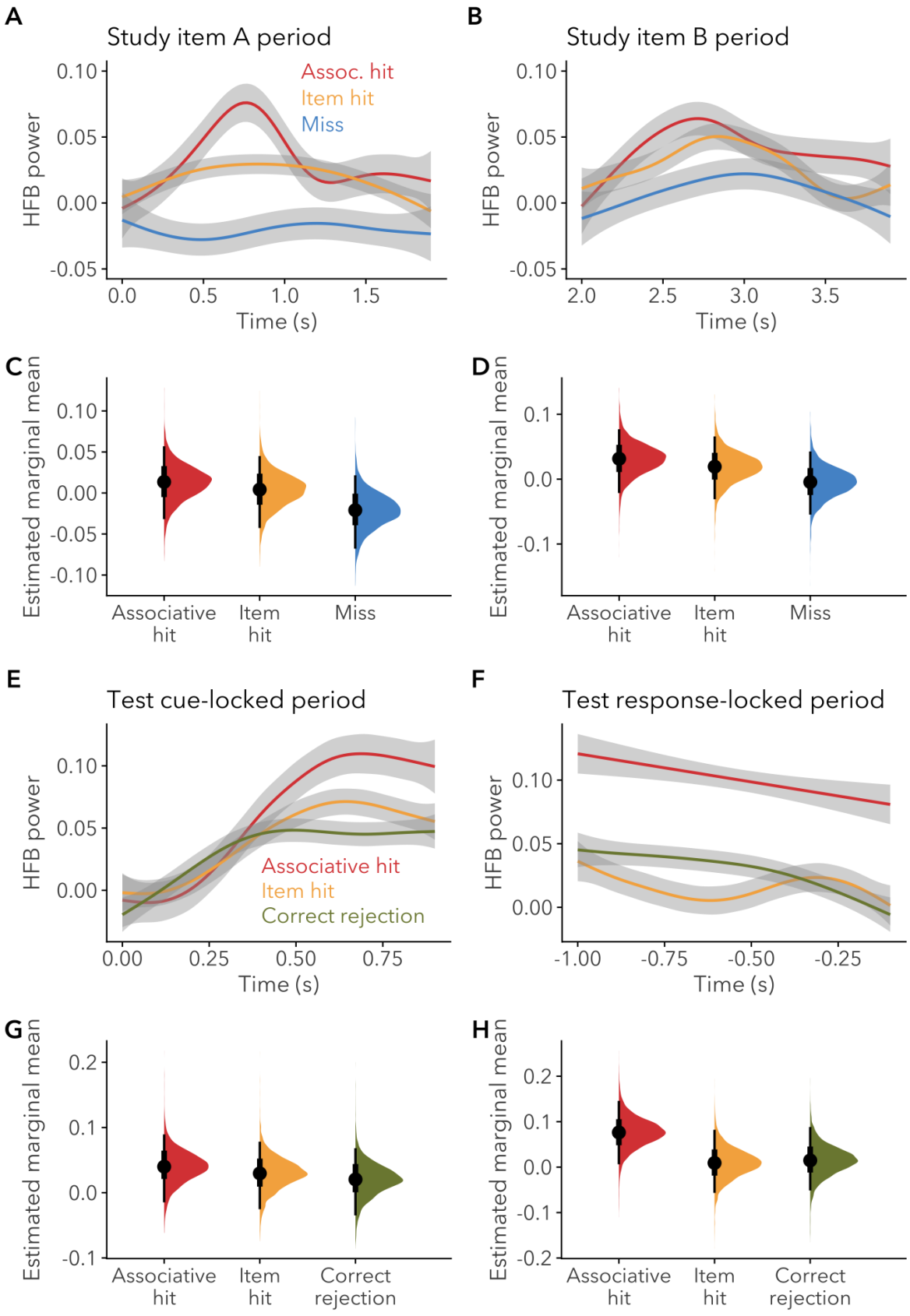

A
Study item A period
Assoc. hit
Item hit
Miss
HFB power
Time (s)
B
Study item B period
C
Estimated marginal mean
Associative hit
Item hit
Miss
D
E
Test cue-locked period
Associative hit
Item hit
Correct rejection
F
Test response-locked period
G
H

Fig S6

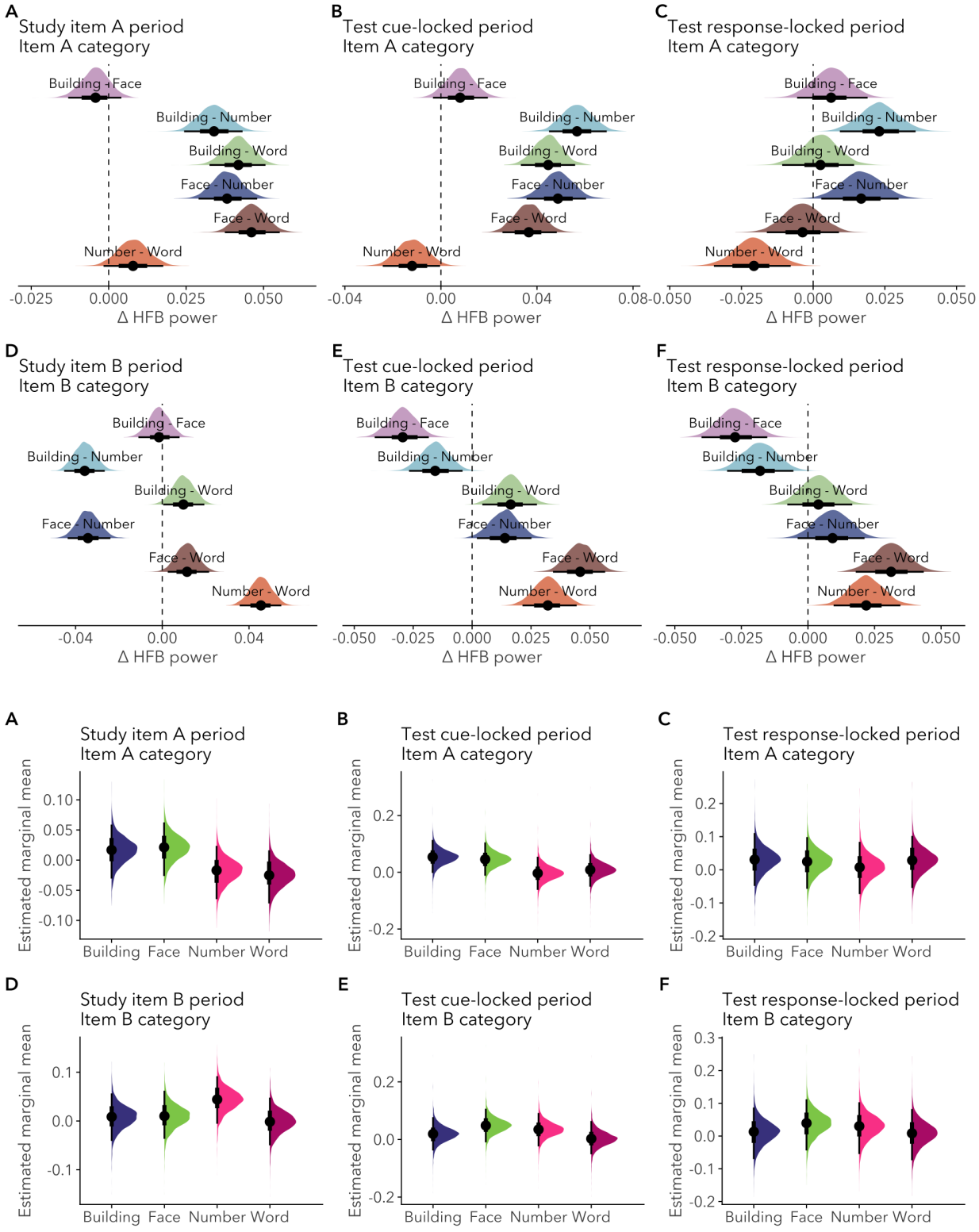
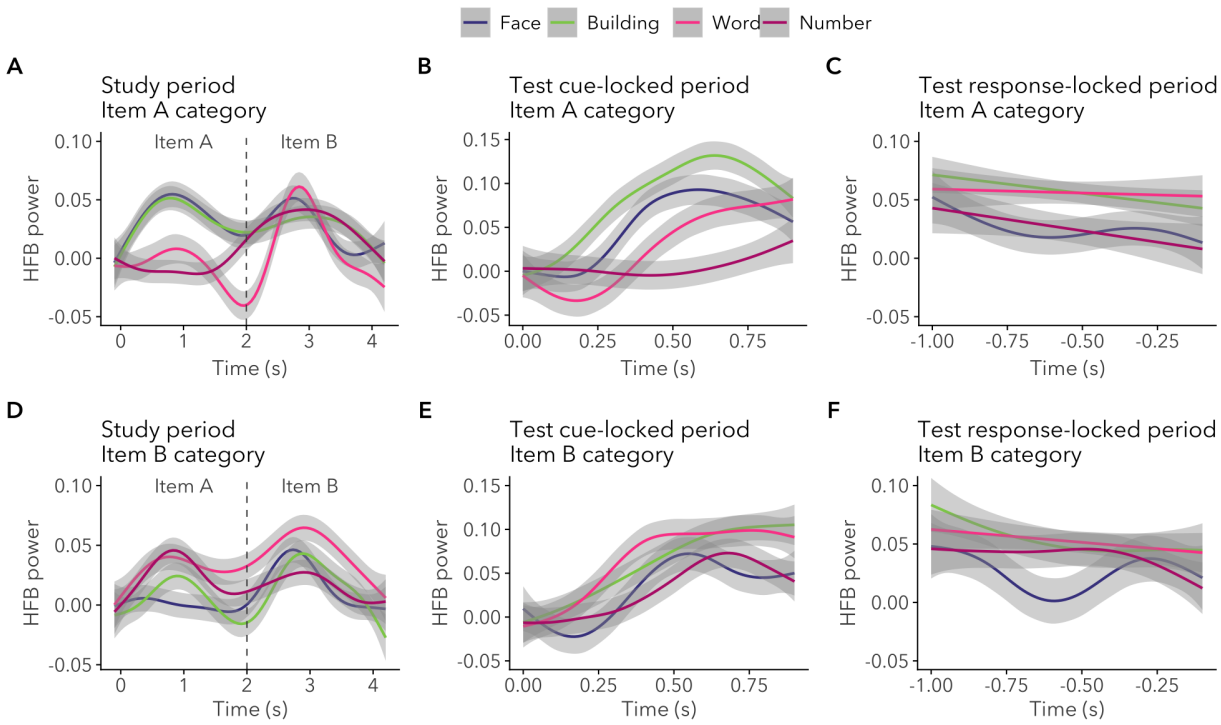
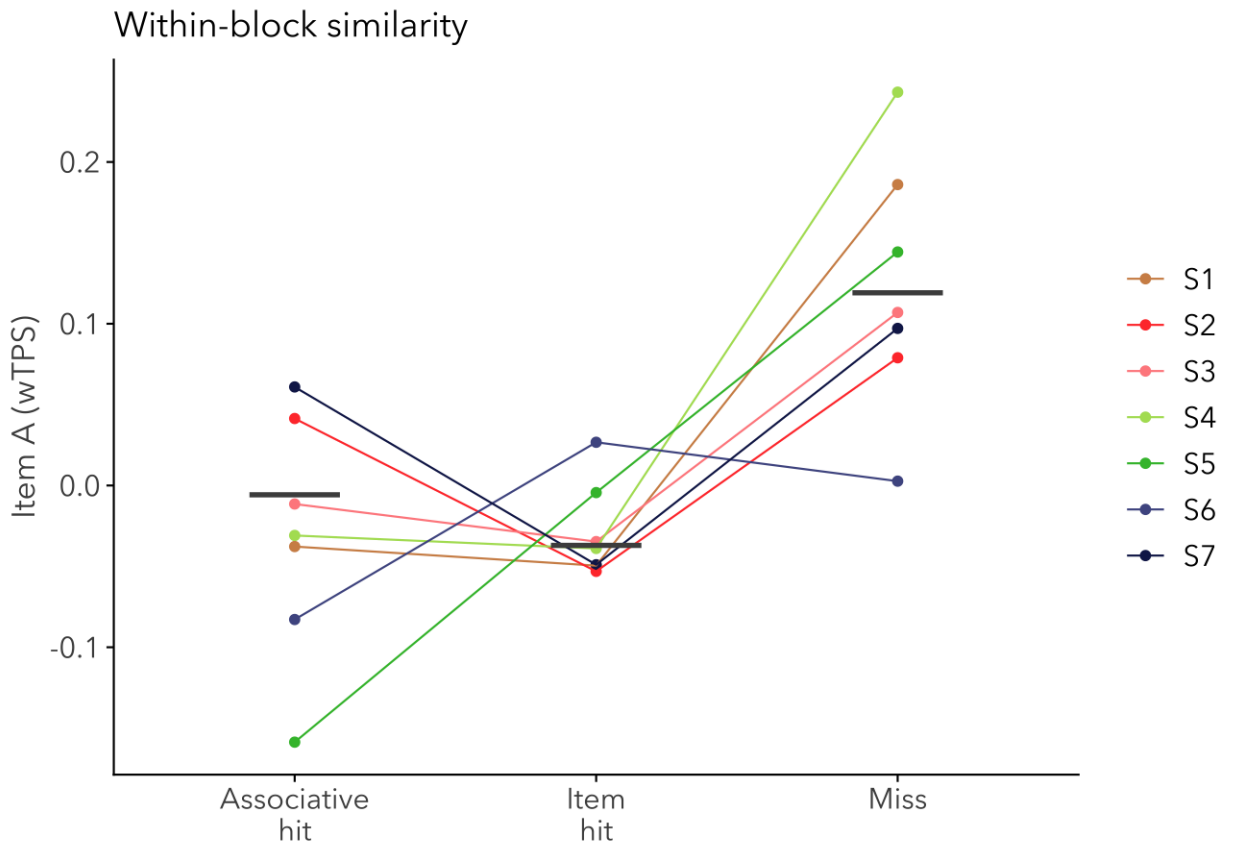

A
Study item A period
Item A category
B
Test cue-locked period
Item A category
C
Test response-locked period
Item A category
D
Study item B period
Item B category
E
Test cue-locked period
Item B category
F
Test response-locked period
Item B category
Building - Face
Building - Number
Building - Word
Face - Number
Face - Word
Number - Word
Δ HFB power
Estimated marginal mean
Building
Face
Number
Word

Fig S7


Face
Building
Word
Number
A
Study period
Item A category
Item A
Item B
HFB power
Time (s)
B
Test cue-locked period
Item A category
C
Test response-locked period
Item A category
D
Study period
Item B category
E
Test cue-locked period
Item B category
F
Test response-locked period
Item B category

Pattern similarity

~ Fig 3A




```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: enc_corr_c1 ~ 1 + cue1_category + trial_type_combo + HFB.mean + (1 | subj/chan)
##    Data: . (Number of observations: 2520)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 7)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.02      0.02     0.00     0.08 1.00     2689     2696
##
## ~subj:chan (Number of levels: 30)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.02      0.01     0.00     0.05 1.00     2782     1494
##
## Regression Coefficients:
```

```
##                          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## Intercept                   -0.02      0.05    -0.13     0.08 1.00     3248
## cue1_categoryBuilding        0.10      0.05    -0.01     0.20 1.00     3448
## cue1_categoryWord           -0.03      0.05    -0.14     0.08 1.00     3166
## cue1_categoryNumber          0.01      0.06    -0.11     0.12 1.00     3135
## trial_type_comboitemhit     -0.04      0.05    -0.13     0.05 1.00     4333
## trial_type_combomiss         0.14      0.06     0.02     0.26 1.00     3915
## HFB.mean                     0.04      0.08    -0.11     0.19 1.00     7116
##                          Tail_ESS
## Intercept                    3076
## cue1_categoryBuilding        2962
## cue1_categoryWord            2696
## cue1_categoryNumber          2871
## trial_type_comboitemhit      3210
## trial_type_combomiss         3099
## HFB.mean                     2878
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.93      0.01     0.91     0.96 1.00     7550     2891
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
## enc_corr_c1 ~ 1 + cue1_category + trial_type_combo + HFB.mean + (1 | subj/chan)
```

```
## enc_corr_c1 ~ 1 + cue1_category + HFB.mean + (1 | subj/chan)
```

```
##            elpd_diff se_diff elpd_loo se_elpd_loo p_loo   se_p_loo looic
## wTPS.A.mod     0.0       0.0 -3406.7     29.7         9.6     0.2   6813.3
## mod.simple    -3.3       3.1 -3409.9     29.7         7.7     0.2   6819.8
##            se_looic
## wTPS.A.mod    59.3
## mod.simple    59.5
```

```
## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'wTPS.A.mod' is the best model (ELPD =
## -3406.66), followed by 'mod.simple' (diff-ELPD = -3.26 +- 3.14, p = 0.299)
```

Table 50: study A emmeans

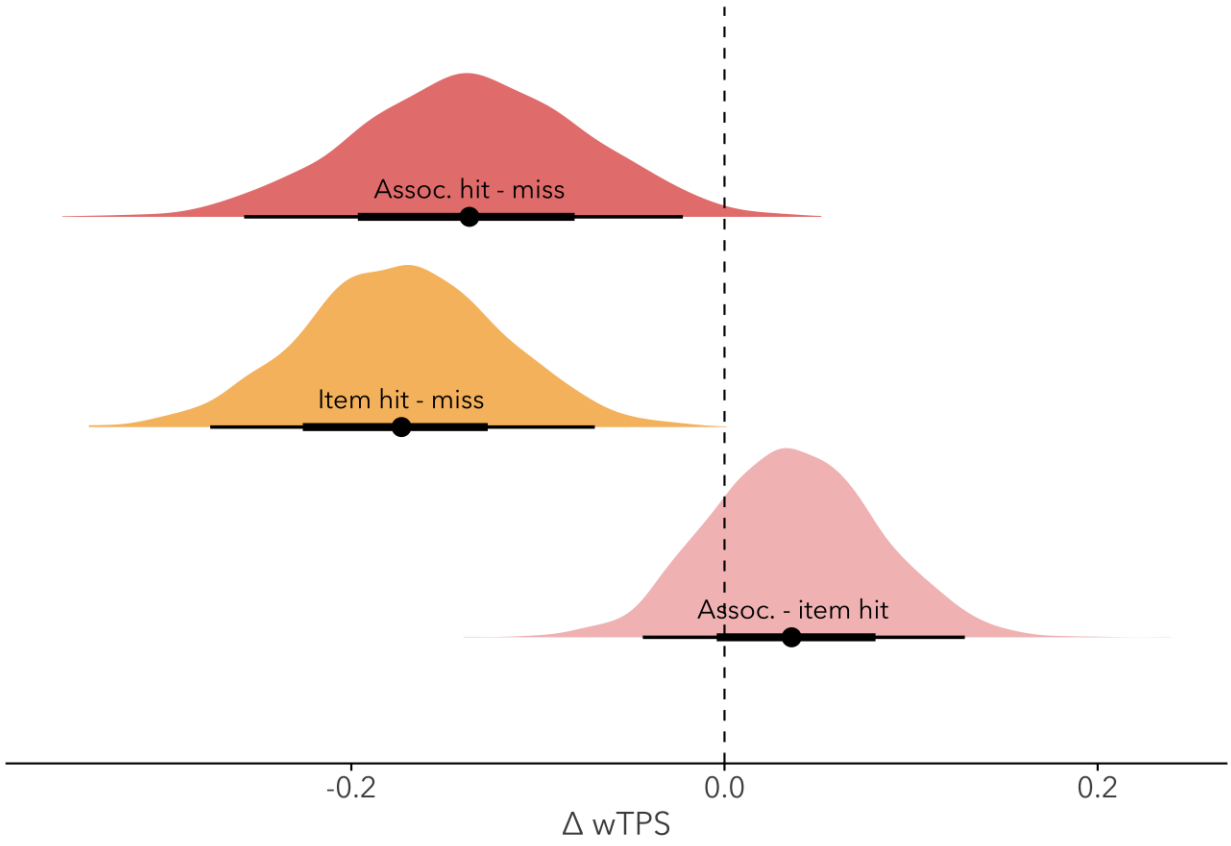
term	contrast	null.value	estimate	lower.HPD	upper.HPD
trial_type_combo	item hit - assoc hit	0	-0.036	-0.128	0.044
trial_type_combo	miss - assoc hit	0	0.137	0.022	0.257
trial_type_combo	miss - item hit	0	0.173	0.069	0.275

Table 51: study A emmeans

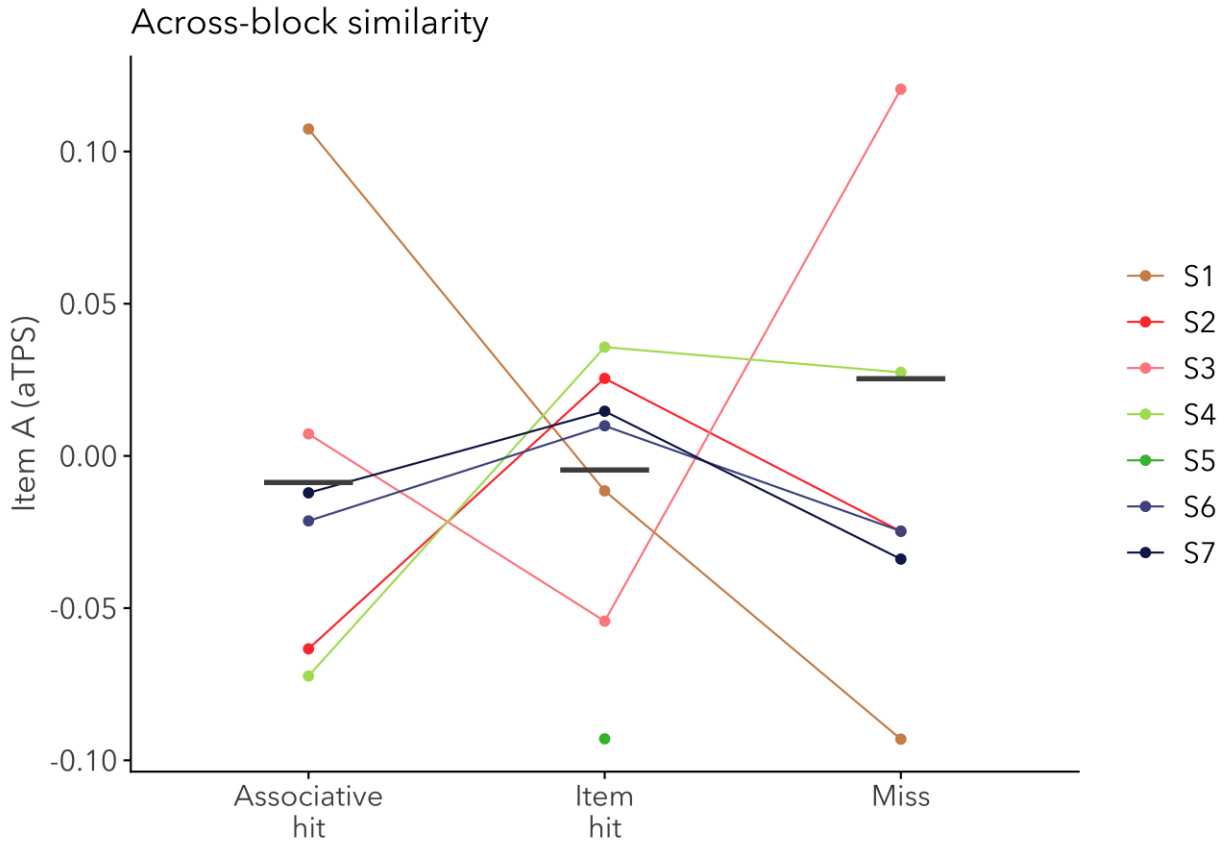
trial_type_combo	estimate	lower.HPD	upper.HPD
assoc hit	-0.006	-0.083	0.077
item hit	-0.042	-0.096	0.016
miss	0.132	0.035	0.223

Table 52: study A emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
trial_type_combo	assoc hit - item hit	0	0.036	-0.044	0.128
trial_type_combo	assoc hit - miss	0	-0.137	-0.257	-0.022
trial_type_combo	item hit - miss	0	-0.173	-0.275	-0.069



~ **Fig 3B**




```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: enc_corr_c1_future ~ 1 + cue1_category + trial_type_combo + HFB.mean + (1 | subj/chan)
##    Data: . (Number of observations: 2526)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 7)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.02      0.02     0.00     0.08 1.00     1929     2063
##
## ~subj:chan (Number of levels: 30)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.02      0.01     0.00     0.05 1.00     3115     1607
##
## Regression Coefficients:
##                         Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## Intercept                  -0.08      0.05    -0.18     0.02 1.00     3200
```

```
## cue1_categoryBuilding       0.05      0.05    -0.05     0.14 1.00     3575
## cue1_categoryWord           0.13      0.05     0.03     0.23 1.00     3384
## cue1_categoryNumber         0.13      0.06     0.02     0.24 1.00     3159
## trial_type_comboitemhit     0.02      0.05    -0.07     0.10 1.00     3772
## trial_type_combomiss       -0.01      0.06    -0.13     0.11 1.00     3696
## HFB.mean                    0.03      0.08    -0.12     0.17 1.00     5589
##                          Tail_ESS
## Intercept                    3293
## cue1_categoryBuilding        3273
## cue1_categoryWord            3366
## cue1_categoryNumber          3007
## trial_type_comboitemhit      3157
## trial_type_combomiss         3109
## HFB.mean                     3046
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.93      0.01     0.91     0.96 1.00     6430     2826
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
## enc_corr_c1_future ~ 1 + cue1_category + trial_type_combo + HFB.mean + (1 | subj/chan)
```

```
## enc_corr_c1_future ~ 1 + cue1_category + HFB.mean + (1 | subj/chan)
```

```
##            elpd_diff se_diff elpd_loo se_elpd_loo p_loo   se_p_loo looic
## mod.simple     0.0       0.0 -3414.4     28.8         7.7     0.2   6828.7
## aTPS.A.mod    -1.7       0.5 -3416.1     28.8         9.5     0.2   6832.2
##            se_looic
## mod.simple    57.7
## aTPS.A.mod    57.6
```

```
## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'mod.simple' is the best model (ELPD =
## -3414.37), followed by 'aTPS.A.mod' (diff-ELPD = -1.74 +- 0.48, p < .001)
```

Table 53: study A emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
trial_type_combo	item hit - assoc hit	0	0.016	-0.069	0.106
trial_type_combo	miss - assoc hit	0	-0.008	-0.124	0.113
trial_type_combo	miss - item hit	0	-0.022	-0.131	0.082

Table 54: study A emmeans

trial_type_combo	estimate	lower.HPD	upper.HPD
assoc hit	-0.006	-0.080	0.078
item hit	0.010	-0.043	0.070
miss	-0.012	-0.105	0.079

Table 55: study A emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
trial_type_combo	assoc hit - item hit	0	-0.016	-0.106	0.069
trial_type_combo	assoc hit - miss	0	0.008	-0.113	0.124
trial_type_combo	item hit - miss	0	0.022	-0.082	0.131

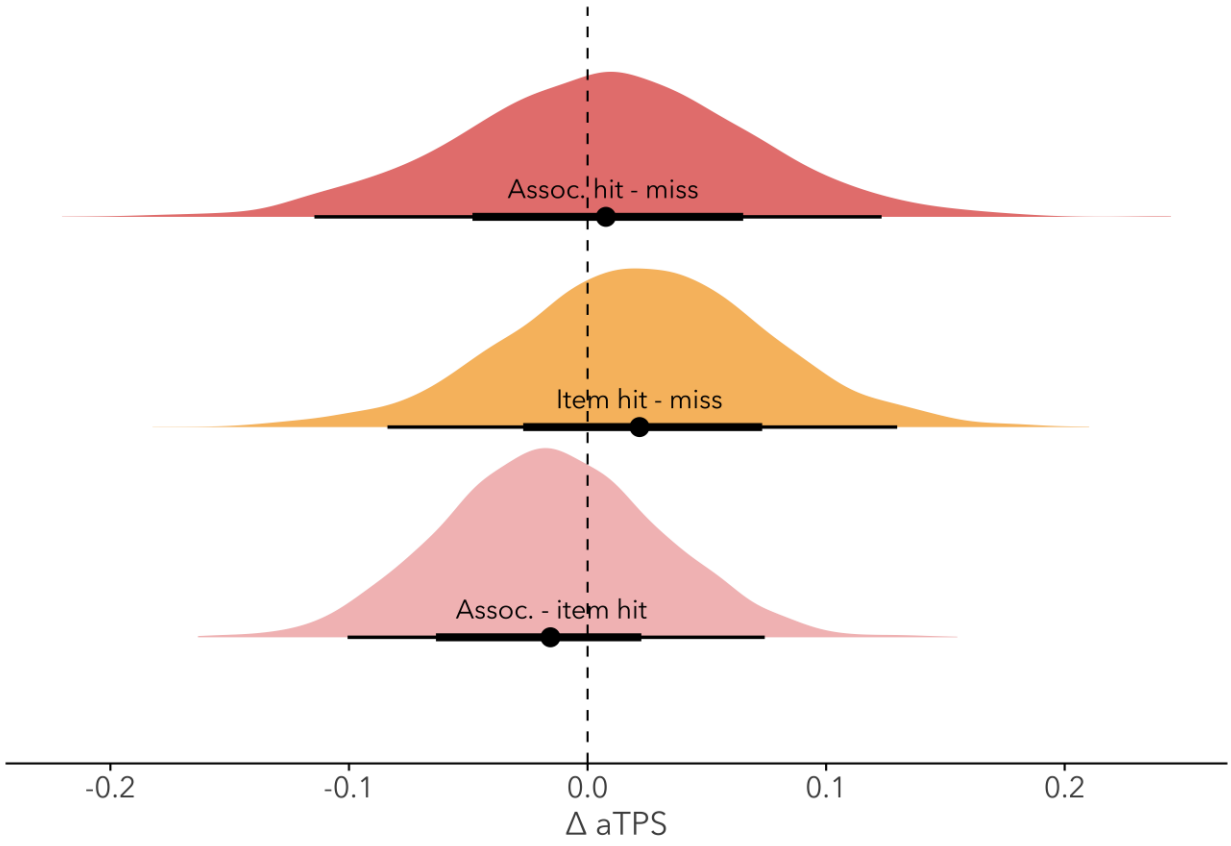
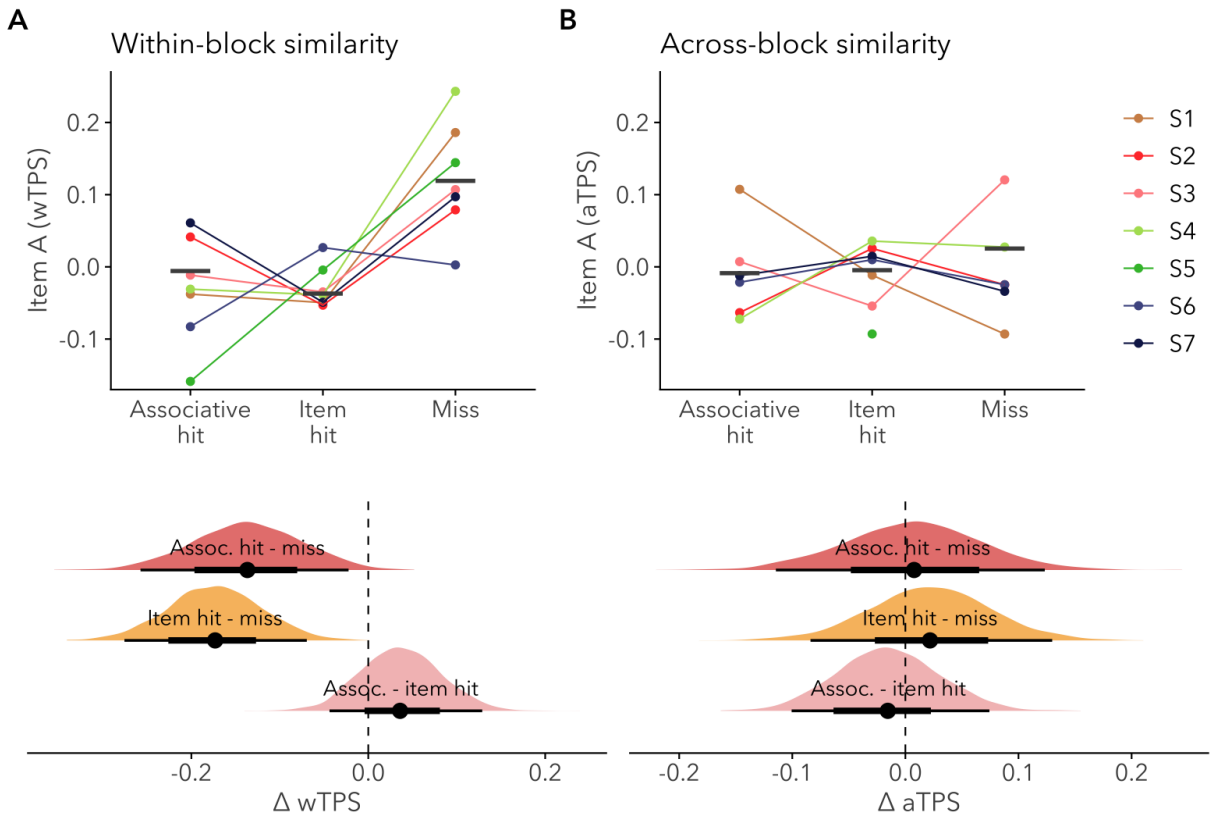


Fig 3



Interaction: TPS x memory outcome

```
## [1] "adapt_delta: 0.99"

## [1] "max_treedepth: 10"

##  Family: gaussian
##   Links: mu = identity
## Formula: similarity ~ type + trial_type_combo + type:trial_type_combo + cue1_category + HFB.mean + (
##    Data: . (Number of observations: 5046)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 7)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.02      0.02     0.00     0.06 1.00     2245     1781
##
## ~subj:chan (Number of levels: 30)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.01      0.01     0.00     0.03 1.00     3423     1779
##
```

```
## Regression Coefficients:
##                                       Estimate Est.Error l-95% CI u-95% CI Rhat
## Intercept                                -0.03      0.05    -0.14     0.07 1.00
## typewithin                                0.09      0.06    -0.03     0.21 1.00
## trial_type_comboassoc\nhit               -0.02      0.06    -0.14     0.10 1.00
## trial_type_comboitemhit                  -0.01      0.05    -0.11     0.09 1.00
## cue1_categoryBuilding                     0.07      0.04    -0.00     0.14 1.00
## cue1_categoryWord                         0.05      0.04    -0.02     0.12 1.00
## cue1_categoryNumber                       0.07      0.04    -0.01     0.15 1.00
## HFB.mean                                  0.03      0.05    -0.07     0.14 1.00
## typewithin:trial_type_comboassoc\nhit    -0.09      0.08    -0.25     0.06 1.00
## typewithin:trial_type_comboitemhit       -0.12      0.07    -0.26     0.01 1.00
##                                       Bulk_ESS Tail_ESS
## Intercept                                 2494     2713
## typewithin                                2136     2750
## trial_type_comboassoc\nhit                2624     2946
## trial_type_comboitemhit                   2495     2830
## cue1_categoryBuilding                     4185     3489
## cue1_categoryWord                         4611     3324
## cue1_categoryNumber                       4063     3244
## HFB.mean                                  7131     2836
## typewithin:trial_type_comboassoc\nhit     2477     3133
## typewithin:trial_type_comboitemhit        2089     2801
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.93      0.01     0.92     0.95 1.00     7528     2746
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

effect	component	term	estimate	std.error	conf.low	conf.high
fixed	cond	(Intercept)	-0.032	0.054	-0.145	0.069
fixed	cond	typewithin	0.091	0.060	-0.026	0.206
fixed	cond	trial_type_comboassoc hit	| -0.02	| 0.05	| -0.13	| 0.09
fixed	cond	trial_type_comboitemhit	-0.015	0.052	-0.115	0.091
fixed	cond	cue1_categoryBuilding	0.070	0.037	-0.002	0.143
fixed	cond	cue1_categoryWord	0.051	0.038	-0.023	0.123
fixed	cond	cue1_categoryNumber	0.069	0.040	-0.009	0.148
fixed	cond	HFB.mean	0.033	0.055	-0.075	0.142
fixed	cond	typewithin:trial_type_comboassoc hit	| -0.08	| 0.07	| -0.24	| 0.06
fixed	cond	typewithin:trial_type_comboitemhit	-0.124	0.070	-0.260	0.014

```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
## similarity ~ type + trial_type_combo + type:trial_type_combo + cue1_category + HFB.mean + (1 | subj/
```

```
## similarity ~ type + trial_type_combo + cue1_category + HFB.mean + (1 | subj/chan)
```

```
##            elpd_diff se_diff elpd_loo se_elpd_loo p_loo   se_p_loo looic
## mod.simple     0.0       0.0 -6821.0     41.4        10.8     0.2  13641.9
## mod           -0.3       1.8 -6821.2     41.4        12.7     0.2  13642.4
##            se_looic
## mod.simple    82.8
## mod           82.7

## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'mod.simple' is the best model (ELPD =
## -6820.96), followed by 'mod' (diff-ELPD = -0.25 +- 1.78, p = 0.887)

##  trial_type_combo_pairwise type_pairwise   estimate lower.HPD upper.HPD
##  miss - assoc\nhit         across - within  -0.0886    -0.248   0.05888
##  miss - item hit           across - within  -0.1234    -0.262   0.00883
##  assoc\nhit - item hit     across - within  -0.0354    -0.160   0.08605
##
## Results are averaged over the levels of: cue1_category
## Point estimate displayed: median
## HPD interval probability: 0.95
```

Exclude S5

~ Fig 3A

```
## [1] "adapt_delta: 0.8"

## [1] "max_treedepth: 10"

##  Family: gaussian
##   Links: mu = identity
## Formula: enc_corr_c1 ~ 1 + cue1_category + trial_type_combo + HFB.mean + (1 | subj/chan)
##    Data: . (Number of observations: 2456)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 6)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.03      0.03     0.00     0.09 1.00     2379     2268
##
## ~subj:chan (Number of levels: 28)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.02      0.01     0.00     0.05 1.00     3361     1724
##
## Regression Coefficients:
##                          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## Intercept                   -0.03      0.05    -0.14     0.07 1.00     4246
## cue1_categoryBuilding        0.11      0.05     0.00     0.21 1.00     5130
## cue1_categoryWord           -0.02      0.05    -0.13     0.08 1.00     4737
## cue1_categoryNumber          0.01      0.06    -0.11     0.12 1.00     4266
## trial_type_comboitemhit     -0.04      0.05    -0.13     0.05 1.00     4839
```

```
## trial_type_combomiss        0.14      0.06     0.01     0.26 1.00     4649
## HFB.mean                    0.03      0.08    -0.12     0.18 1.00     6796
##                          Tail_ESS
## Intercept                    2572
## cue1_categoryBuilding        3431
## cue1_categoryWord            3179
## cue1_categoryNumber          3306
## trial_type_comboitemhit      2713
## trial_type_combomiss         3347
## HFB.mean                     2484
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.94      0.01     0.91     0.96 1.00     7934     2849
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
## [1] "adapt_delta: 0.8"
```

```
## [1] "max_treedepth: 10"
```

```
## enc_corr_c1 ~ 1 + cue1_category + trial_type_combo + HFB.mean + (1 | subj/chan)
```

```
## enc_corr_c1 ~ 1 + cue1_category + HFB.mean + (1 | subj/chan)
```

```
##            elpd_diff se_diff elpd_loo se_elpd_loo p_loo   se_p_loo looic
## wTPS.A.mod     0.0       0.0 -3323.2     29.3         9.4     0.2   6646.4
## mod.simple    -3.3       3.2 -3326.5     29.4         7.6     0.2   6653.0
##            se_looic
## wTPS.A.mod    58.7
## mod.simple    58.9
```

```
## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'wTPS.A.mod' is the best model (ELPD =
## -3323.18), followed by 'mod.simple' (diff-ELPD = -3.34 +- 3.17, p = 0.293)
```

Table 57: study A emmeans

trial_type_combo	estimate	lower.HPD	upper.HPD
assoc hit	-0.006	-0.090	0.075
item hit	-0.043	-0.100	0.015
miss	0.134	0.039	0.232

Table 58: study A emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
trial_type_combo	assoc hit - item hit	0	0.039	-0.055	0.126

trial_type_combo	assoc hit - miss	0	-0.140	-0.258	-0.013
trial_type_combo	item hit - miss	0	-0.178	-0.283	-0.071

Table 59: study A emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
trial_type_combo	item hit - assoc hit	0	-0.039	-0.126	0.055
trial_type_combo	miss - assoc hit	0	0.140	0.013	0.258
trial_type_combo	miss - item hit	0	0.178	0.071	0.283

~ Fig 3B

```
## [1] "adapt_delta: 0.8"

## [1] "max_treedepth: 10"

##  Family: gaussian
##   Links: mu = identity
## Formula: enc_corr_c1_future ~ 1 + cue1_category + trial_type_combo + HFB.mean + (1 | subj/chan)
##    Data: . (Number of observations: 2454)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 6)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.03      0.02     0.00     0.09 1.00     2714     2769
##
## ~subj:chan (Number of levels: 28)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.02      0.01     0.00     0.05 1.00     3268     2210
##
## Regression Coefficients:
##                          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## Intercept                   -0.11      0.05    -0.21    -0.00 1.00     4517
## cue1_categoryBuilding        0.07      0.05    -0.04     0.18 1.00     5557
## cue1_categoryWord            0.16      0.06     0.05     0.27 1.00     4777
## cue1_categoryNumber          0.15      0.06     0.03     0.26 1.00     4455
## trial_type_comboitemhit      0.03      0.05    -0.06     0.12 1.00     5998
## trial_type_combomiss        -0.00      0.06    -0.12     0.12 1.00     4818
## HFB.mean                     0.02      0.08    -0.14     0.17 1.00     7031
##                          Tail_ESS
## Intercept                    3160
## cue1_categoryBuilding        2956
## cue1_categoryWord            3095
## cue1_categoryNumber          3369
## trial_type_comboitemhit      3209
## trial_type_combomiss         3007
## HFB.mean                     2675
##
## Further Distributional Parameters:
```

```
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.94      0.01     0.91     0.96 1.00     8223     3034
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
## [1] "adapt_delta: 0.8"
```

```
## [1] "max_treedepth: 10"
```

```
## enc_corr_c1_future ~ 1 + cue1_category + trial_type_combo + HFB.mean + (1 | subj/chan)
```

```
## enc_corr_c1_future ~ 1 + cue1_category + HFB.mean + (1 | subj/chan)
```

```
##            elpd_diff se_diff elpd_loo se_elpd_loo p_loo   se_p_loo looic
## mod.simple     0.0       0.0 -3321.5     28.5         7.7     0.2   6643.0
## aTPS.A.mod    -1.8       0.8 -3323.3     28.5         9.8     0.2   6646.6
##            se_looic
## mod.simple    57.0
## aTPS.A.mod    57.1
```

```
## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'mod.simple' is the best model (ELPD =
## -3321.52), followed by 'aTPS.A.mod' (diff-ELPD = -1.78 +- 0.82, p = 0.030)
```

Table 60: study A emmeans

trial_type_combo	estimate	lower.HPD	upper.HPD
assoc hit	-0.013	-0.099	0.070
item hit	0.016	-0.043	0.074
miss	-0.019	-0.115	0.075

Table 61: study A emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
trial_type_combo	assoc hit - item hit	0	-0.031	-0.122	0.064
trial_type_combo	assoc hit - miss	0	0.004	-0.117	0.128
trial_type_combo	item hit - miss	0	0.035	-0.070	0.138

Table 62: study A emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
trial_type_combo	item hit - assoc hit	0	0.031	-0.064	0.122
trial_type_combo	miss - assoc hit	0	-0.004	-0.128	0.117
trial_type_combo	miss - item hit	0	-0.035	-0.138	0.070

Interaction: TPS x memory outcome

```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: similarity ~ type + trial_type_combo + type:trial_type_combo + cue1_category + HFB.mean + (
##    Data: . (Number of observations: 4910)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 6)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.02      0.02     0.00     0.07 1.00     2256     2241
##
## ~subj:chan (Number of levels: 28)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.01      0.01     0.00     0.03 1.00     3655     2129
##
## Regression Coefficients:
##                                      Estimate Est.Error l-95% CI u-95% CI Rhat
## Intercept                               -0.05      0.06    -0.16     0.06 1.00
## typewithin                               0.10      0.06    -0.02     0.22 1.00
## trial_type_comboassoc\nhit              -0.02      0.06    -0.14     0.10 1.00
## trial_type_comboitemhit                 -0.00      0.05    -0.10     0.11 1.00
## cue1_categoryBuilding                    0.09      0.04     0.02     0.16 1.00
## cue1_categoryWord                        0.07      0.04    -0.01     0.14 1.00
## cue1_categoryNumber                      0.08      0.04    -0.00     0.16 1.00
## HFB.mean                                 0.02      0.06    -0.08     0.13 1.00
## typewithin:trial_type_comboassoc\nhit   -0.09      0.08    -0.25     0.07 1.00
## typewithin:trial_type_comboitemhit      -0.14      0.07    -0.28    -0.00 1.00
##                                      Bulk_ESS Tail_ESS
## Intercept                                2047     2660
## typewithin                               1771     2426
## trial_type_comboassoc\nhit               2241     2703
## trial_type_comboitemhit                  1948     2821
## cue1_categoryBuilding                    3600     3480
## cue1_categoryWord                        3539     3114
## cue1_categoryNumber                      3472     3255
## HFB.mean                                 5712     3116
## typewithin:trial_type_comboassoc\nhit    1963     2701
## typewithin:trial_type_comboitemhit       1901     2541
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.94      0.01     0.92     0.95 1.00     6061     3158
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

effect	component	term	estimate	std.error	conf.low	conf.high
fixed	cond	(Intercept)	-0.051	0.057	-0.164	0.061
fixed	cond	typewithin	0.101	0.061	-0.021	0.223
fixed	cond	trial_type_comboassoc hit	| -0.02	| 0.06	| -0.14	| 0.10
fixed	cond	trial_type_comboitemhit	-0.001	0.053	-0.104	0.106
fixed	cond	cue1_categoryBuilding	0.089	0.037	0.016	0.160
fixed	cond	cue1_categoryWord	0.067	0.038	-0.008	0.143
fixed	cond	cue1_categoryNumber	0.077	0.041	-0.004	0.158
fixed	cond	HFB.mean	0.022	0.055	-0.085	0.128
fixed	cond	typewithin:trial_type_comboassoc hit	| -0.09	| 0.08	| -0.25	| 0.06
fixed	cond	typewithin:trial_type_comboitemhit	-0.139	0.072	-0.279	-0.003

```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
## similarity ~ type + trial_type_combo + type:trial_type_combo + cue1_category + HFB.mean + (1 | subj/
```

```
## similarity ~ type + trial_type_combo + cue1_category + HFB.mean + (1 | subj/chan)
```

```
##            elpd_diff se_diff elpd_loo se_elpd_loo p_loo   se_p_loo looic
## mod.simple     0.0       0.0 -6644.8     41.0        10.6     0.2  13289.7
## mod           -0.2       2.0 -6645.0     40.9        12.6     0.2  13290.0
##            se_looic
## mod.simple    81.9
## mod           81.9
```

```
## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'mod.simple' is the best model (ELPD =
## -6644.84), followed by 'mod' (diff-ELPD = -0.18 +- 1.99, p = 0.930)
```

```
##  trial_type_combo_pairwise type_pairwise   estimate lower.HPD upper.HPD
##  miss - assoc\nhit         across - within  -0.0913    -0.258   0.05921
##  miss - item hit           across - within  -0.1375    -0.273   0.00307
##  assoc\nhit - item hit     across - within  -0.0483    -0.170   0.07566
##
## Results are averaged over the levels of: cue1_category
## Point estimate displayed: median
## HPD interval probability: 0.95
```

Exclude S2 and S5

~ Fig 3A

```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: enc_corr_c1 ~ 1 + cue1_category + trial_type_combo + HFB.mean + (1 | subj/chan)
##    Data: . (Number of observations: 2205)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 5)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.03      0.04     0.00     0.13 1.00     1920     2253
##
## ~subj:chan (Number of levels: 24)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.02      0.01     0.00     0.05 1.00     2568     1501
##
## Regression Coefficients:
##                          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## Intercept                   -0.02      0.06    -0.13     0.09 1.00     2745
## cue1_categoryBuilding        0.08      0.06    -0.03     0.19 1.00     3726
## cue1_categoryWord           -0.03      0.06    -0.14     0.08 1.00     3579
## cue1_categoryNumber          0.00      0.06    -0.12     0.12 1.00     3700
## trial_type_comboitemhit     -0.03      0.05    -0.13     0.06 1.00     4113
## trial_type_combomiss         0.15      0.07     0.01     0.28 1.00     4138
## HFB.mean                     0.03      0.09    -0.14     0.19 1.00     5692
##                          Tail_ESS
## Intercept                    2466
## cue1_categoryBuilding        2963
## cue1_categoryWord            3063
## cue1_categoryNumber          3214
## trial_type_comboitemhit      2899
## trial_type_combomiss         3140
## HFB.mean                     2613
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.94      0.01     0.91     0.96 1.00     6121     3099
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
## enc_corr_c1 ~ 1 + cue1_category + trial_type_combo + HFB.mean + (1 | subj/chan)
```

```
## enc_corr_c1 ~ 1 + cue1_category + HFB.mean + (1 | subj/chan)
```

```
##            elpd_diff se_diff elpd_loo se_elpd_loo p_loo   se_p_loo looic
## wTPS.A.mod     0.0       0.0 -2990.5     28.1         9.5     0.3   5981.0
## mod.simple    -2.7       2.9 -2993.1     28.1         7.7     0.2   5986.3
```

```
##            se_looic
## wTPS.A.mod    56.2
## mod.simple    56.3
```

```
## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'wTPS.A.mod' is the best model (ELPD =
## -2990.50), followed by 'mod.simple' (diff-ELPD = -2.65 +- 2.92, p = 0.364)
```

Table 64: study A emmeans

trial_type_combo	estimate	lower.HPD	upper.HPD
assoc hit	-0.004	-0.097	0.081
item hit	-0.040	-0.111	0.028
miss	0.142	0.034	0.257

Table 65: study A emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
trial_type_combo	assoc hit - item hit	0	0.035	-0.053	0.130
trial_type_combo	assoc hit - miss	0	-0.148	-0.279	-0.015
trial_type_combo	item hit - miss	0	-0.182	-0.309	-0.072

Table 66: study A emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
trial_type_combo	item hit - assoc hit	0	-0.035	-0.130	0.053
trial_type_combo	miss - assoc hit	0	0.148	0.015	0.279
trial_type_combo	miss - item hit	0	0.182	0.072	0.309

~ Fig 3B

```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: enc_corr_c1_future ~ 1 + cue1_category + trial_type_combo + HFB.mean + (1 | subj/chan)
##    Data: . (Number of observations: 2203)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 5)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.03      0.04     0.00     0.13 1.00     1976     1993
##
```

```
## ~subj:chan (Number of levels: 24)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.02      0.01     0.00     0.05 1.00     3198     1868
##
## Regression Coefficients:
##                          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## Intercept                   -0.14      0.06    -0.25    -0.02 1.00     2621
## cue1_categoryBuilding        0.10      0.06    -0.01     0.21 1.00     3530
## cue1_categoryWord            0.20      0.06     0.09     0.32 1.00     3127
## cue1_categoryNumber          0.20      0.06     0.08     0.33 1.00     3512
## trial_type_comboitemhit      0.03      0.05    -0.06     0.12 1.00     4234
## trial_type_combomiss        -0.02      0.07    -0.15     0.11 1.00     3998
## HFB.mean                     0.01      0.08    -0.15     0.16 1.00     6451
##                          Tail_ESS
## Intercept                    2489
## cue1_categoryBuilding        2991
## cue1_categoryWord            2690
## cue1_categoryNumber          3531
## trial_type_comboitemhit      2733
## trial_type_combomiss         2983
## HFB.mean                     3147
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.94      0.01     0.91     0.96 1.00     5314     3016
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
## enc_corr_c1_future ~ 1 + cue1_category + trial_type_combo + HFB.mean + (1 | subj/chan)
```

```
## enc_corr_c1_future ~ 1 + cue1_category + HFB.mean + (1 | subj/chan)
```

```
##            elpd_diff se_diff elpd_loo se_elpd_loo p_loo   se_p_loo looic
## mod.simple     0.0       0.0 -2984.2     27.4         7.7     0.2   5968.4
## aTPS.A.mod    -1.5       0.9 -2985.7     27.4         9.6     0.3   5971.5
##            se_looic
## mod.simple    54.8
## aTPS.A.mod    54.8
```

```
## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'mod.simple' is the best model (ELPD =
## -2984.22), followed by 'aTPS.A.mod' (diff-ELPD = -1.52 +- 0.90, p = 0.092)
```

Table 67: study A emmeans

trial_type_combo	estimate	lower.HPD	upper.HPD
assoc hit	-0.011	-0.092	0.079
item hit	0.018	-0.052	0.089
miss	-0.032	-0.147	0.082

Table 68: study A emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
trial_type_combo	assoc hit - item hit	0	-0.029	-0.115	0.067
trial_type_combo	assoc hit - miss	0	0.020	-0.117	0.146
trial_type_combo	item hit - miss	0	0.050	-0.068	0.168

Table 69: study A emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
trial_type_combo	item hit - assoc hit	0	0.029	-0.067	0.115
trial_type_combo	miss - assoc hit	0	-0.020	-0.146	0.117
trial_type_combo	miss - item hit	0	-0.050	-0.168	0.068

Interaction: TPS x memory outcome

```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: similarity ~ type + trial_type_combo + type:trial_type_combo + cue1_category + HFB.mean + (
##    Data: . (Number of observations: 4408)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 5)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.02      0.03     0.00     0.09 1.00     2088     2097
##
## ~subj:chan (Number of levels: 24)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.01      0.01     0.00     0.04 1.00     3447     1672
##
## Regression Coefficients:
##                                     Estimate Est.Error l-95% CI u-95% CI Rhat
## Intercept                              -0.06      0.06    -0.19     0.06 1.00
## typewithin                              0.10      0.07    -0.03     0.23 1.00
## trial_type_comboassoc\nhit             -0.02      0.07    -0.14     0.12 1.00
## trial_type_comboitemhit                -0.00      0.06    -0.11     0.11 1.00
## cue1_categoryBuilding                   0.09      0.04     0.02     0.17 1.00
## cue1_categoryWord                       0.09      0.04     0.00     0.17 1.00
```

```
## cue1_categoryNumber                         0.10      0.04     0.02     0.19 1.00
## HFB.mean                                    0.02      0.06    -0.09     0.13 1.00
## typewithin:trial_type_comboassoc\nhit      -0.09      0.09    -0.27     0.07 1.00
## typewithin:trial_type_comboitemhit         -0.13      0.08    -0.28     0.02 1.00
##                                         Bulk_ESS Tail_ESS
## Intercept                                   1655     2654
## typewithin                                  2006     2992
## trial_type_comboassoc\nhit                  2039     2786
## trial_type_comboitemhit                     2122     3131
## cue1_categoryBuilding                       4440     3438
## cue1_categoryWord                           4139     3097
## cue1_categoryNumber                         3662     3133
## HFB.mean                                    7086     2428
## typewithin:trial_type_comboassoc\nhit       2360     2956
## typewithin:trial_type_comboitemhit          2219     3008
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.94      0.01     0.92     0.96 1.00     7531     3004
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

effect	component	term	estimate	std.error	conf.low	conf.high
fixed	cond	(Intercept)	-0.064	0.063	-0.186	0.060
fixed	cond	typewithin	0.098	0.067	-0.030	0.230
fixed	cond	trial_type_comboassoc hit	| -0.01	| 0.06	| -0.14	| 0.11
fixed	cond	trial_type_comboitemhit	-0.001	0.058	-0.113	0.113
fixed	cond	cue1_categoryBuilding	0.093	0.040	0.015	0.170
fixed	cond	cue1_categoryWord	0.086	0.041	0.005	0.166
fixed	cond	cue1_categoryNumber	0.103	0.044	0.015	0.193
fixed	cond	HFB.mean	0.017	0.058	-0.094	0.135
fixed	cond	typewithin:trial_type_comboassoc hit	| -0.09	| 0.08	| -0.26	| 0.07
fixed	cond	typewithin:trial_type_comboitemhit	-0.131	0.077	-0.281	0.017

```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
## similarity ~ type + trial_type_combo + type:trial_type_combo + cue1_category + HFB.mean + (1 | subj/
```

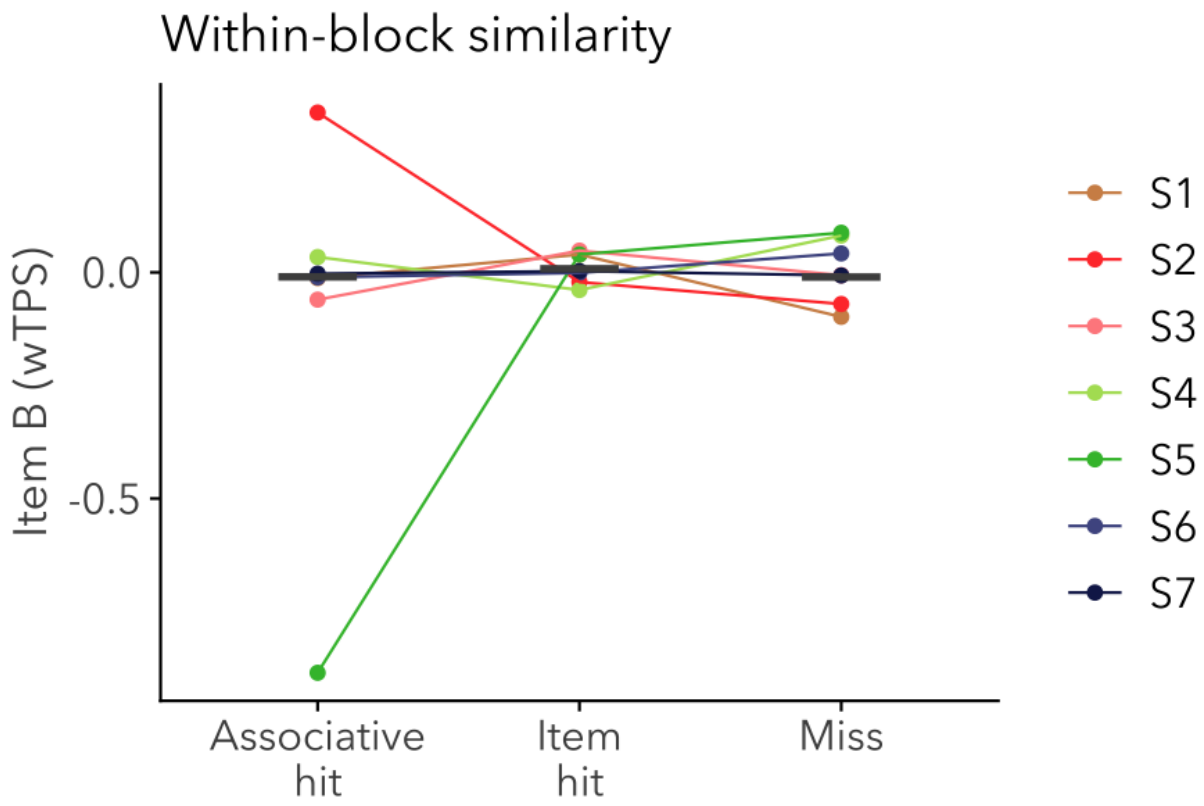
```
## similarity ~ type + trial_type_combo + cue1_category + HFB.mean + (1 | subj/chan)
```

```
##            elpd_diff se_diff elpd_loo se_elpd_loo p_loo   se_p_loo looic
## mod.simple     0.0       0.0 -5975.2     39.2        10.5     0.2  11950.3
## mod           -0.7       1.7 -5975.9     39.2        12.6     0.2  11951.7
##            se_looic
## mod.simple    78.4
## mod           78.4
```

```
## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'mod.simple' is the best model (ELPD =
## -5975.17), followed by 'mod' (diff-ELPD = -0.69 +- 1.70, p = 0.683)
```

```
##  trial_type_combo_pairwise type_pairwise   estimate lower.HPD upper.HPD
##  miss - assoc\nhit         across - within  -0.0906    -0.260    0.0807
##  miss - item hit           across - within  -0.1297    -0.281    0.0173
##  assoc\nhit - item hit     across - within  -0.0381    -0.165    0.0952
##
## Results are averaged over the levels of: cue1_category
## Point estimate displayed: median
## HPD interval probability: 0.95
```

Fig S8




```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: enc_corr_c2 ~ 1 + cue1_category + cue2_category + trial_type_combo + HFB.mean.A + HFB.mean.
##    Data: . (Number of observations: 2520)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
```

```
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 7)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.02      0.02     0.00     0.09 1.00     2011     2060
##
## ~subj:chan (Number of levels: 30)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.02      0.01     0.00     0.05 1.00     3255     1949
##
## Regression Coefficients:
##                          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## Intercept                   -0.05      0.06    -0.17     0.07 1.00     3417
## cue1_categoryBuilding        0.04      0.06    -0.07     0.15 1.00     3408
## cue1_categoryWord            0.05      0.06    -0.06     0.16 1.00     3943
## cue1_categoryNumber          0.07      0.06    -0.05     0.18 1.00     3679
## cue2_categoryFace           -0.01      0.05    -0.11     0.11 1.00     2871
## cue2_categoryNumber          0.01      0.06    -0.10     0.12 1.00     2982
## cue2_categoryWord           -0.00      0.06    -0.11     0.11 1.00     3166
## trial_type_comboitemhit      0.02      0.05    -0.07     0.12 1.00     4377
## trial_type_combomiss        -0.03      0.06    -0.15     0.10 1.00     3930
## HFB.mean.A                  -0.29      0.14    -0.57    -0.01 1.00     3367
## HFB.mean.B                   0.22      0.13    -0.04     0.48 1.00     3443
##                          Tail_ESS
## Intercept                    2848
## cue1_categoryBuilding        2424
## cue1_categoryWord            3207
## cue1_categoryNumber          2968
## cue2_categoryFace            3000
## cue2_categoryNumber          3063
## cue2_categoryWord            3012
## trial_type_comboitemhit      2603
## trial_type_combomiss         3029
## HFB.mean.A                   2937
## HFB.mean.B                   2864
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.94      0.01     0.91     0.96 1.00     5710     3046
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).

## [1] "adapt_delta: 0.99"

## [1] "max_treedepth: 10"

## enc_corr_c2 ~ 1 + cue1_category + cue2_category + trial_type_combo + HFB.mean.A + HFB.mean.B + (1 | s

## enc_corr_c2 ~ 1 + cue1_category + cue2_category + HFB.mean.A + HFB.mean.B + (1 | subj/chan)

##            elpd_diff se_diff elpd_loo se_elpd_loo p_loo   se_p_loo looic
```

```
## mod.simple     0.0       0.0 -3414.1     29.5        11.5     0.3   6828.1
## wTPS.B.mod    -1.7       0.9 -3415.8     29.5        13.5     0.3   6831.5
##            se_looic
## mod.simple    58.9
## wTPS.B.mod    58.9

## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'mod.simple' is the best model (ELPD =
## -3414.07), followed by 'wTPS.B.mod' (diff-ELPD = -1.70 +- 0.87, p = 0.050)
```

Table 71: study B emmeans

trial_type_combo	estimate	lower.HPD	upper.HPD
assoc hit	-0.003	-0.094	0.074
item hit	0.014	-0.043	0.074
miss	-0.032	-0.129	0.067

Table 72: study B emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
trial_type_combo	assoc hit - item hit	0	-0.018	-0.109	0.078
trial_type_combo	assoc hit - miss	0	0.027	-0.101	0.150
trial_type_combo	item hit - miss	0	0.046	-0.061	0.157

Fig S9

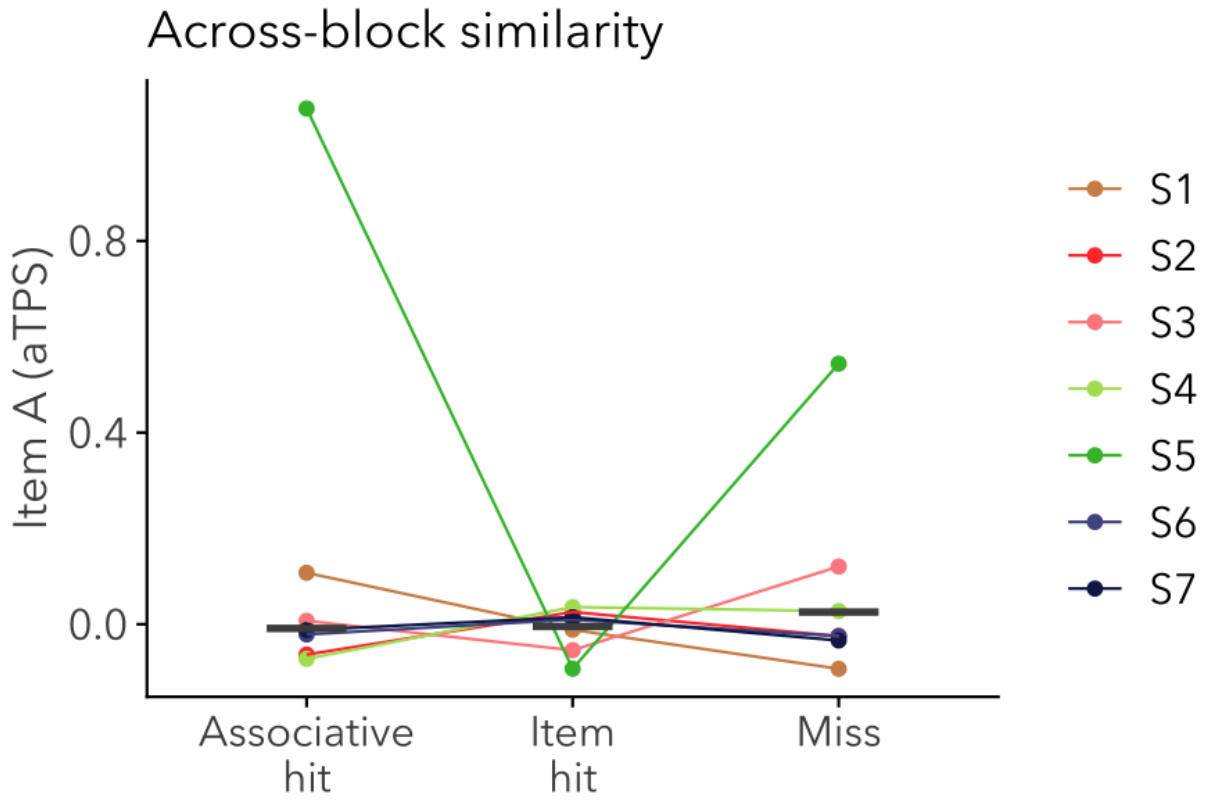


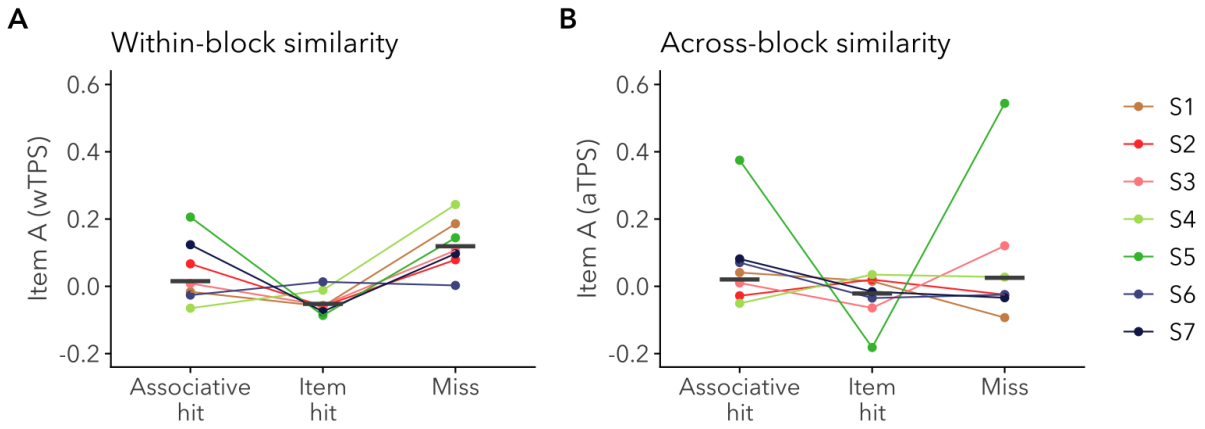
Fig 11

we recomputed the critical univariate and multivariate memory analyses by grouping memory response categories (2) and (3) into the "item hit" condition and response categories (4) and (5) into the "associative hit" condition

~ **Fig S11A**

~ **Fig S11B**

Fig S11



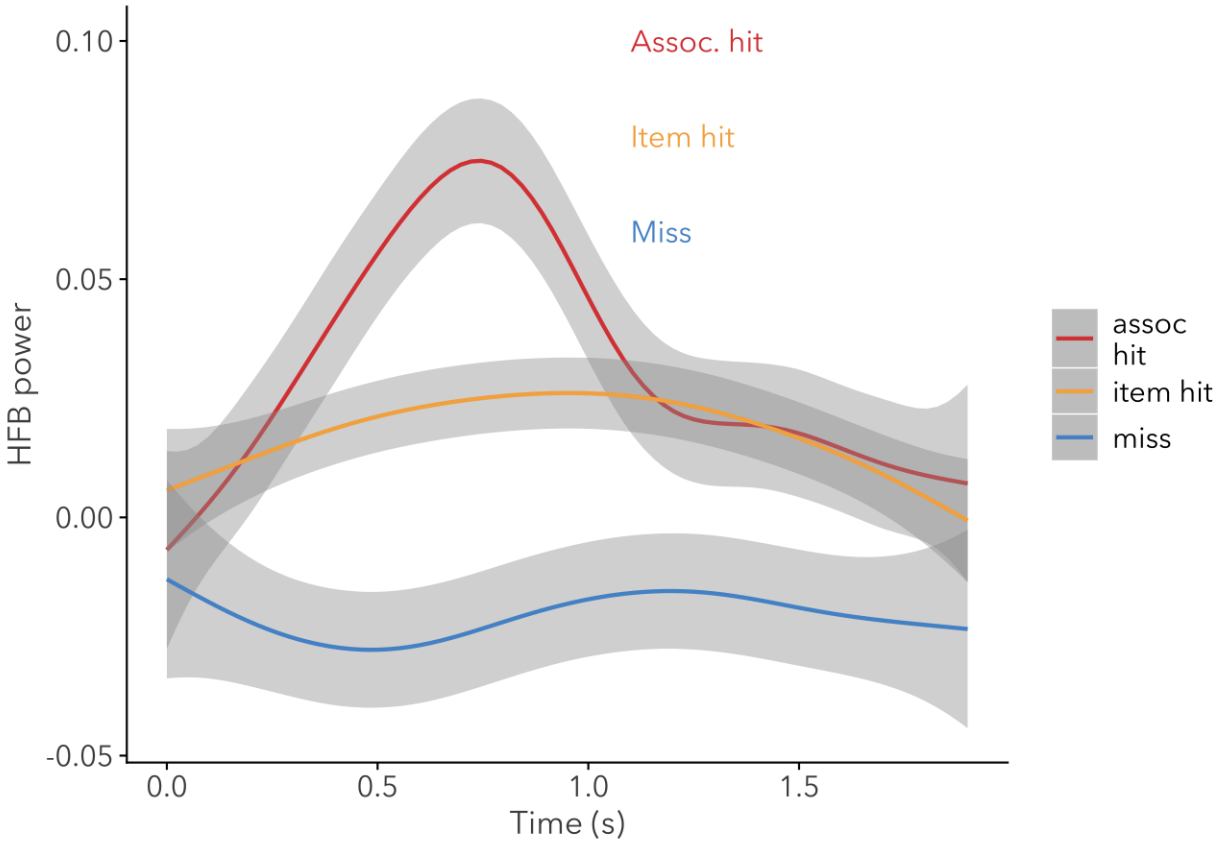
Alternative memory coding

HFB

Study For all univariate HFB analyses, HFB was first divided into 100-ms time bins separately for each electrode.

For study phase item A univariate analyses, the models included time-bin, item A category, and subsequent memory outcome as predictors and controls.

For study phase item B univariate analyses, the models included time-bin, both item A and item B categories, and subsequent memory outcome as predictors and controls.



Item A

```
## [1] "adapt_delta: 0.999"
```

```
## [1] "max_treedepth: 10"
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: value ~ 1 + time + cue1_category + trial_type_combo_alt + (1 | subj/chan)
##    Data: . (Number of observations: 50680)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 7)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.05      0.03     0.02     0.12 1.00      953     1430
##
## ~subj:chan (Number of levels: 30)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.03      0.01     0.02     0.04 1.00     1114     1877
##
## Regression Coefficients:
##                             Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## Intercept                       0.04      0.02    -0.01     0.08 1.00      895
## time                           -0.00      0.00    -0.01     0.00 1.00     6637
## cue1_categoryBuilding          -0.00      0.00    -0.01     0.00 1.00     4404
```

```
## cue1_categoryWord              -0.05      0.00    -0.05    -0.04 1.00     4356
## cue1_categoryNumber            -0.04      0.00    -0.05    -0.03 1.00     3959
## trial_type_combo_altitemhit    -0.01      0.00    -0.02    -0.00 1.00     4884
## trial_type_combo_altmiss       -0.03      0.01    -0.04    -0.02 1.00     4249
##                             Tail_ESS
## Intercept                       1523
## time                            2477
## cue1_categoryBuilding           3241
## cue1_categoryWord               3138
## cue1_categoryNumber             2903
## trial_type_combo_altitemhit     3343
## trial_type_combo_altmiss        2799
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.35      0.00     0.35     0.36 1.00     6893     2839
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
## [1] "adapt_delta: 0.999"
```

```
## [1] "max_treedepth: 10"
```

```
## value ~ 1 + time + cue1_category + trial_type_combo_alt + (1 | subj/chan)
```

```
## value ~ 1 + time + cue1_category + (1 | subj/chan)
```

```
##             elpd_diff se_diff  elpd_loo se_elpd_loo p_loo    se_p_loo looic
## study.mod.A      0.0       0.0 -19332.1    187.8        35.3      0.3  38664.1
## mod.simple     -18.7       6.4 -19350.7    187.6        33.5      0.3  38701.5
##             se_looic
## study.mod.A    375.6
## mod.simple     375.2
```

```
## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'study.mod.A' is the best model (ELPD =
## -19332.07), followed by 'mod.simple' (diff-ELPD = -18.67 +- 6.36, p = 0.003)
```

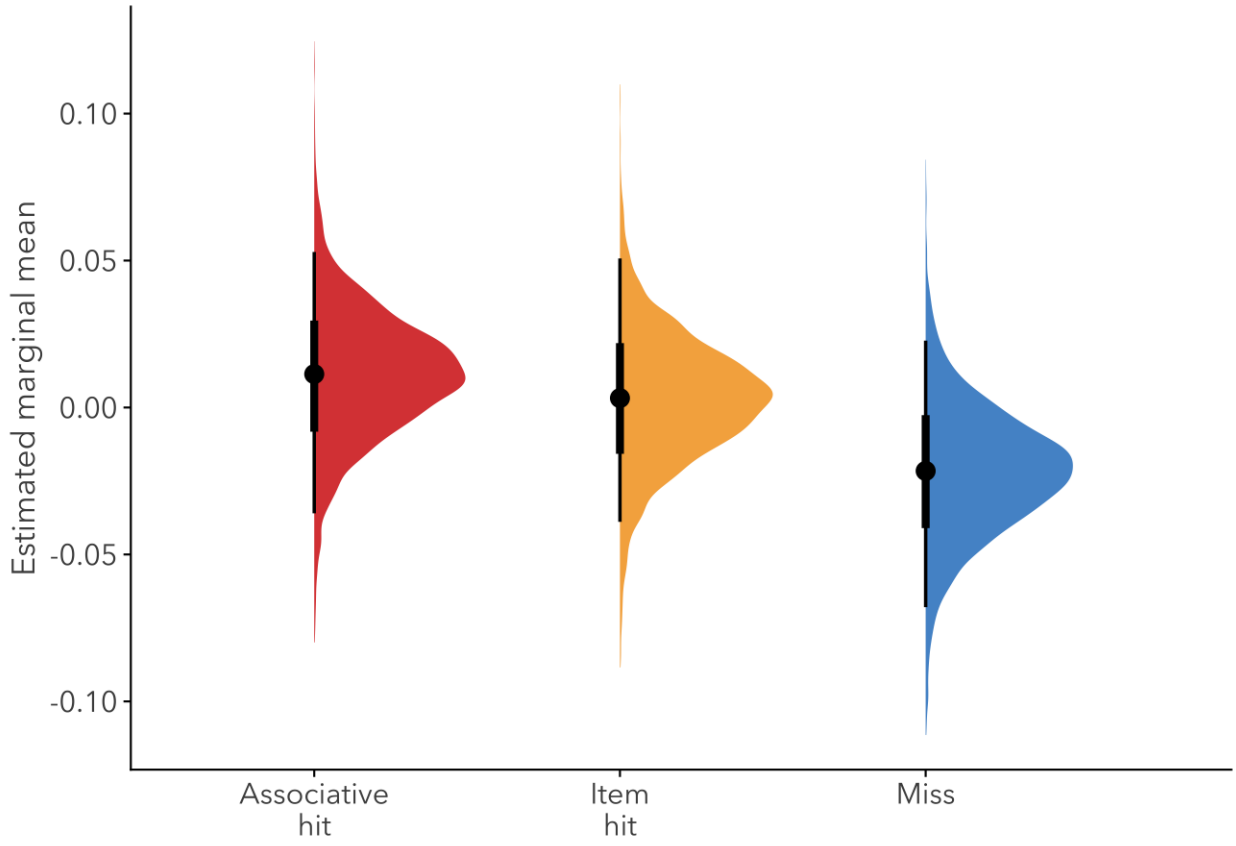
effect	component	term	estimate	std.error	conf.low	conf.high
fixed	cond	(Intercept)	0.0368341	0.0224960	-0.0096822	0.0816710
fixed	cond	time	-0.0037995	0.0027370	-0.0090342	0.0016581
fixed	cond	cue1_categoryBuilding	-0.0040001	0.0044619	-0.0125883	0.0047083
fixed	cond	cue1_categoryWord	-0.0459186	0.0045442	-0.0548600	-0.0371296
fixed	cond	cue1_categoryNumber	-0.0377383	0.0048507	-0.0472956	-0.0282262
fixed	cond	trial_type_combo_altitemhit	-0.0084523	0.0038294	-0.0159401	-0.0008404
fixed	cond	trial_type_combo_altmiss	-0.0333238	0.0052167	-0.0436779	-0.0229483

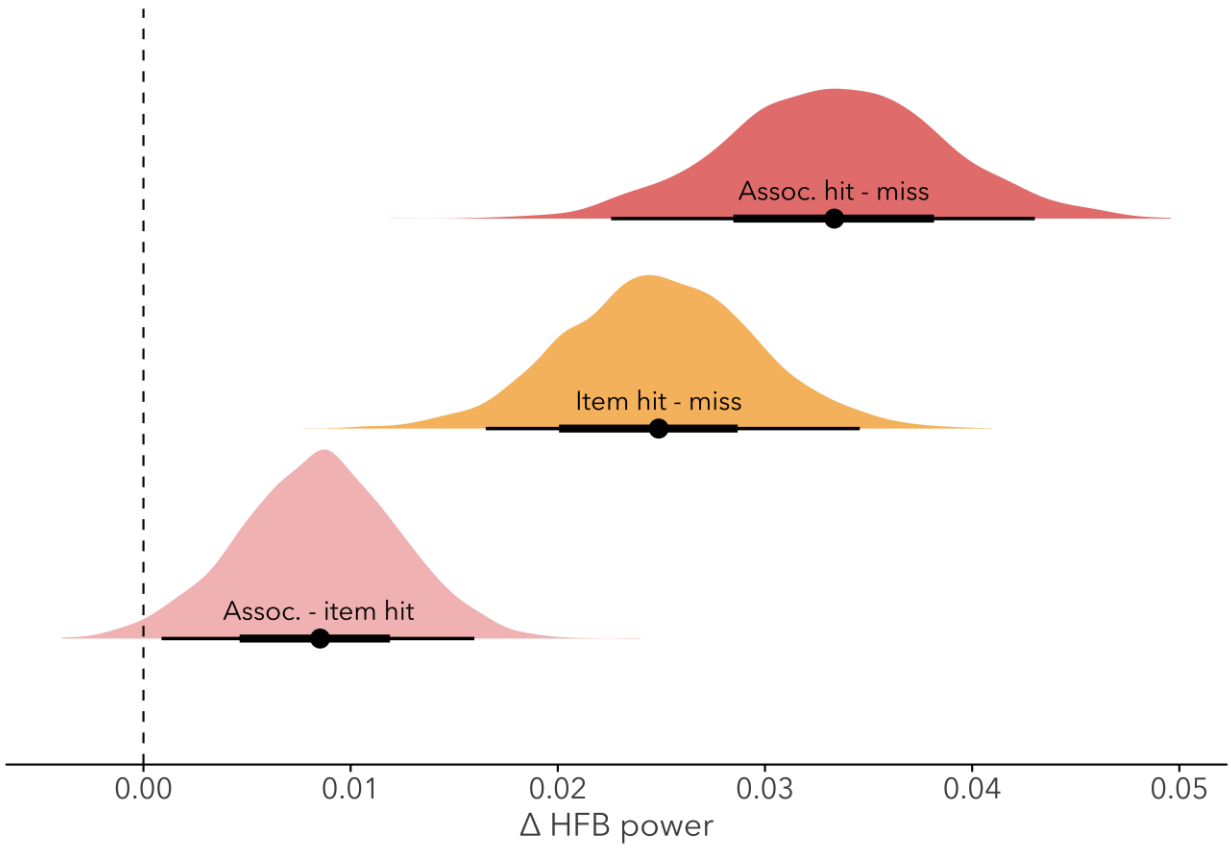
Table 74: study A emmeans

trial_type_combo_alt	estimate	lower.HPD	upper.HPD
assoc hit	0.011	-0.036	0.053
item hit	0.003	-0.044	0.045
miss	-0.022	-0.070	0.021

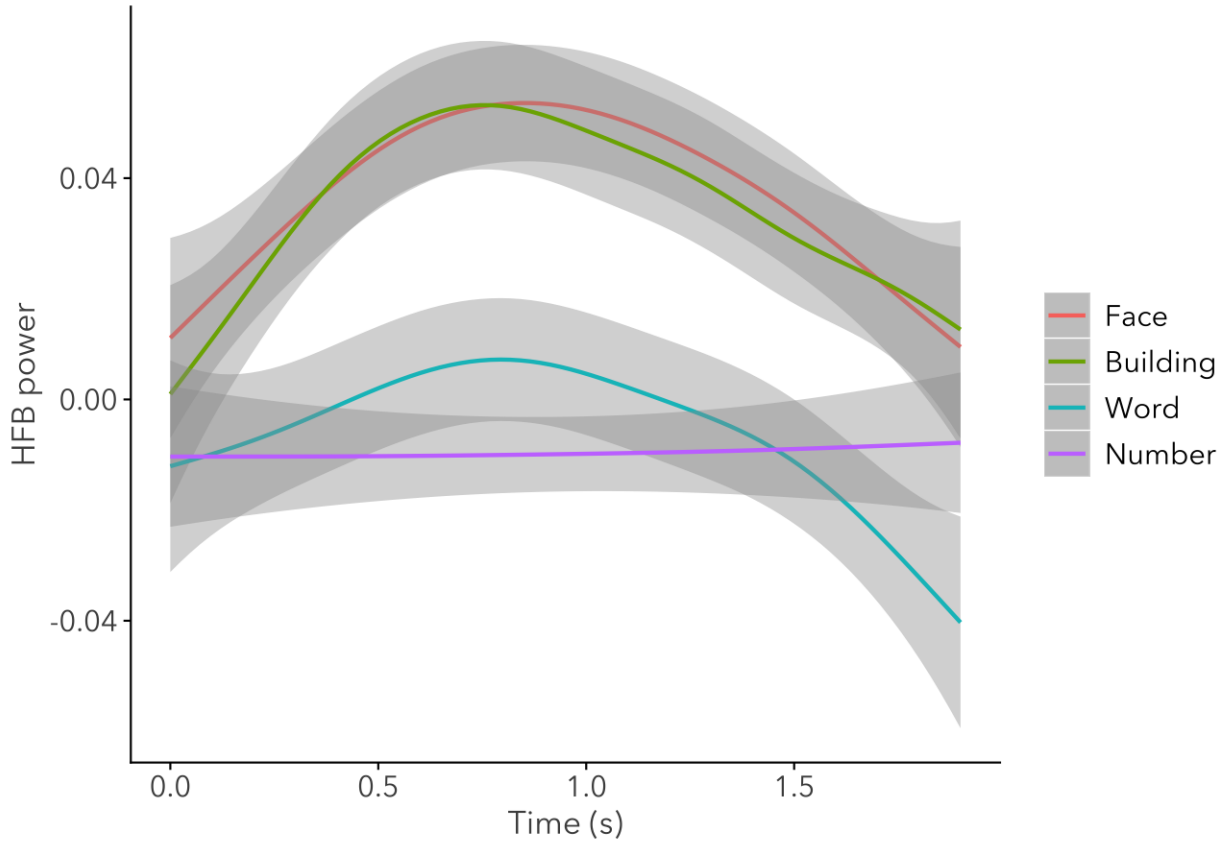
Table 75: study A emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
trial_type_combo_alt	assoc hit - item hit	0	0.009	0.001	0.016
trial_type_combo_alt	assoc hit - miss	0	0.033	0.023	0.043
trial_type_combo_alt	item hit - miss	0	0.025	0.017	0.035





Stimulus category




```
## [1] "adapt_delta: 0.8"
```

```
## [1] "max_treedepth: 10"
```

```
## value ~ 1 + time + cue1_category + trial_type_combo_alt + (1 | subj/chan)
```

```
## value ~ 1 + time + trial_type_combo_alt + (1 | subj/chan)
```

```
##             elpd_diff se_diff  elpd_loo se_elpd_loo p_loo    se_p_loo looic
## study.mod.A      0.0       0.0 -19332.1    187.8        35.3      0.3  38664.1
## mod.simple     -71.3      12.2 -19403.4    186.9        32.4      0.3  38806.7
##             se_looic
## study.mod.A    375.6
## mod.simple     373.8
```

```
## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'study.mod.A' is the best model (ELPD =
## -19332.07), followed by 'mod.simple' (diff-ELPD = -71.29 +- 12.25, p < .001)
```

Table 76: study A emmeans

cue1_category	estimate	lower.HPD	upper.HPD

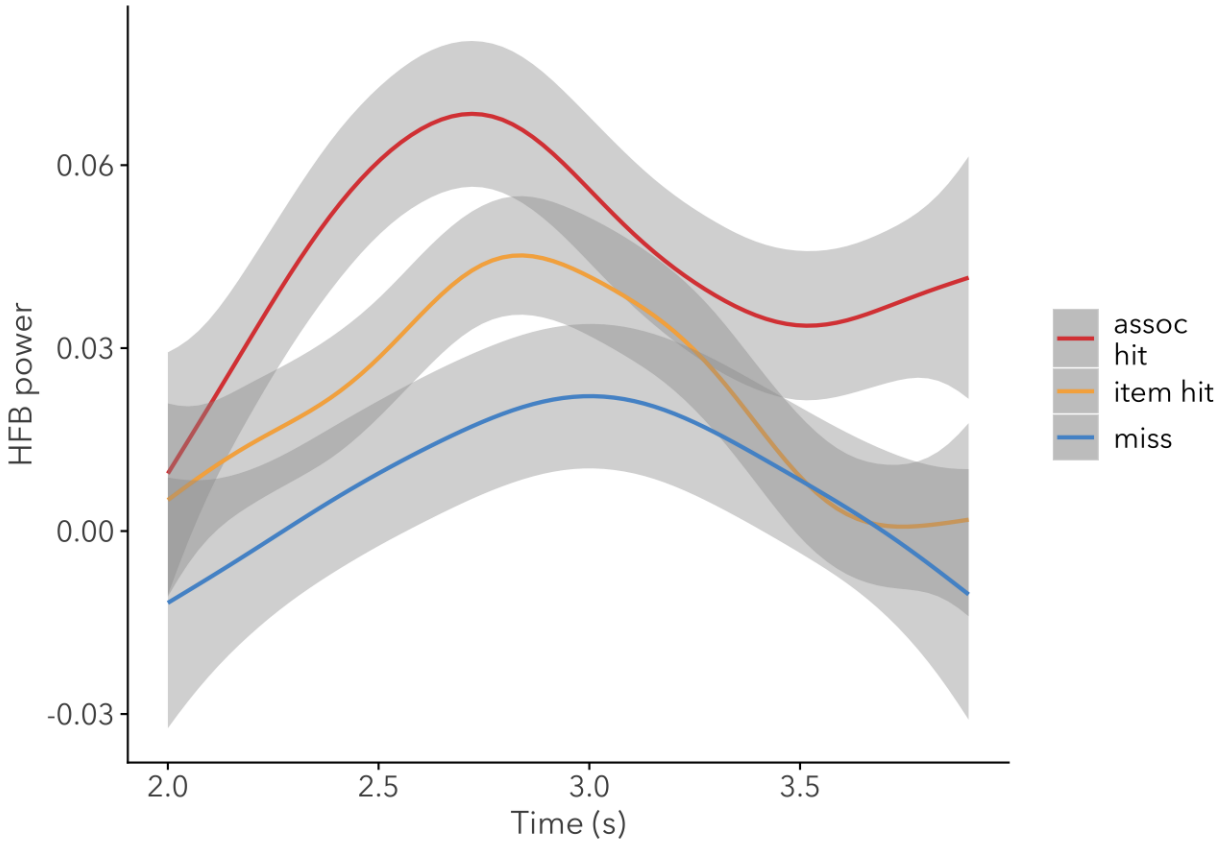
Face	0.020	-0.028	0.062
Building	0.016	-0.030	0.060
Word	-0.026	-0.072	0.018
Number	-0.018	-0.064	0.025

Table 77: study A emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
cue1_category	Face - Building	0	0.004	-0.004	0.013
cue1_category	Face - Word	0	0.046	0.037	0.055
cue1_category	Face - Number	0	0.038	0.028	0.047
cue1_category	Building - Word	0	0.042	0.033	0.050
cue1_category	Building - Number	0	0.034	0.024	0.043
cue1_category	Word - Number	0	-0.008	-0.018	0.001

Table 78: study A emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
cue1_category	Building - Face	0	-0.004	-0.013	0.004
cue1_category	Word - Face	0	-0.046	-0.055	-0.037
cue1_category	Word - Building	0	-0.042	-0.050	-0.033
cue1_category	Number - Face	0	-0.038	-0.047	-0.028
cue1_category	Number - Building	0	-0.034	-0.043	-0.024
cue1_category	Number - Word	0	0.008	-0.001	0.018



Item B

```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: value ~ 1 + time + cue1_category + cue2_category + trial_type_combo_alt + (1 | subj/chan)
##    Data: . (Number of observations: 50680)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 7)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.06      0.03     0.02     0.12 1.00     1032     1469
##
## ~subj:chan (Number of levels: 30)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.03      0.01     0.02     0.04 1.00      968     1660
##
## Regression Coefficients:
##                             Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## Intercept                       0.04      0.03    -0.01     0.09 1.00     1525
## time                           -0.00      0.00    -0.01     0.00 1.00     6854
## cue1_categoryBuilding           0.00      0.00    -0.01     0.01 1.00     3566
```

```
## cue1_categoryWord              -0.02      0.01    -0.03    -0.01 1.00     3902
## cue1_categoryNumber             0.03      0.01     0.01     0.04 1.00     3748
## cue2_categoryFace               0.00      0.00    -0.01     0.01 1.00     3528
## cue2_categoryNumber             0.03      0.00     0.02     0.04 1.00     3734
## cue2_categoryWord              -0.01      0.00    -0.02    -0.00 1.00     3548
## trial_type_combo_altitemhit    -0.02      0.00    -0.03    -0.01 1.00     4418
## trial_type_combo_altmiss       -0.04      0.01    -0.05    -0.03 1.00     3941
##                             Tail_ESS
## Intercept                       1678
## time                            3040
## cue1_categoryBuilding           2996
## cue1_categoryWord               2980
## cue1_categoryNumber             3389
## cue2_categoryFace               2960
## cue2_categoryNumber             3394
## cue2_categoryWord               3478
## trial_type_combo_altitemhit     3078
## trial_type_combo_altmiss        2885
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.37      0.00     0.36     0.37 1.00     6425     2682
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).

## [1] "adapt_delta: 0.99"

## [1] "max_treedepth: 10"

## value ~ 1 + time + cue1_category + cue2_category + trial_type_combo_alt + (1 | subj/chan)

## value ~ 1 + time + cue1_category + cue2_category + (1 | subj/chan)

##             elpd_diff se_diff  elpd_loo se_elpd_loo p_loo    se_p_loo looic
## study.mod.B      0.0       0.0 -21088.4    201.7        38.5      0.4  42176.9
## mod.simple     -25.6       7.4 -21114.0    201.3        36.4      0.4  42228.1
##             se_looic
## study.mod.B    403.4
## mod.simple     402.6

## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'study.mod.B' is the best model (ELPD =
## -21088.43), followed by 'mod.simple' (diff-ELPD = -25.60 +- 7.36, p < .001)
```

effect	component	term	estimate	std.error	conf.low	conf.high
fixed	cond	(Intercept)	0.0377938	0.0274998	-0.0147200	0.0928854
fixed	cond	time	-0.0033926	0.0028771	-0.0090023	0.0023121
fixed	cond	cue1_categoryBuilding	0.0005480	0.0048569	-0.0091602	0.0098303

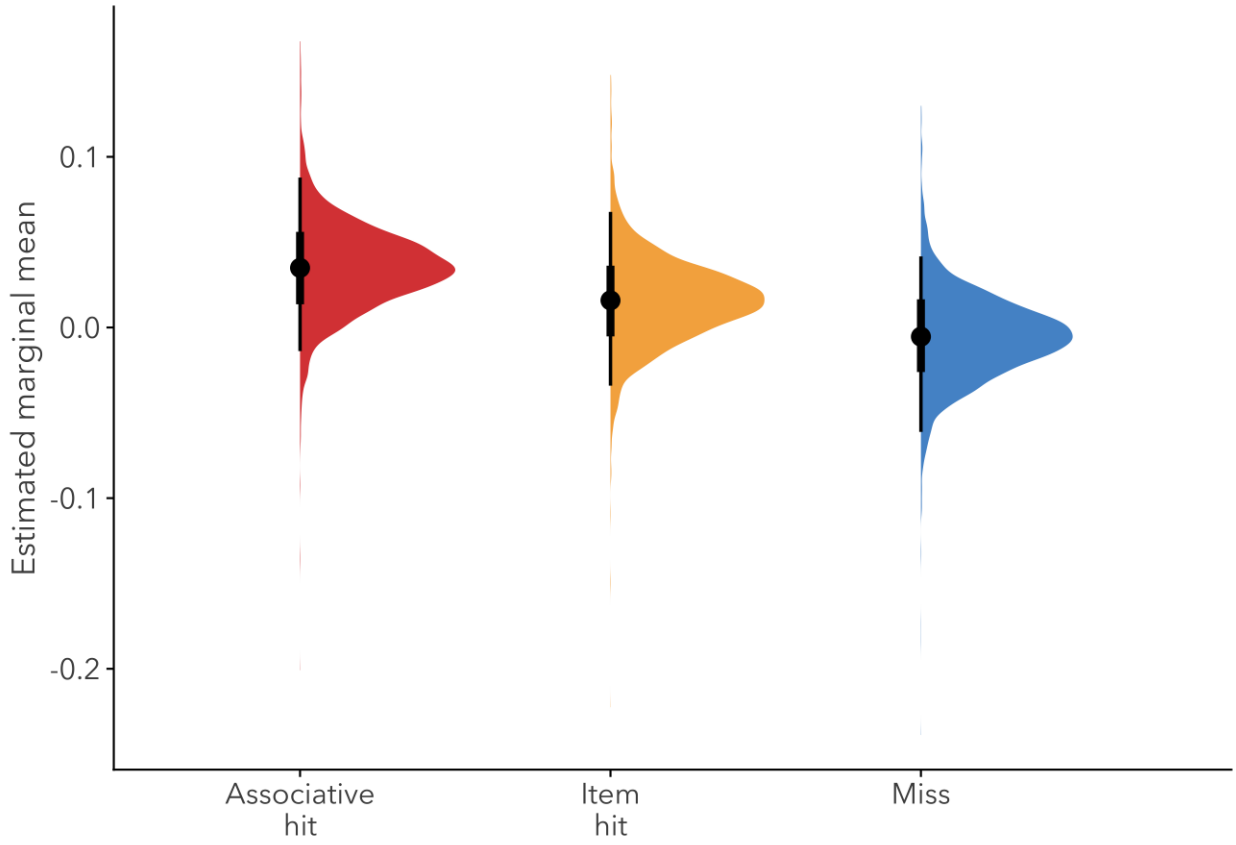
fixed	cond	cue1_categoryWord	-0.0221222	0.0050044	-0.0318977	-0.0126008
fixed	cond	cue1_categoryNumber	0.0251362	0.0053210	0.0146441	0.0353063
fixed	cond	cue2_categoryFace	0.0011283	0.0048131	-0.0082601	0.0105119
fixed	cond	cue2_categoryNumber	0.0343582	0.0049215	0.0248130	0.0438984
fixed	cond	cue2_categoryWord	-0.0104237	0.0048162	-0.0199374	-0.0008769
fixed	cond	trial_type_combo_altitemhit	-0.0192201	0.0039192	-0.0268204	-0.0118394
fixed	cond	trial_type_combo_altmiss	-0.0404332	0.0055646	-0.0510315	-0.0291665

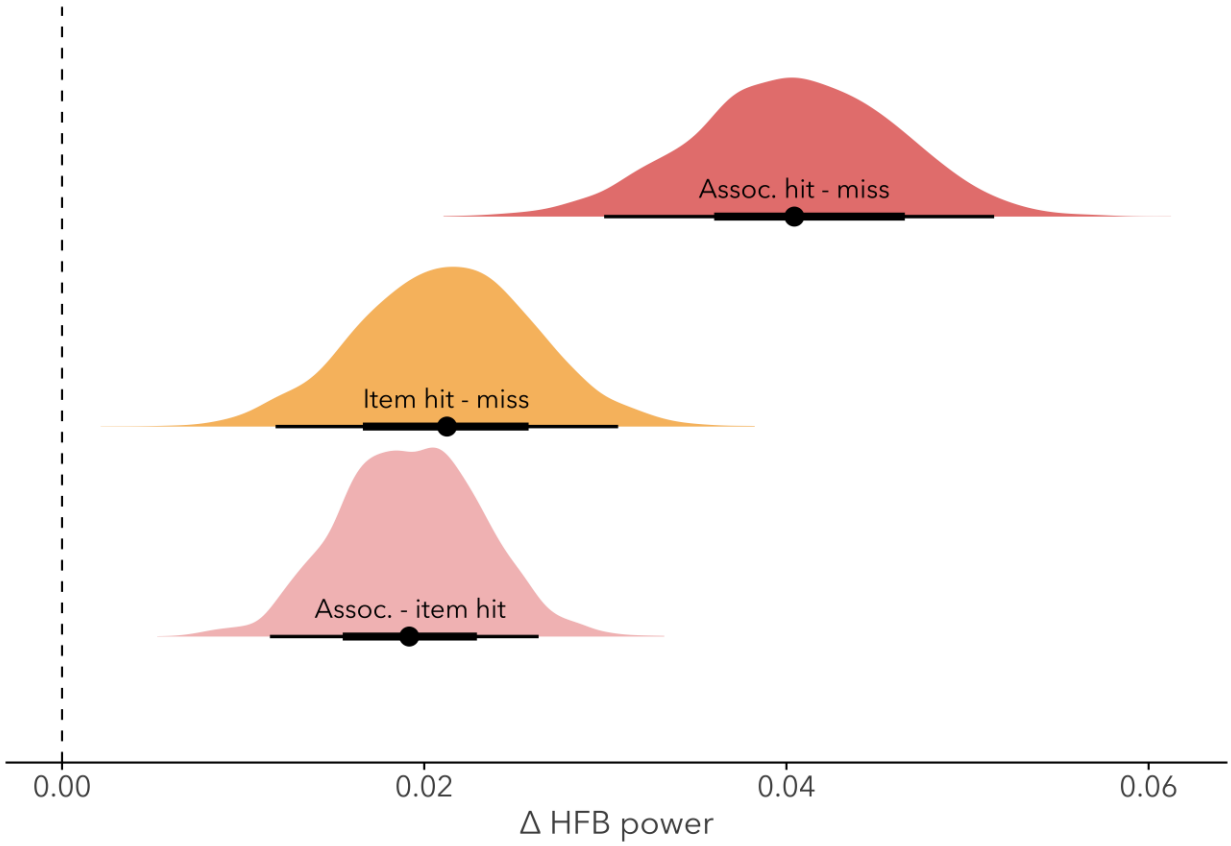
Table 80: study B emmeans

trial_type_combo_alt	estimate	lower.HPD	upper.HPD
assoc hit	0.035	-0.014	0.088
item hit	0.016	-0.033	0.068
miss	-0.005	-0.061	0.042

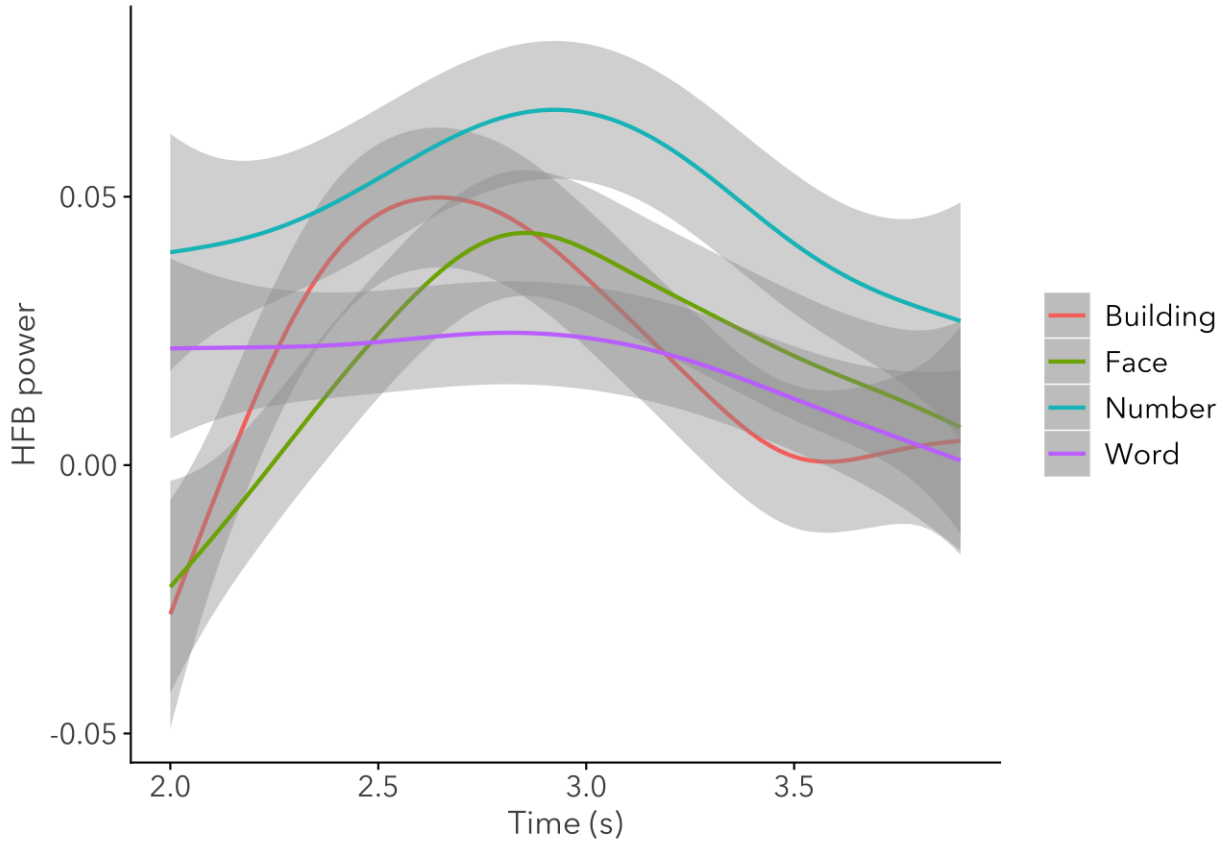
Table 81: study B emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
trial_type_combo_alt	assoc hit - item hit	0	0.019	0.011	0.026
trial_type_combo_alt	assoc hit - miss	0	0.040	0.030	0.051
trial_type_combo_alt	item hit - miss	0	0.021	0.012	0.031





Stimulus category




```
## [1] "adapt_delta: 0.8"
```

```
## [1] "max_treedepth: 10"
```

```
## value ~ 1 + time + cue1_category + cue2_category + trial_type_combo_alt + (1 | subj/chan)
```

```
## value ~ 1 + time + cue1_category + trial_type_combo_alt + (1 | subj/chan)
```

```
##             elpd_diff se_diff  elpd_loo se_elpd_loo p_loo    se_p_loo looic
## study.mod.B      0.0       0.0 -21088.4    201.7        38.5      0.4  42176.9
## mod.simple     -42.1       9.8 -21130.5    203.5        35.1      0.4  42261.1
##             se_looic
## study.mod.B    403.4
## mod.simple     407.0
```

```
## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'study.mod.B' is the best model (ELPD =
## -21088.43), followed by 'mod.simple' (diff-ELPD = -42.11 +- 9.79, p < .001)
```

Table 82: study B emmeans

cue2_category	estimate	lower.HPD	upper.HPD

Building	0.009	-0.042	0.061
Face	0.010	-0.039	0.061
Number	0.043	-0.010	0.092
Word	-0.001	-0.054	0.049

Table 83: study B emmeans

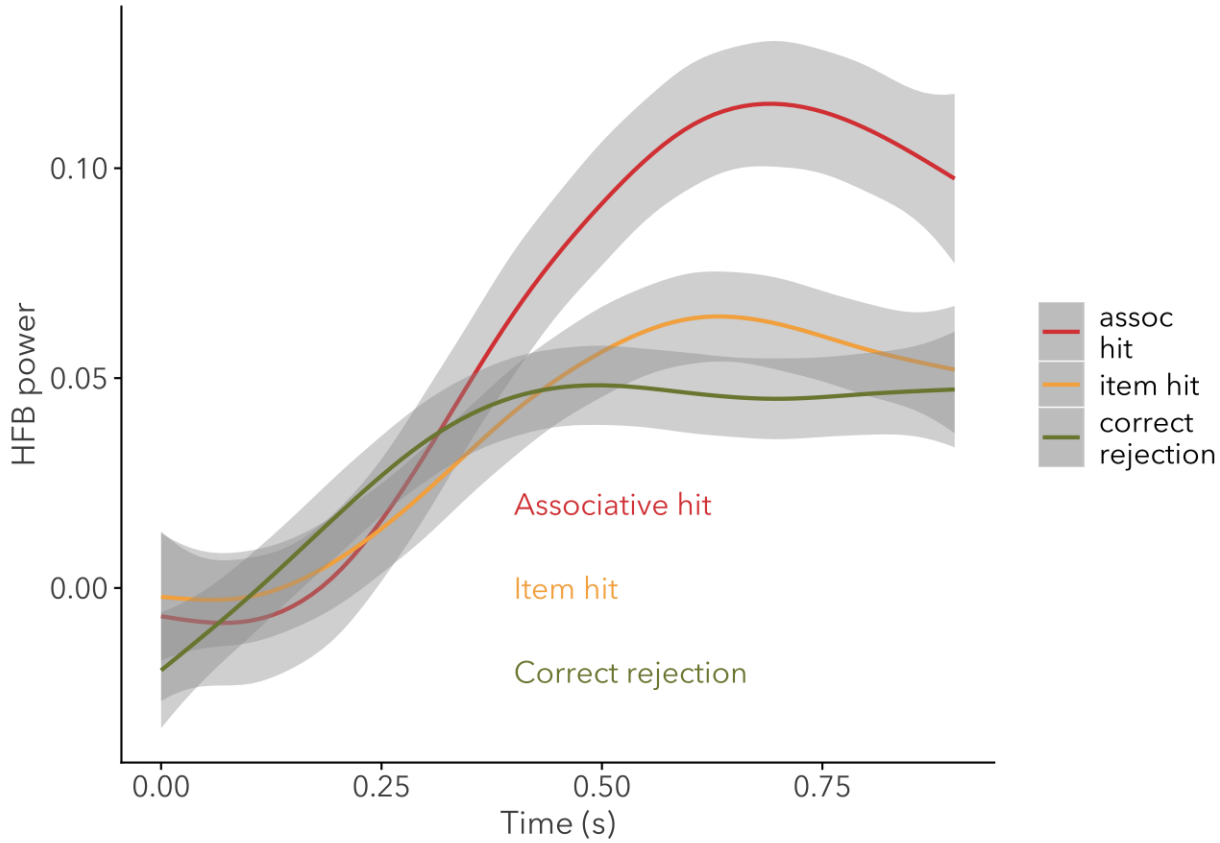
term	contrast	null.value	estimate	lower.HPD	upper.HPD
cue2_category	Building - Face	0	-0.001	-0.011	0.008
cue2_category	Building - Number	0	-0.034	-0.044	-0.025
cue2_category	Building - Word	0	0.010	0.001	0.020
cue2_category	Face - Number	0	-0.033	-0.044	-0.024
cue2_category	Face - Word	0	0.011	0.001	0.020
cue2_category	Number - Word	0	0.045	0.035	0.054

Table 84: study B emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
cue2_category	Face - Building	0	0.001	-0.008	0.011
cue2_category	Number - Building	0	0.034	0.025	0.044
cue2_category	Number - Face	0	0.033	0.024	0.044
cue2_category	Word - Building	0	-0.010	-0.020	-0.001
cue2_category	Word - Face	0	-0.011	-0.020	-0.001
cue2_category	Word - Number	0	-0.045	-0.054	-0.035

Test cue For test phase univariate analyses on the effects of item categories, *the lure trials were excluded*, and the model included time-bin, item A and item B categories, and subsequent memory outcome as predictors and controls.

For test phase univariate analyses on the effects of memory outcome, *all trials* were included, and time-bin and item A category were included as control variables.




```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: value ~ 1 + time + cue1_category + trial_type_combo_alt + (1 | subj/chan)
##    Data: . (Number of observations: 48610)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 7)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.06      0.03     0.03     0.14 1.00     1133     1847
##
## ~subj:chan (Number of levels: 30)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.03      0.01     0.02     0.04 1.00     1223     2192
##
## Regression Coefficients:
##                                      Estimate Est.Error l-95% CI u-95% CI
## Intercept                                0.01      0.03    -0.05     0.06
## time                                     0.08      0.01     0.07     0.09
## cue1_categoryBuilding                    0.02      0.00     0.01     0.03
```

```
## cue1_categoryWord                          -0.01      0.00    -0.01     0.00
## cue1_categoryNumber                        -0.04      0.00    -0.05    -0.03
## trial_type_combo_altitemhit                -0.01      0.00    -0.02    -0.01
## trial_type_combo_altcorrect\nrejection     -0.02      0.00    -0.03    -0.01
## trial_type_combo_altmiss                   -0.03      0.01    -0.04    -0.01
## trial_type_combo_altfa                     -0.01      0.01    -0.03     0.01
##                                         Rhat Bulk_ESS Tail_ESS
## Intercept                               1.00     1348     1538
## time                                    1.00     5078     2574
## cue1_categoryBuilding                   1.00     4573     3260
## cue1_categoryWord                       1.00     4363     3014
## cue1_categoryNumber                     1.00     3913     2733
## trial_type_combo_altitemhit             1.00     3528     3247
## trial_type_combo_altcorrect\nrejection 1.00     4174     3359
## trial_type_combo_altmiss                1.00     4025     3076
## trial_type_combo_altfa                  1.00     3781     3073
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.33      0.00     0.33     0.34 1.00     5820     2880
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).

## [1] "adapt_delta: 0.99"

## [1] "max_treedepth: 10"

## value ~ 1 + time + cue1_category + trial_type_combo_alt + (1 | subj/chan)

## value ~ 1 + time + cue1_category + (1 | subj/chan)

##              elpd_diff se_diff  elpd_loo se_elpd_loo p_loo    se_p_loo looic
## test.mod.cue      0.0       0.0 -15784.8    167.0        37.4      0.3  31569.7
## mod.simple      -11.6       5.5 -15796.5    167.0        33.7      0.2  31592.9
##              se_looic
## test.mod.cue    333.9
## mod.simple      334.0

## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'test.mod.cue' is the best model (ELPD =
## -15784.84), followed by 'mod.simple' (diff-ELPD = -11.61 +- 5.54, p = 0.036)
```

effect	component	term	estimate	std.error	conf.low	conf.high
fixed	cond	(Intercept)	0.0095326	0.0273286	-0.0470999	0.0606412
fixed	cond	time	0.0801012	0.0053593	0.0697281	0.0906679
fixed	cond	cue1_categoryBuilding	0.0213057	0.0042290	0.0126533	0.0292828
fixed	cond	cue1_categoryWord	-0.0053563	0.0044393	-0.0142633	0.0032443
fixed	cond	cue1_categoryNumber	-0.0370704	0.0046630	-0.0462152	-0.0278316

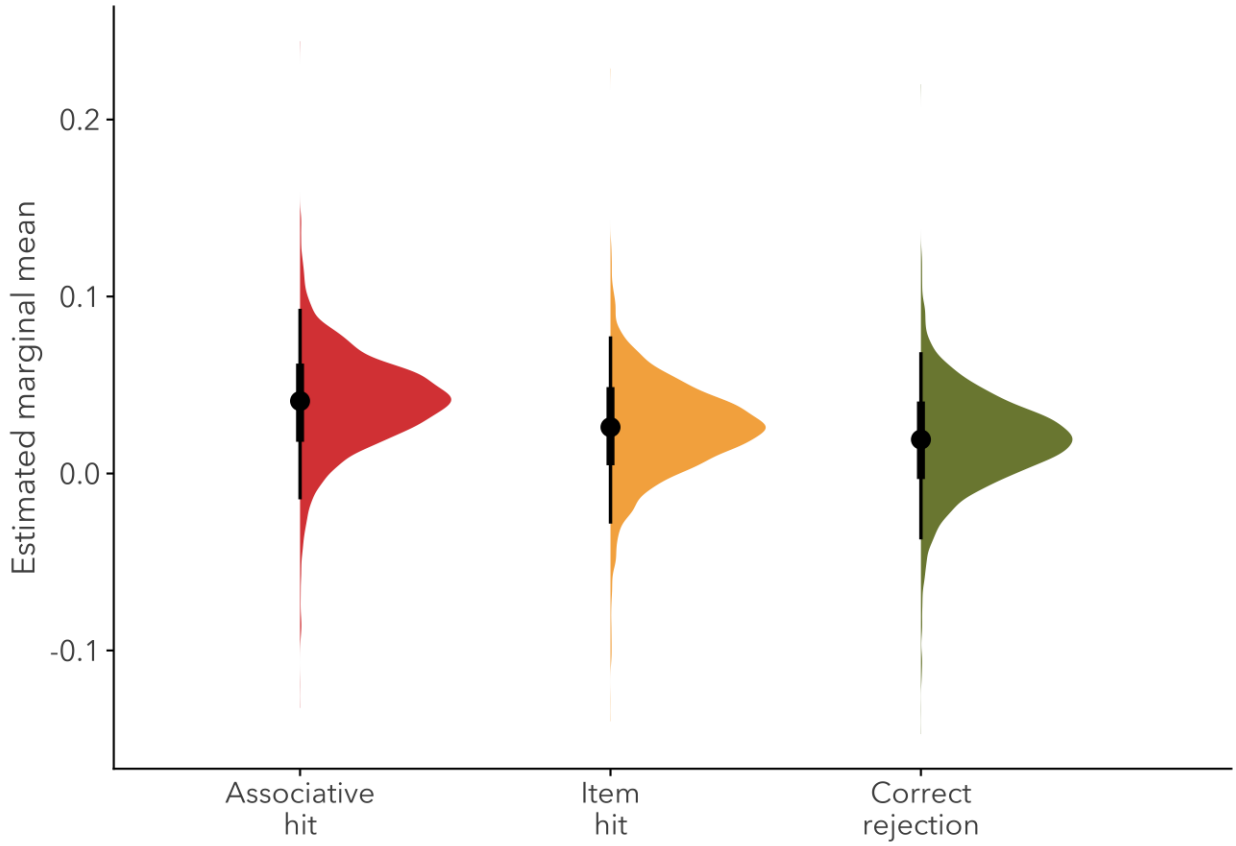
fixed	cond	trial_type_combo_altitemhit	-0.0148432	0.0045608	-0.0238358	-0.0058391
fixed	cond	trial_type_combo_altcorrect rejection	-0.0217989	0.0043766	-0.0302625	-0.0129664
fixed	cond	trial_type_combo_altmiss	-0.0270663	0.0061650	-0.0391726	-0.0149291
fixed	cond	trial_type_combo_altfa	-0.0070950	0.0097868	-0.0258003	0.0121196

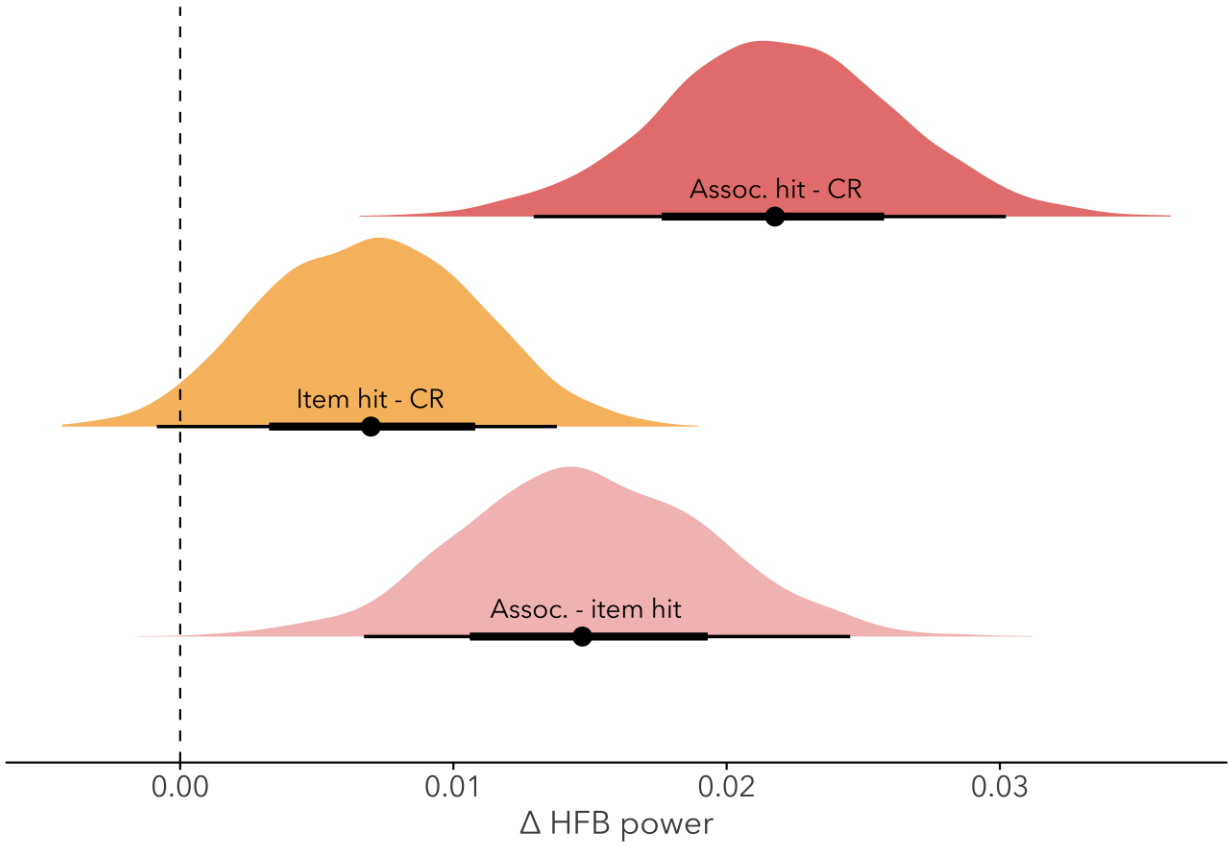
Table 86: test cue emmeans

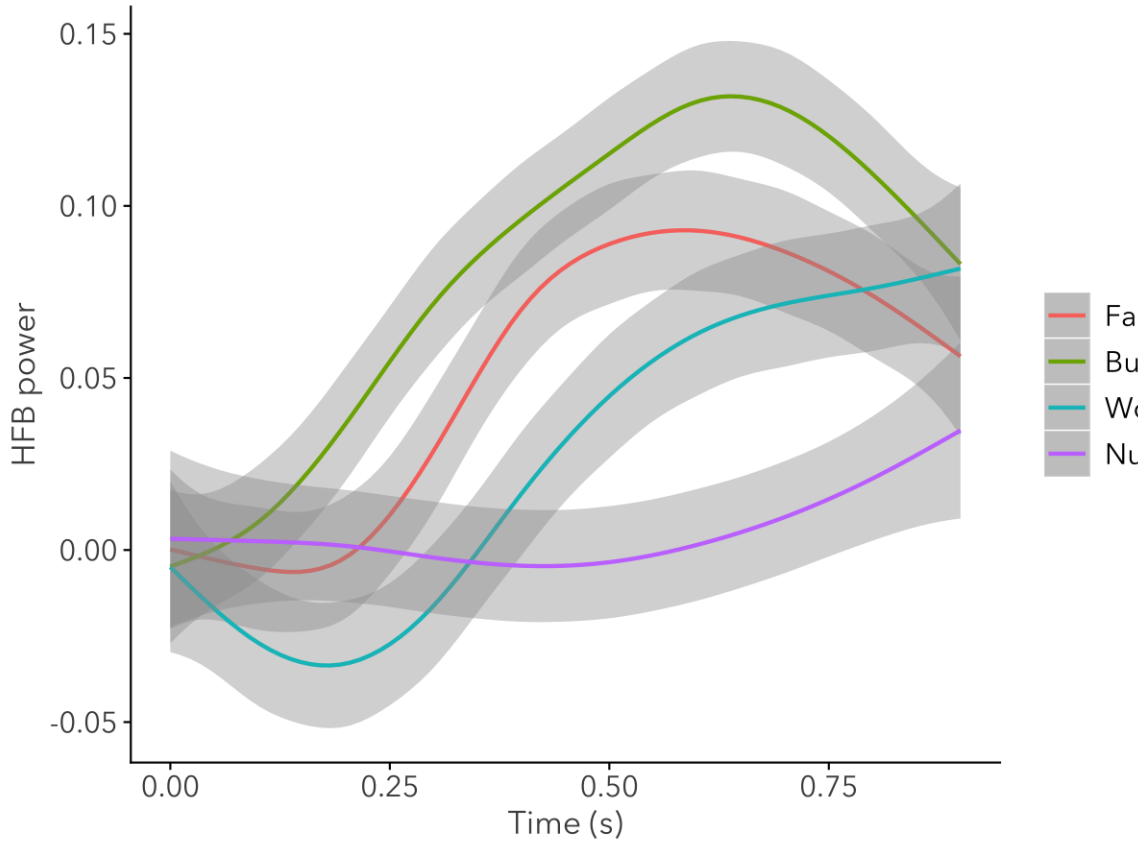
trial_type_combo_alt	estimate	lower.HPD	upper.HPD
assoc hit	0.041	-0.015	0.092
item hit	0.026	-0.024	0.081
correct rejection	0.019	-0.037	0.069
miss	0.014	-0.040	0.069
fa	0.034	-0.025	0.086

Table 87: test cue emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
trial_type_combo_alt	assoc hit - item hit	0	0.015	0.007	0.024
trial_type_combo_alt	assoc hit - correct rejection	0	0.022	0.013	0.030
trial_type_combo_alt	assoc hit - miss	0	0.027	0.015	0.039
trial_type_combo_alt	assoc hit - fa	0	0.007	-0.012	0.026
trial_type_combo_alt	item hit - correct rejection	0	0.007	-0.001	0.014
trial_type_combo_alt	item hit - miss	0	0.012	0.001	0.023
trial_type_combo_alt	item hit - fa	0	-0.008	-0.027	0.010
trial_type_combo_alt	correct rejection - miss	0	0.005	-0.005	0.016
trial_type_combo_alt	correct rejection - fa	0	-0.015	-0.033	0.003
trial_type_combo_alt	miss - fa	0	-0.020	-0.038	0.000


Estimated marginal mean
0.2
0.1
0.0
-0.1
Associative hit
Item hit
Correct rejection


Assoc. hit - CR
Item hit - CR
Assoc. - item hit
0.00
0.01
0.02
0.03
Δ HFB power



Item category effects

```
## [1] "adapt_delta: 0.999"
```

```
## [1] "max_treedepth: 10"
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: value ~ 1 + time + cue1_category + cue2_category + trial_type_combo_alt + (1 | subj/chan)
##    Data: . (Number of observations: 29670)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 7)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.07      0.03     0.02     0.15 1.00      863      985
##
## ~subj:chan (Number of levels: 30)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.04      0.01     0.03     0.05 1.00     1006     1710
##
## Regression Coefficients:
##                             Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## Intercept                       0.01      0.03    -0.05     0.07 1.00     1419
## time                            0.09      0.01     0.08     0.10 1.00     5402
## cue1_categoryBuilding           0.01      0.01    -0.00     0.02 1.00     3620
```

```
## cue1_categoryWord              -0.04      0.01    -0.05    -0.03 1.00     3791
## cue1_categoryNumber            -0.05      0.01    -0.06    -0.04 1.00     3171
## cue2_categoryFace               0.03      0.01     0.02     0.04 1.00     3660
## cue2_categoryNumber             0.01      0.01     0.00     0.03 1.00     3836
## cue2_categoryWord              -0.02      0.01    -0.03    -0.00 1.00     3850
## trial_type_combo_altitemhit    -0.02      0.00    -0.03    -0.01 1.00     5334
## trial_type_combo_altmiss       -0.03      0.01    -0.04    -0.02 1.00     4498
##                             Tail_ESS
## Intercept                       1694
## time                            2750
## cue1_categoryBuilding           3187
## cue1_categoryWord               3268
## cue1_categoryNumber             2993
## cue2_categoryFace               3187
## cue2_categoryNumber             2487
## cue2_categoryWord               3050
## trial_type_combo_altitemhit     3424
## trial_type_combo_altmiss        3206
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.33      0.00     0.33     0.34 1.00     6759     2903
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
## value ~ 1 + time + cue1_category + cue2_category + trial_type_combo_alt + (1 | subj/chan)
```

```
## value ~ 1 + time + cue1_category + cue2_category + (1 | subj/chan)
```

```
##              elpd_diff se_diff elpd_loo se_elpd_loo p_loo   se_p_loo looic
## test.mod.cue     0.0       0.0 -9526.6    128.8        38.5     0.4  19053.2
## mod.simple      -9.9       5.0 -9536.5    128.9        36.1     0.3  19072.9
##              se_looic
## test.mod.cue   257.7
## mod.simple     257.8
```

```
## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'test.mod.cue' is the best model (ELPD =
## -9526.60), followed by 'mod.simple' (diff-ELPD = -9.86 +- 5.00, p = 0.048)
```

Stimulus cat A

```
## [1] "adapt_delta: 0.8"
```

```
## [1] "max_treedepth: 10"
```

```
## value ~ 1 + time + cue1_category + cue2_category + trial_type_combo_alt + (1 | subj/chan)

## value ~ 1 + time + cue2_category + trial_type_combo_alt + (1 | subj/chan)

##              elpd_diff se_diff elpd_loo se_elpd_loo p_loo   se_p_loo looic
## test.mod.cue     0.0       0.0 -9526.6    128.8        38.5     0.4  19053.2
## mod.simple     -56.6      10.8 -9583.2    129.1        35.1     0.3  19166.5
##              se_looic
## test.mod.cue   257.7
## mod.simple     258.2

## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'test.mod.cue' is the best model (ELPD =
## -9526.60), followed by 'mod.simple' (diff-ELPD = -56.65 +- 10.83, p < .001)
```

Table 88: test cue emmeans

cue1_category	estimate	lower.HPD	upper.HPD
Face	0.045	-0.011	0.106
Building	0.053	-0.002	0.116
Word	0.008	-0.048	0.069
Number	-0.003	-0.061	0.057

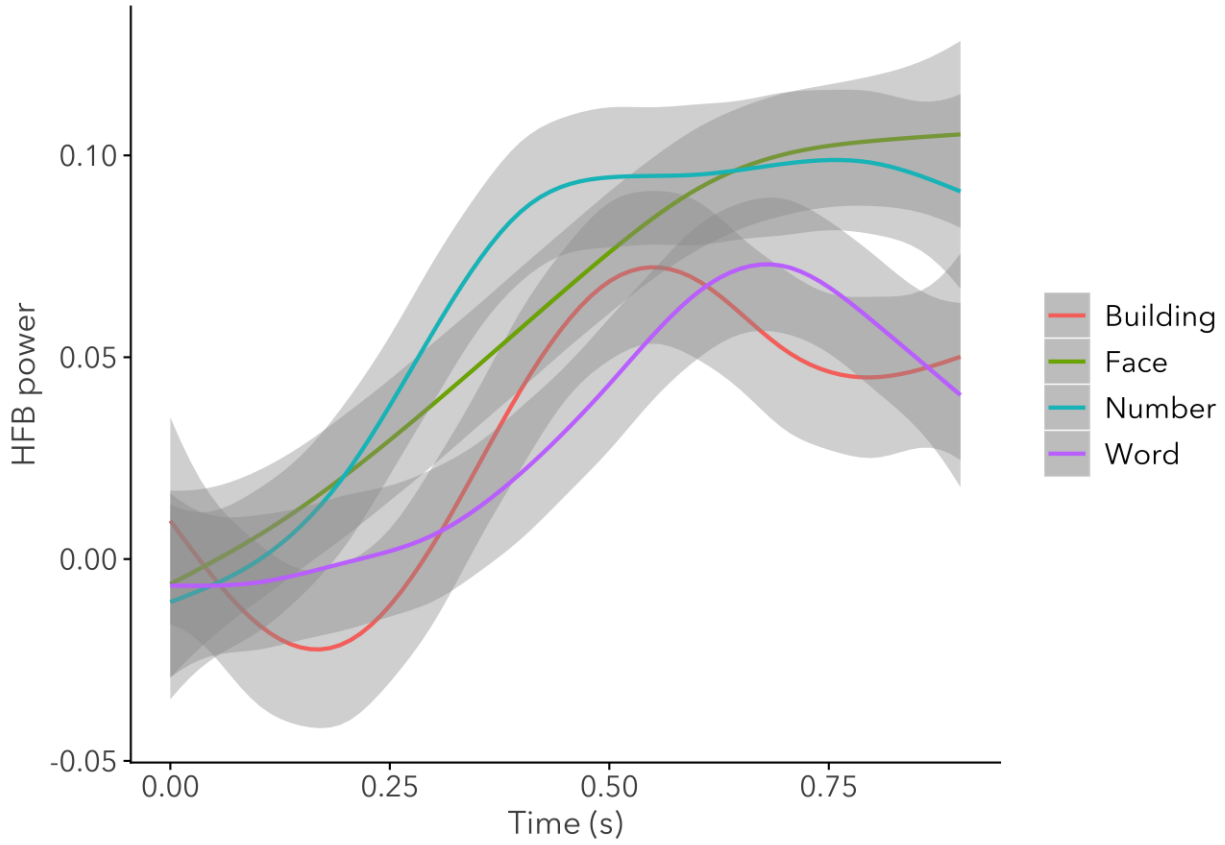
Table 89: test cue emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
cue1_category	Face - Building	0	-0.008	-0.019	0.004
cue1_category	Face - Word	0	0.037	0.025	0.049
cue1_category	Face - Number	0	0.048	0.036	0.060
cue1_category	Building - Word	0	0.045	0.034	0.056
cue1_category	Building - Number	0	0.056	0.044	0.068
cue1_category	Word - Number	0	0.011	-0.001	0.024

Table 90: test cue emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
cue1_category	Building - Face	0	0.008	-0.004	0.019
cue1_category	Word - Face	0	-0.037	-0.049	-0.025
cue1_category	Word - Building	0	-0.045	-0.056	-0.034
cue1_category	Number - Face	0	-0.048	-0.060	-0.036
cue1_category	Number - Building	0	-0.056	-0.068	-0.044
cue1_category	Number - Word	0	-0.011	-0.024	0.001

Stimulus cat B




```
## [1] "adapt_delta: 0.8"
```

```
## [1] "max_treedepth: 10"
```

```
## value ~ 1 + time + cue1_category + cue2_category + trial_type_combo_alt + (1 | subj/chan)
```

```
## value ~ 1 + time + cue1_category + trial_type_combo_alt + (1 | subj/chan)
```

```
##              elpd_diff se_diff elpd_loo se_elpd_loo p_loo   se_p_loo looic
## test.mod.cue     0.0       0.0 -9526.6    128.8        38.5     0.4  19053.2
## mod.simple     -30.8       8.2 -9557.4    128.9        34.9     0.3  19114.9
##              se_looic
## test.mod.cue   257.7
## mod.simple     257.9
```

```
## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'test.mod.cue' is the best model (ELPD =
## -9526.60), followed by 'mod.simple' (diff-ELPD = -30.84 +- 8.25, p < .001)
```

Table 91: test cue emmeans

cue2_category	estimate	lower.HPD	upper.HPD

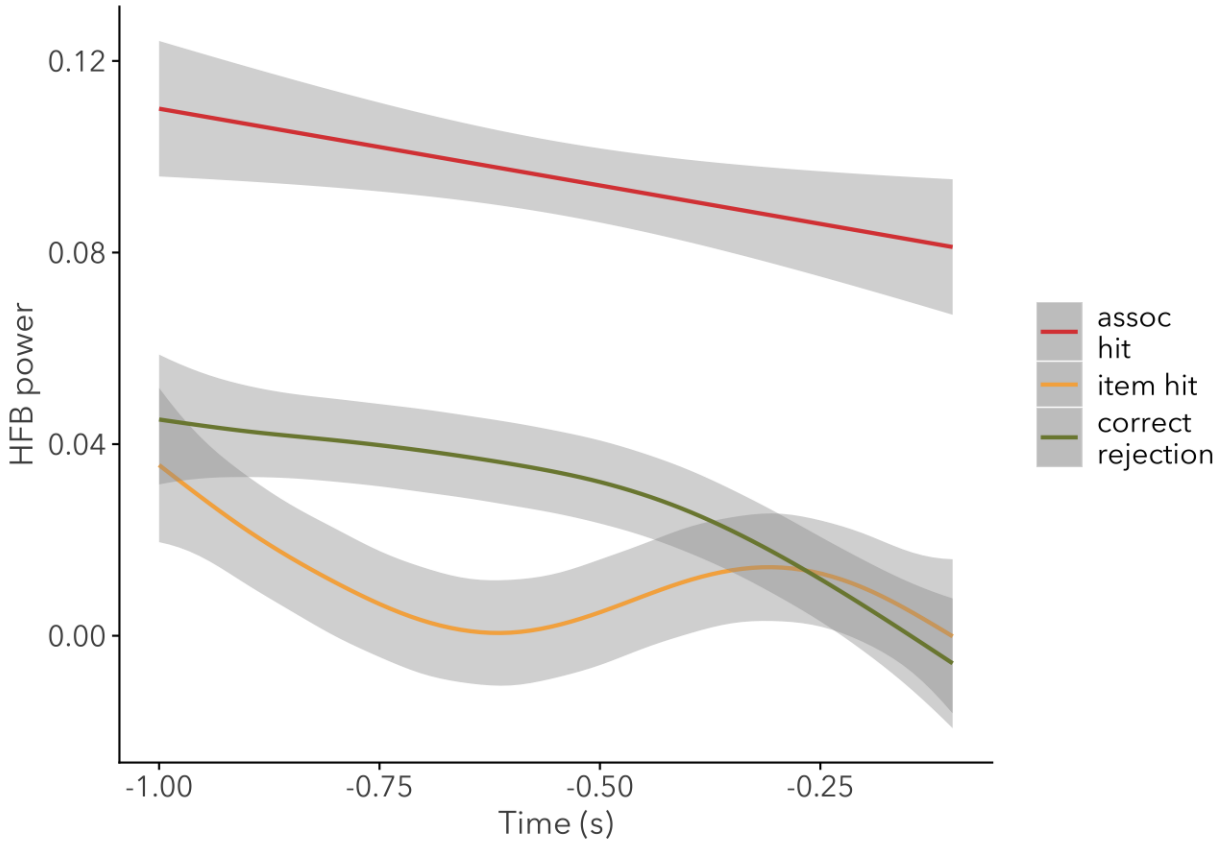
Building	0.019	-0.043	0.075
Face	0.048	-0.011	0.106
Number	0.033	-0.025	0.093
Word	0.002	-0.057	0.061

Table 92: test cue emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
cue2_category	Building - Face	0	-0.029	-0.041	-0.018
cue2_category	Building - Number	0	-0.015	-0.027	-0.004
cue2_category	Building - Word	0	0.016	0.005	0.028
cue2_category	Face - Number	0	0.015	0.003	0.026
cue2_category	Face - Word	0	0.046	0.035	0.057
cue2_category	Number - Word	0	0.031	0.020	0.043

Table 93: test cue emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
cue2_category	Face - Building	0	0.029	0.018	0.041
cue2_category	Number - Building	0	0.015	0.004	0.027
cue2_category	Number - Face	0	-0.015	-0.026	-0.003
cue2_category	Word - Building	0	-0.016	-0.028	-0.005
cue2_category	Word - Face	0	-0.046	-0.057	-0.035
cue2_category	Word - Number	0	-0.031	-0.043	-0.020



Test response

```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: value ~ 1 + time + cue1_category + trial_type_combo_alt + (1 | subj/chan)
##    Data: . (Number of observations: 48610)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 7)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.08      0.04     0.04     0.18 1.00      902     1532
##
## ~subj:chan (Number of levels: 30)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.03      0.01     0.03     0.05 1.00     1228     1998
##
## Regression Coefficients:
##                                      Estimate Est.Error l-95% CI u-95% CI
## Intercept                                0.04      0.03    -0.03     0.10
## time                                    -0.04      0.01    -0.05    -0.03
## cue1_categoryBuilding                    0.02      0.00     0.01     0.03
```

```
## cue1_categoryWord                          0.02      0.00     0.01     0.03
## cue1_categoryNumber                       -0.01      0.00    -0.02     0.00
## trial_type_combo_altitemhit               -0.07      0.00    -0.08    -0.06
## trial_type_combo_altcorrect\nrejection    -0.06      0.00    -0.07    -0.05
## trial_type_combo_altmiss                  -0.08      0.01    -0.09    -0.06
## trial_type_combo_altfa                    -0.04      0.01    -0.06    -0.02
##                                        Rhat Bulk_ESS Tail_ESS
## Intercept                              1.00     1440     1567
## time                                   1.00     4000     2823
## cue1_categoryBuilding                  1.00     3877     3219
## cue1_categoryWord                      1.00     3631     2942
## cue1_categoryNumber                    1.00     3566     3126
## trial_type_combo_altitemhit            1.00     3313     2553
## trial_type_combo_altcorrect\nrejection 1.00     3328     2779
## trial_type_combo_altmiss               1.00     3825     3410
## trial_type_combo_altfa                 1.00     4012     3284
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.36      0.00     0.36     0.37 1.00     6382     3055
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).

## [1] "adapt_delta: 0.99"

## [1] "max_treedepth: 10"

## value ~ 1 + time + cue1_category + trial_type_combo_alt + (1 | subj/chan)

## value ~ 1 + time + cue1_category + (1 | subj/chan)

##               elpd_diff se_diff  elpd_loo se_elpd_loo p_loo    se_p_loo
## test.mod.resp      0.0       0.0 -19908.3    163.5        37.6      0.3
## mod.simple      -105.3      14.7 -20013.6    163.3        33.8      0.2
##               looic    se_looic
## test.mod.resp  39816.6    327.1
## mod.simple     40027.2    326.5

## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'test.mod.resp' is the best model (ELPD =
## -19908.32), followed by 'mod.simple' (diff-ELPD = -105.27 +- 14.68, p < .001)
```

effect	component	term	estimate	std.error	conf.low	conf.high
fixed	cond	(Intercept)	0.0374882	0.0339566	-0.0287143	0.1032960
fixed	cond	time	-0.0425663	0.0056132	-0.0531596	-0.0316224
fixed	cond	cue1_categoryBuilding	0.0206054	0.0045894	0.0114812	0.0296863
fixed	cond	cue1_categoryWord	0.0203120	0.0048566	0.0106342	0.0296150
fixed	cond	cue1_categoryNumber	-0.0084879	0.0049797	-0.0182515	0.0013334

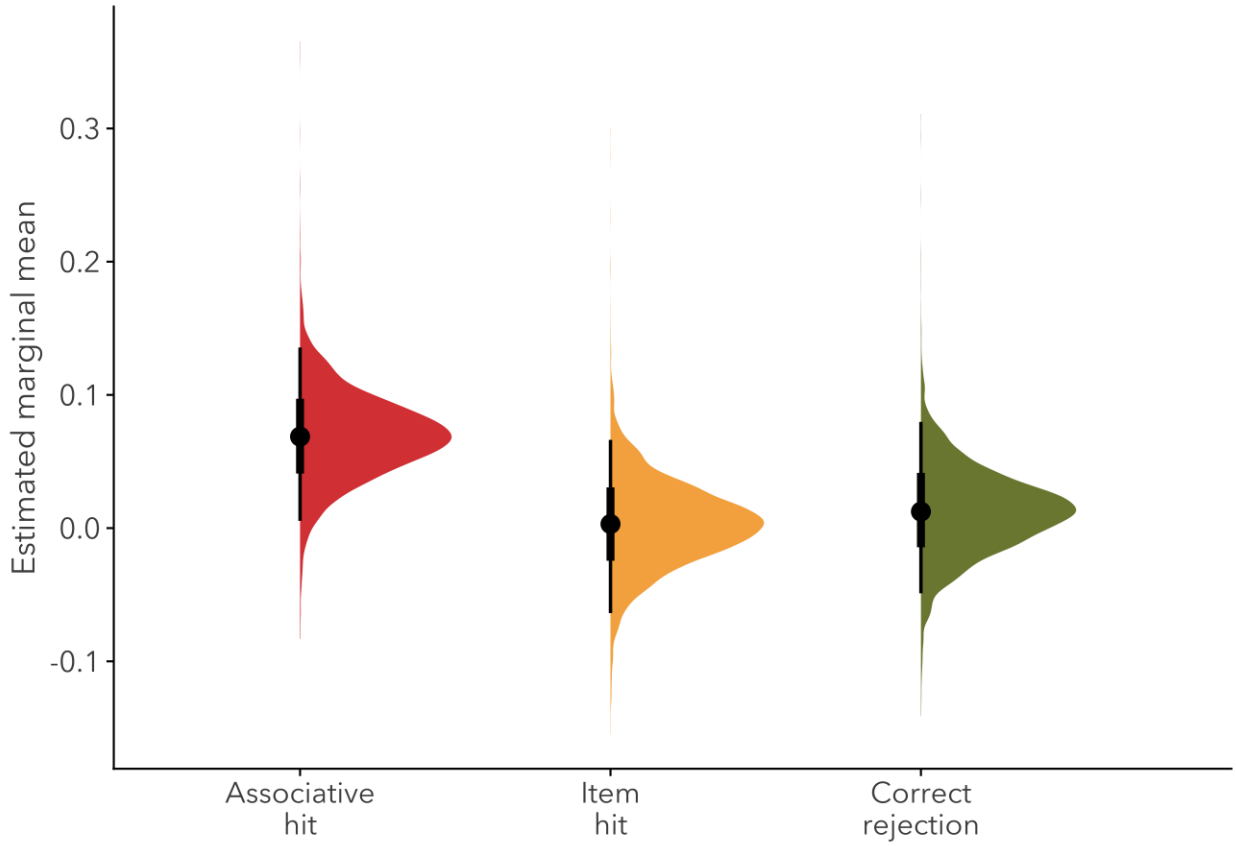
fixed	cond	trial_type_combo_altitemhit	-0.0659198	0.0049619	-0.0756511	-0.0562480
fixed	cond	trial_type_combo_altcorrect rejection	-0.0565832	0.0047931	-0.0659418	-0.0470991
fixed	cond	trial_type_combo_altmiss	-0.0779513	0.0068991	-0.0910149	-0.0645719
fixed	cond	trial_type_combo_altfa	-0.0370475	0.0107661	-0.0578186	-0.0156126

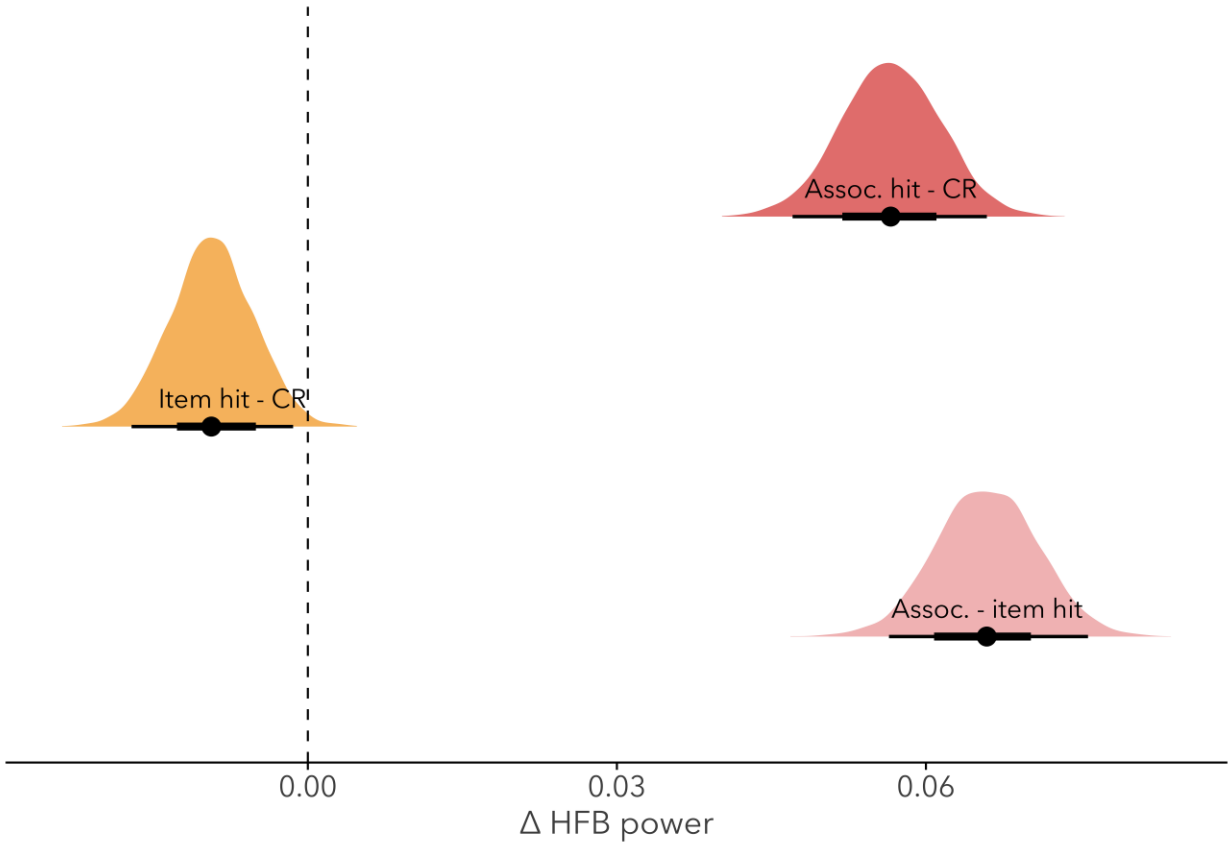
Table 95: test resp emmeans

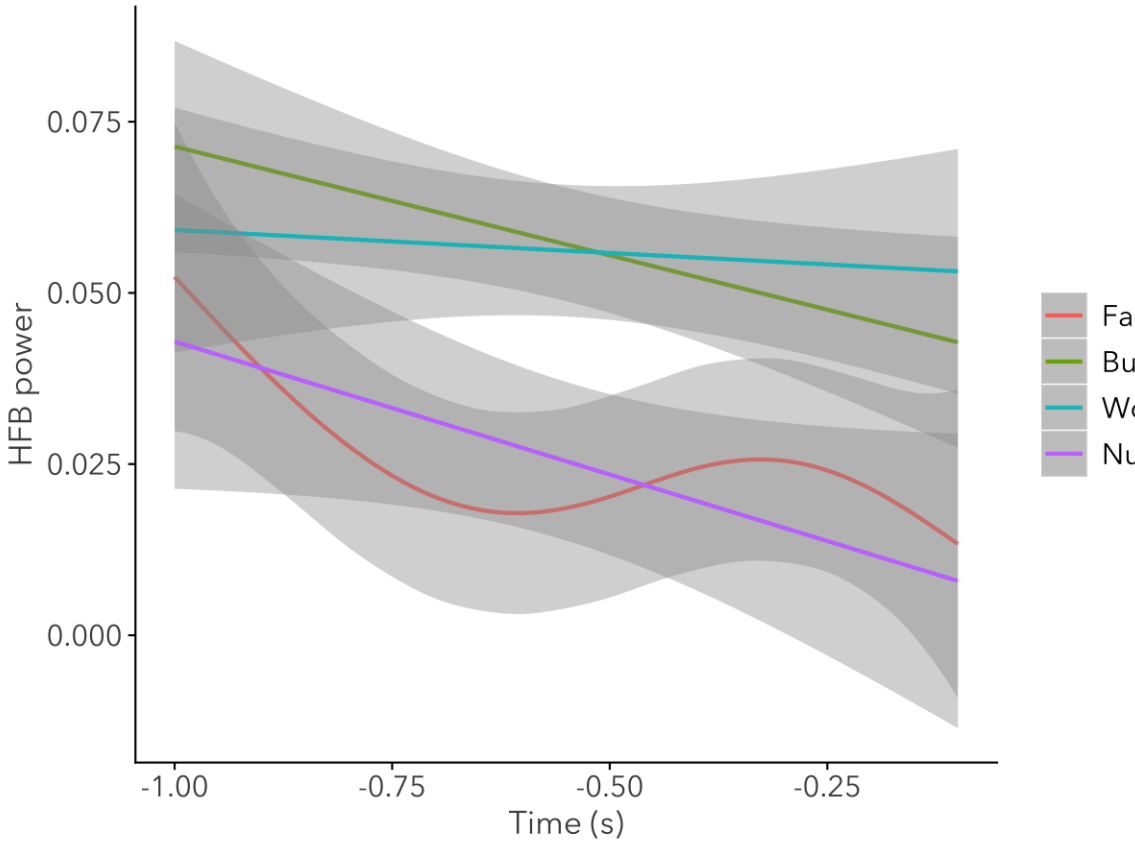
trial_type_combo_alt	estimate	lower.HPD	upper.HPD
assoc hit	0.069	0.004	0.134
item hit	0.003	-0.061	0.069
correct rejection	0.012	-0.049	0.080
miss	-0.009	-0.072	0.056
fa	0.032	-0.032	0.102

Table 96: test resp emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
trial_type_combo_alt	assoc hit - item hit	0	0.066	0.057	0.076
trial_type_combo_alt	assoc hit - correct rejection	0	0.057	0.047	0.066
trial_type_combo_alt	assoc hit - miss	0	0.078	0.064	0.091
trial_type_combo_alt	assoc hit - fa	0	0.037	0.017	0.059
trial_type_combo_alt	item hit - correct rejection	0	-0.009	-0.017	-0.001
trial_type_combo_alt	item hit - miss	0	0.012	0.000	0.023
trial_type_combo_alt	item hit - fa	0	-0.029	-0.050	-0.010
trial_type_combo_alt	correct rejection - miss	0	0.021	0.009	0.033
trial_type_combo_alt	correct rejection - fa	0	-0.019	-0.038	0.001
trial_type_combo_alt	miss - fa	0	-0.041	-0.063	-0.021


Estimated marginal mean
0.3
0.2
0.1
0.0
-0.1
Associative hit
Item hit
Correct rejection


Assoc. hit - CR
Item hit - CR
Assoc. - item hit
0.00
0.03
0.06
Δ HFB power



Item category effects

```
## [1] "adapt_delta: 0.999"
```

```
## [1] "max_treedepth: 20"
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: value ~ 1 + time + cue1_category + cue2_category + trial_type_combo_alt + (1 | subj/chan)
##    Data: . (Number of observations: 29670)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 7)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.09      0.04     0.04     0.20 1.00     1043     1621
##
## ~subj:chan (Number of levels: 30)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.04      0.01     0.03     0.06 1.00     1201     2105
##
## Regression Coefficients:
##                             Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## Intercept                       0.04      0.04    -0.04     0.12 1.01     1076
## time                           -0.04      0.01    -0.05    -0.02 1.00     4785
## cue1_categoryBuilding           0.01      0.01    -0.00     0.02 1.00     4018
```

```
## cue1_categoryWord               0.00      0.01    -0.01     0.02 1.00     4419
## cue1_categoryNumber            -0.02      0.01    -0.03    -0.00 1.00     3560
## cue2_categoryFace               0.03      0.01     0.01     0.04 1.00     3659
## cue2_categoryNumber             0.01      0.01     0.00     0.03 1.00     3921
## cue2_categoryWord              -0.00      0.01    -0.02     0.01 1.00     3679
## trial_type_combo_altitemhit    -0.07      0.01    -0.08    -0.06 1.00     4842
## trial_type_combo_altmiss       -0.08      0.01    -0.10    -0.07 1.00     3877
##                             Tail_ESS
## Intercept                       1250
## time                            2946
## cue1_categoryBuilding           3542
## cue1_categoryWord               3586
## cue1_categoryNumber             3379
## cue2_categoryFace               3245
## cue2_categoryNumber             3361
## cue2_categoryWord               3188
## trial_type_combo_altitemhit     3309
## trial_type_combo_altmiss        2672
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.36      0.00     0.36     0.36 1.00     5934     3055
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
## value ~ 1 + time + cue1_category + cue2_category + trial_type_combo_alt + (1 | subj/chan)
```

```
## value ~ 1 + time + cue1_category + cue2_category + (1 | subj/chan)
```

```
##               elpd_diff se_diff  elpd_loo se_elpd_loo p_loo    se_p_loo
## test.mod.resp      0.0       0.0 -11970.8    126.3        38.6      0.4
## mod.simple      -101.9      14.4 -12072.7    125.7        36.8      0.3
##               looic    se_looic
## test.mod.resp  23941.7    252.7
## mod.simple     24145.4    251.4
```

```
## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'test.mod.resp' is the best model (ELPD =
## -11970.83), followed by 'mod.simple' (diff-ELPD = -101.86 +- 14.36, p < .001)
```

Stimulus cat A

```
## [1] "adapt_delta: 0.8"
```

```
## [1] "max_treedepth: 10"
```

```
## value ~ 1 + time + cue1_category + cue2_category + trial_type_combo_alt + (1 | subj/chan)

## value ~ 1 + time + cue2_category + trial_type_combo_alt + (1 | subj/chan)

##               elpd_diff se_diff  elpd_loo se_elpd_loo p_loo    se_p_loo
## test.mod.resp      0.0       0.0 -11970.8    126.3        38.6      0.4
## mod.simple        -3.8       3.6 -11974.7    126.3        35.8      0.3
##               looic    se_looic
## test.mod.resp  23941.7    252.7
## mod.simple     23949.3    252.6

## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'test.mod.resp' is the best model (ELPD =
## -11970.83), followed by 'mod.simple' (diff-ELPD = -3.83 +- 3.63, p = 0.291)
```

Table 97: test resp emmeans

cue1_category	estimate	lower.HPD	upper.HPD
Face	0.020	-0.051	0.097
Building	0.028	-0.042	0.108
Word	0.024	-0.055	0.096
Number	0.005	-0.075	0.075

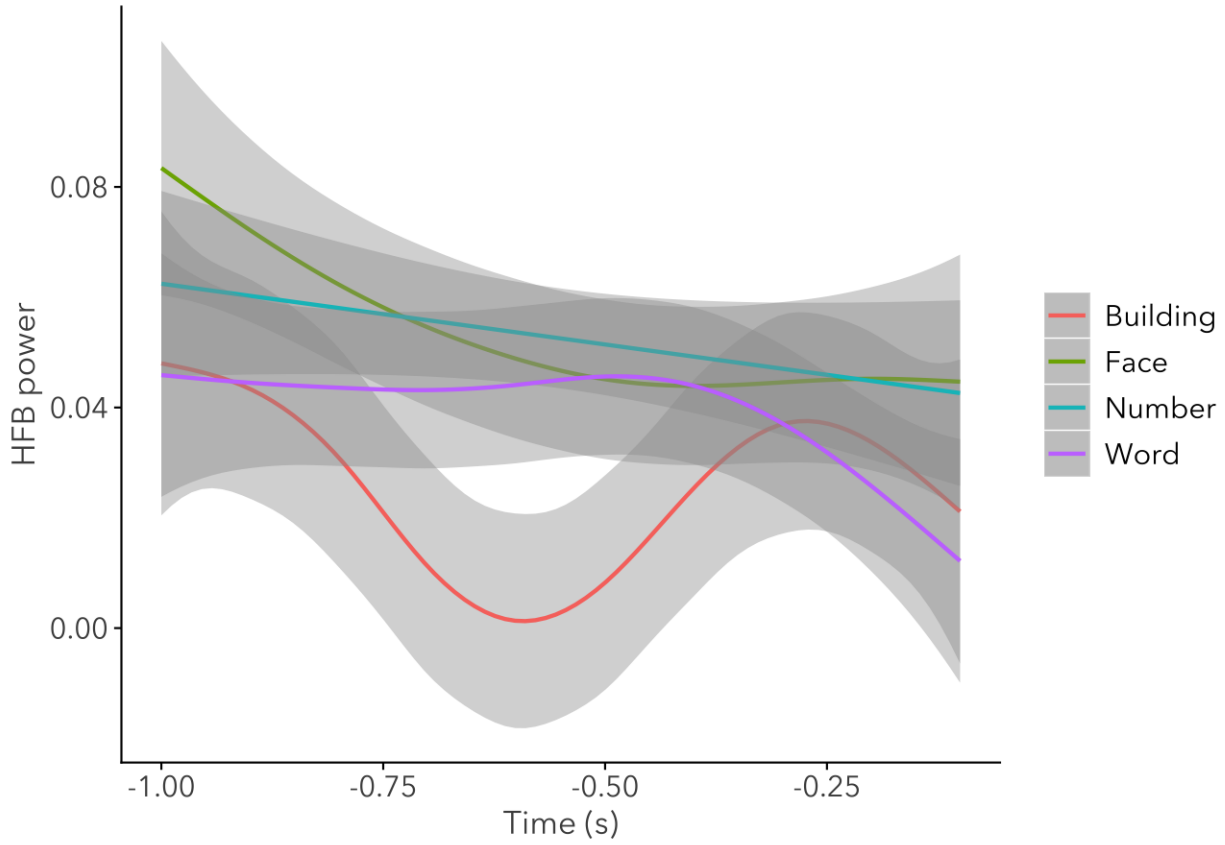
Table 98: test resp emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
cue1_category	Face - Building	0	-0.008	-0.020	0.005
cue1_category	Face - Word	0	-0.005	-0.017	0.008
cue1_category	Face - Number	0	0.015	0.002	0.028
cue1_category	Building - Word	0	0.003	-0.009	0.015
cue1_category	Building - Number	0	0.023	0.010	0.037
cue1_category	Word - Number	0	0.020	0.007	0.033

Table 99: test resp emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
cue1_category	Building - Face	0	0.008	-0.005	0.020
cue1_category	Word - Face	0	0.005	-0.008	0.017
cue1_category	Word - Building	0	-0.003	-0.015	0.009
cue1_category	Number - Face	0	-0.015	-0.028	-0.002
cue1_category	Number - Building	0	-0.023	-0.037	-0.010
cue1_category	Number - Word	0	-0.020	-0.033	-0.007

Stimulus cat B




```
## [1] "adapt_delta: 0.8"
```

```
## [1] "max_treedepth: 10"
```

```
## value ~ 1 + time + cue1_category + cue2_category + trial_type_combo_alt + (1 | subj/chan)
```

```
## value ~ 1 + time + cue1_category + trial_type_combo_alt + (1 | subj/chan)
```

```
##               elpd_diff se_diff  elpd_loo se_elpd_loo p_loo    se_p_loo
## test.mod.resp      0.0       0.0 -11970.8    126.3        38.6      0.4
## mod.simple       -12.1       5.5 -11983.0    126.3        35.7      0.3
##               looic    se_looic
## test.mod.resp  23941.7    252.7
## mod.simple     23965.9    252.6
```

```
## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'test.mod.resp' is the best model (ELPD =
## -11970.83), followed by 'mod.simple' (diff-ELPD = -12.15 +- 5.45, p = 0.026)
```

Table 100: test resp emmeans

cue2_category	estimate	lower.HPD	upper.HPD

Building	0.010	-0.062	0.087
Face	0.037	-0.038	0.112
Number	0.024	-0.053	0.098
Word	0.006	-0.064	0.085

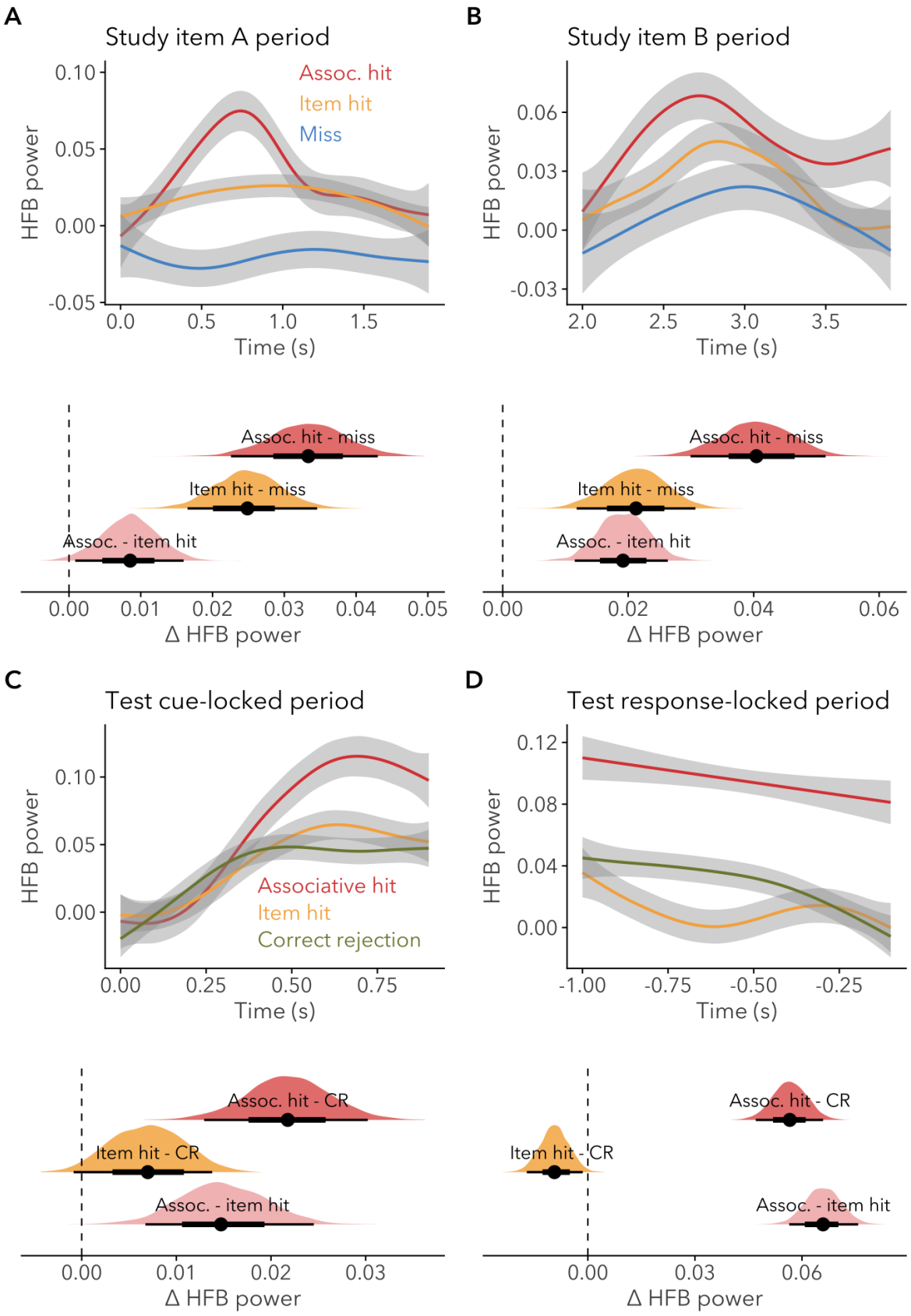
Table 101: test resp emmeans

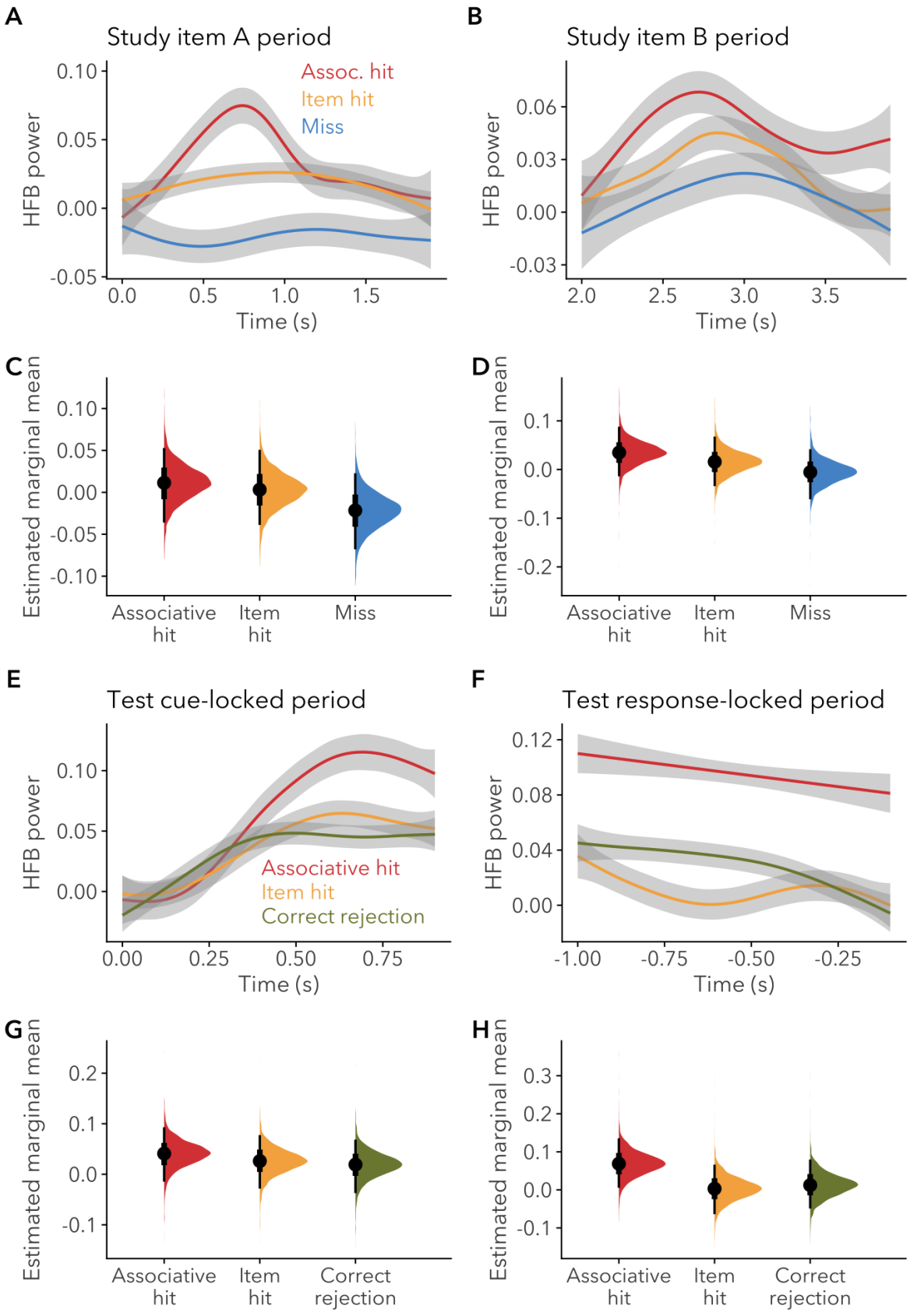
term	contrast	null.value	estimate	lower.HPD	upper.HPD
cue2_category	Building - Face	0	-0.027	-0.039	-0.015
cue2_category	Building - Number	0	-0.014	-0.026	-0.001
cue2_category	Building - Word	0	0.004	-0.008	0.016
cue2_category	Face - Number	0	0.014	0.001	0.026
cue2_category	Face - Word	0	0.031	0.019	0.044
cue2_category	Number - Word	0	0.018	0.006	0.030

Table 102: test resp emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
cue2_category	Face - Building	0	0.027	0.015	0.039
cue2_category	Number - Building	0	0.014	0.001	0.026
cue2_category	Number - Face	0	-0.014	-0.026	-0.001
cue2_category	Word - Building	0	-0.004	-0.016	0.008
cue2_category	Word - Face	0	-0.031	-0.044	-0.019
cue2_category	Word - Number	0	-0.018	-0.030	-0.006

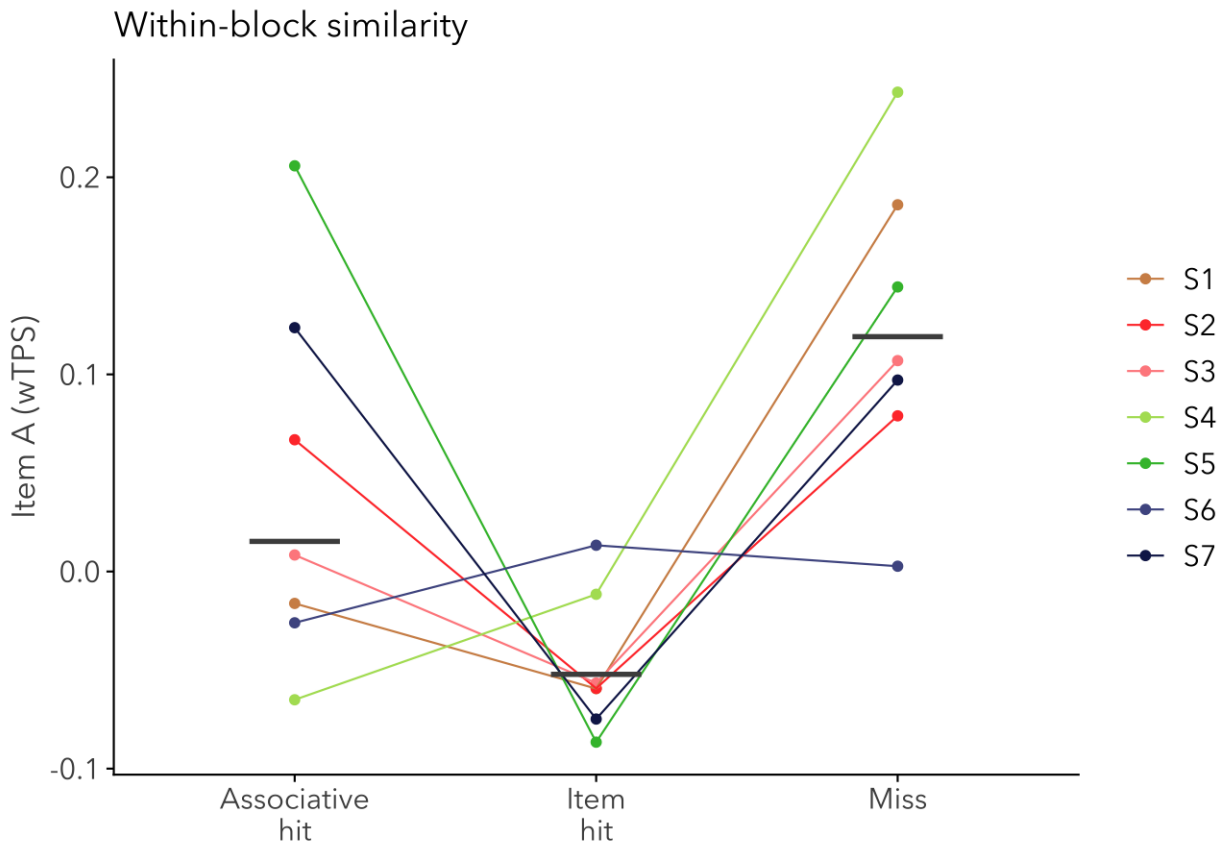
Fig S10


A
Study item A period
Assoc. hit
Item hit
Miss
HFB power
Time (s)
Assoc. hit - miss
Item hit - miss
Assoc. - item hit
Δ HFB power
B
Study item B period
C
Test cue-locked period
Associative hit
Item hit
Correct rejection
Assoc. hit - CR
Item hit - CR
D
Test response-locked period


A
Study item A period
Assoc. hit
Item hit
Miss
HFB power
Time (s)
B
Study item B period
C
Estimated marginal mean
Associative hit
Item hit
Miss
D
E
Test cue-locked period
Associative hit
Item hit
Correct rejection
F
Test response-locked period
G
Correct rejection
H

Pattern similarity

~ Fig S11A




```
## [1] "adapt_delta: 0.8"
```

```
## [1] "max_treedepth: 10"
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: enc_corr_c1 ~ 1 + cue1_category + trial_type_combo_alt + HFB.mean + (1 | subj/chan)
##    Data: . (Number of observations: 2520)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 7)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.02      0.02     0.00     0.09 1.00     2615     2375
##
## ~subj:chan (Number of levels: 30)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.02      0.01     0.00     0.05 1.00     4006     2235
##
## Regression Coefficients:
```

```
##                              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## Intercept                        0.00      0.05    -0.10     0.10 1.00     4785
## cue1_categoryBuilding            0.10      0.05    -0.01     0.20 1.00     5592
## cue1_categoryWord               -0.04      0.05    -0.14     0.07 1.00     4884
## cue1_categoryNumber              0.01      0.06    -0.10     0.12 1.00     4873
## trial_type_combo_altitemhit     -0.08      0.04    -0.17     0.01 1.00     5800
## trial_type_combo_altmiss         0.11      0.06    -0.01     0.23 1.00     4776
## HFB.mean                         0.04      0.08    -0.12     0.19 1.00     8986
##                              Tail_ESS
## Intercept                        2914
## cue1_categoryBuilding            3499
## cue1_categoryWord                3329
## cue1_categoryNumber              3244
## trial_type_combo_altitemhit      2829
## trial_type_combo_altmiss         3039
## HFB.mean                         2804
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.93      0.01     0.91     0.96 1.00     9580     2867
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
## [1] "adapt_delta: 0.8"
```

```
## [1] "max_treedepth: 10"
```

```
## enc_corr_c1 ~ 1 + cue1_category + trial_type_combo_alt + HFB.mean + (1 | subj/chan)
```

```
## enc_corr_c1 ~ 1 + cue1_category + HFB.mean + (1 | subj/chan)
```

```
##            elpd_diff se_diff elpd_loo se_elpd_loo p_loo   se_p_loo looic
## wTPS.A.mod     0.0       0.0 -3405.5     29.7         9.5     0.2   6810.9
## mod.simple    -4.4       3.5 -3409.8     29.7         7.6     0.2   6819.7
##            se_looic
## wTPS.A.mod    59.4
## mod.simple    59.5
```

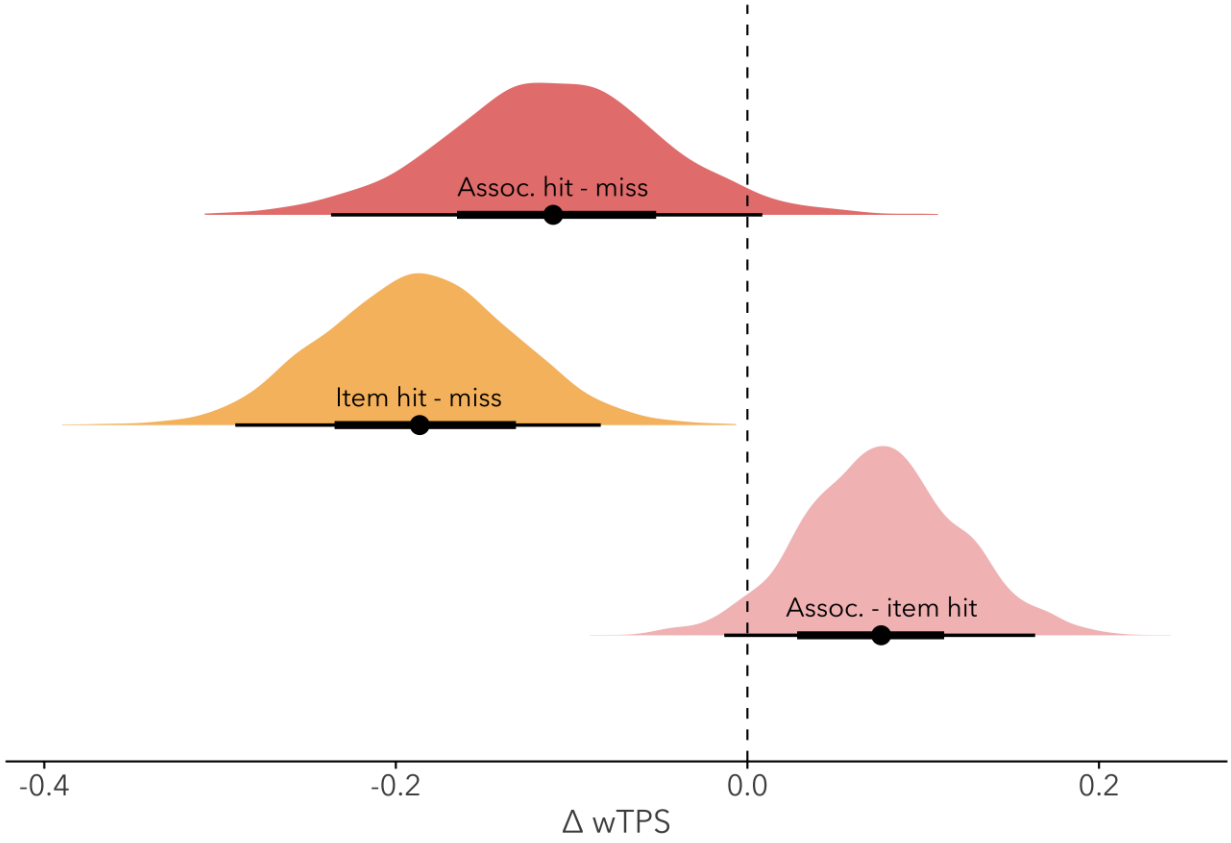
```
## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'wTPS.A.mod' is the best model (ELPD =
## -3405.47), followed by 'mod.simple' (diff-ELPD = -4.36 +- 3.49, p = 0.211)
```

Table 103: study A emmeans

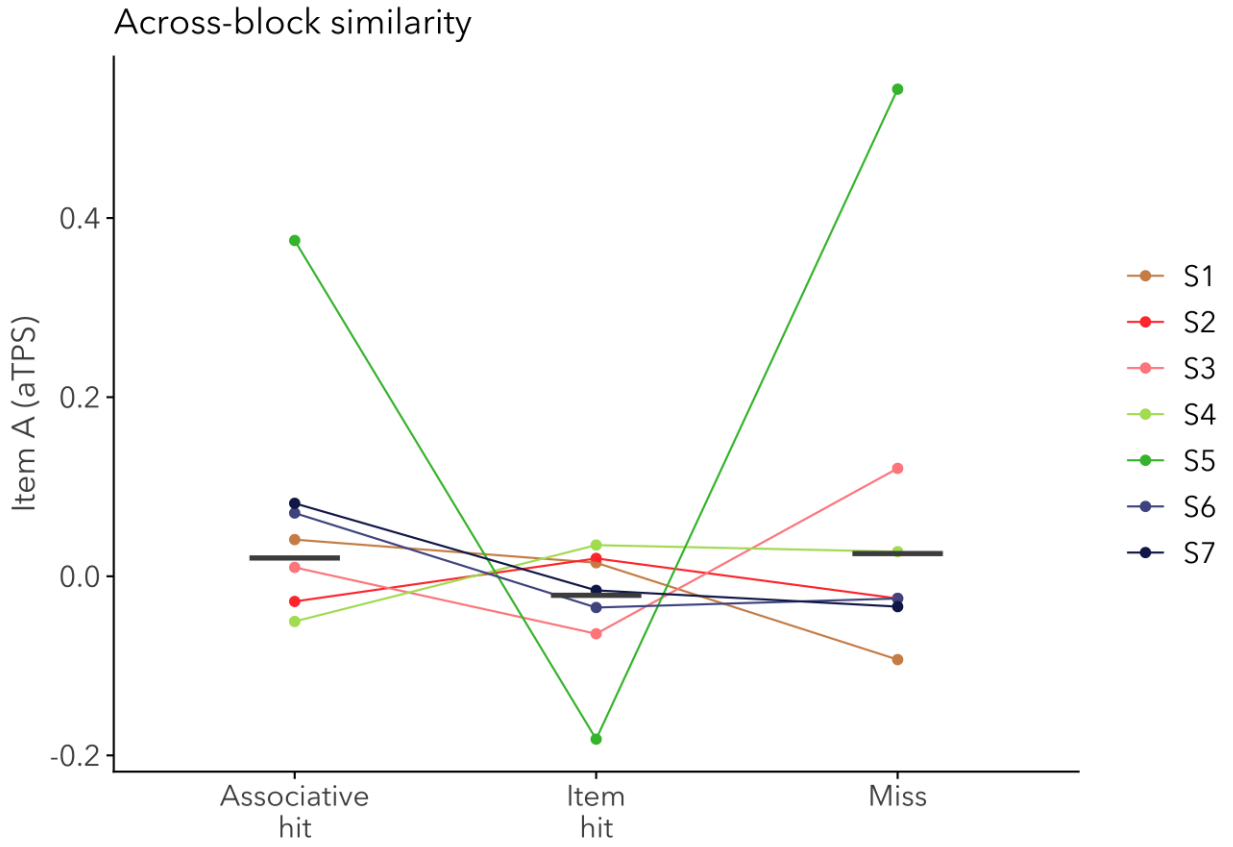
trial_type_combo_alt	estimate	lower.HPD	upper.HPD
assoc hit	0.019	-0.062	0.089
item hit	-0.058	-0.117	-0.001
miss	0.129	0.035	0.224

Table 104: study A emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
trial_type_combo_alt	assoc hit - item hit	0	0.076	-0.013	0.164
trial_type_combo_alt	assoc hit - miss	0	-0.110	-0.234	0.011
trial_type_combo_alt	item hit - miss	0	-0.186	-0.292	-0.084



~ **Fig S11B**




```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: enc_corr_c1_future ~ 1 + cue1_category + trial_type_combo_alt + HFB.mean + (1 | subj/chan)
##    Data: . (Number of observations: 2526)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 7)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.02      0.02     0.00     0.08 1.00     2174     1928
##
## ~subj:chan (Number of levels: 30)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.02      0.01     0.00     0.05 1.00     3268     1878
##
## Regression Coefficients:
##                           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## Intercept                    -0.04      0.05    -0.14     0.05 1.00     3361
```

```
## cue1_categoryBuilding           0.04      0.05    -0.06     0.14 1.00     3792
## cue1_categoryWord               0.12      0.05     0.01     0.22 1.00     3634
## cue1_categoryNumber             0.14      0.06     0.02     0.25 1.00     3551
## trial_type_combo_altitemhit    -0.04      0.04    -0.12     0.05 1.00     4188
## trial_type_combo_altmiss       -0.04      0.06    -0.16     0.08 1.00     3653
## HFB.mean                        0.02      0.08    -0.12     0.17 1.00     4638
##                             Tail_ESS
## Intercept                       3159
## cue1_categoryBuilding           3413
## cue1_categoryWord               3092
## cue1_categoryNumber             2599
## trial_type_combo_altitemhit     3026
## trial_type_combo_altmiss        3209
## HFB.mean                        3091
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.93      0.01     0.91     0.96 1.00     5904     3028
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
## [1] "adapt_delta: 0.8"
```

```
## [1] "max_treedepth: 10"
```

```
## enc_corr_c1_future ~ 1 + cue1_category + trial_type_combo_alt + HFB.mean + (1 | subj/chan)
```

```
## enc_corr_c1_future ~ 1 + cue1_category + HFB.mean + (1 | subj/chan)
```

```
##            elpd_diff se_diff elpd_loo se_elpd_loo p_loo   se_p_loo looic
## mod.simple     0.0       0.0 -3414.5     28.8         7.8     0.2   6829.0
## aTPS.A.mod    -1.4       0.9 -3415.9     28.8         9.6     0.2   6831.8
##            se_looic
## mod.simple    57.7
## aTPS.A.mod    57.6
```

```
## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'mod.simple' is the best model (ELPD =
## -3414.49), followed by 'aTPS.A.mod' (diff-ELPD = -1.43 +- 0.89, p = 0.107)
```

Table 105: study A emmeans

trial_type_combo_alt	estimate	lower.HPD	upper.HPD
assoc hit	0.030	-0.041	0.113
item hit	-0.007	-0.064	0.051
miss	-0.011	-0.101	0.087

Table 106: study A emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
trial_type_combo_alt	assoc hit - item hit	0	0.037	-0.044	0.126
trial_type_combo_alt	assoc hit - miss	0	0.040	-0.078	0.155
trial_type_combo_alt	item hit - miss	0	0.004	-0.101	0.107

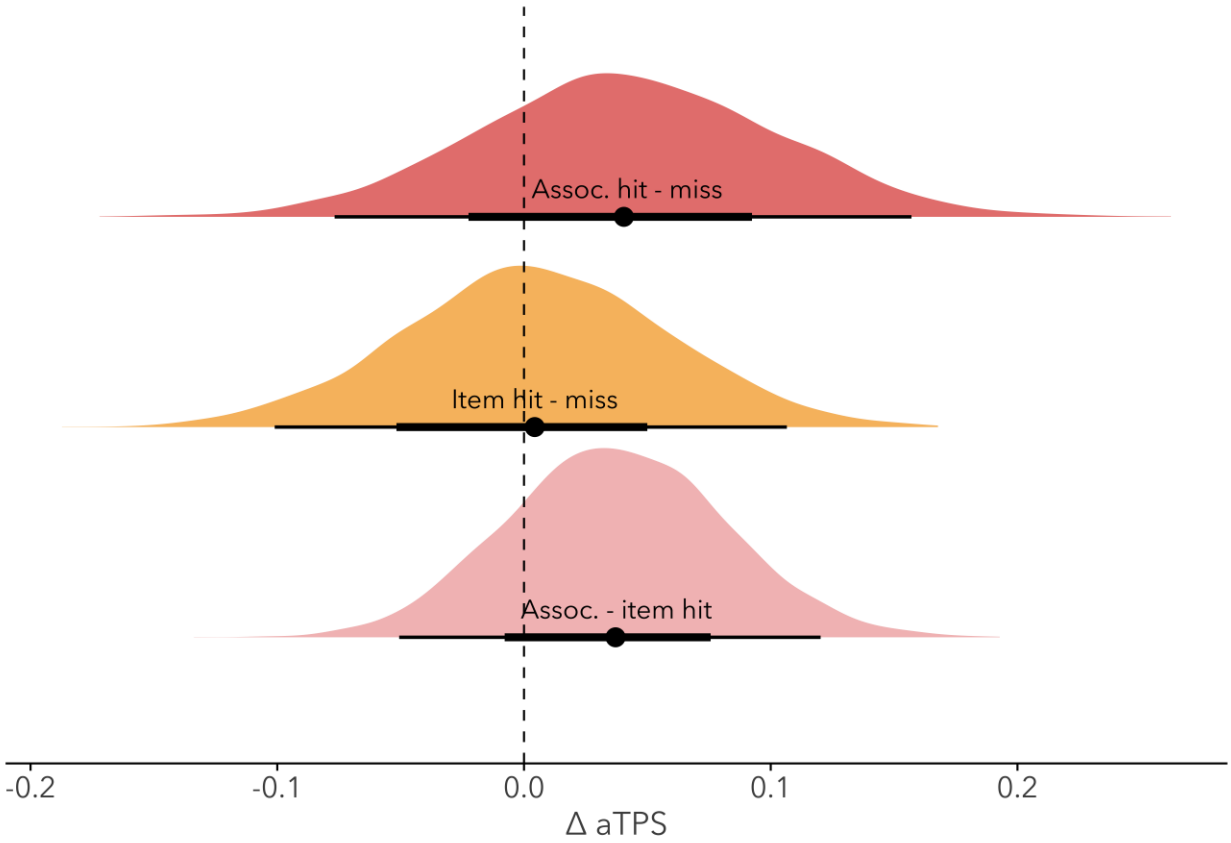
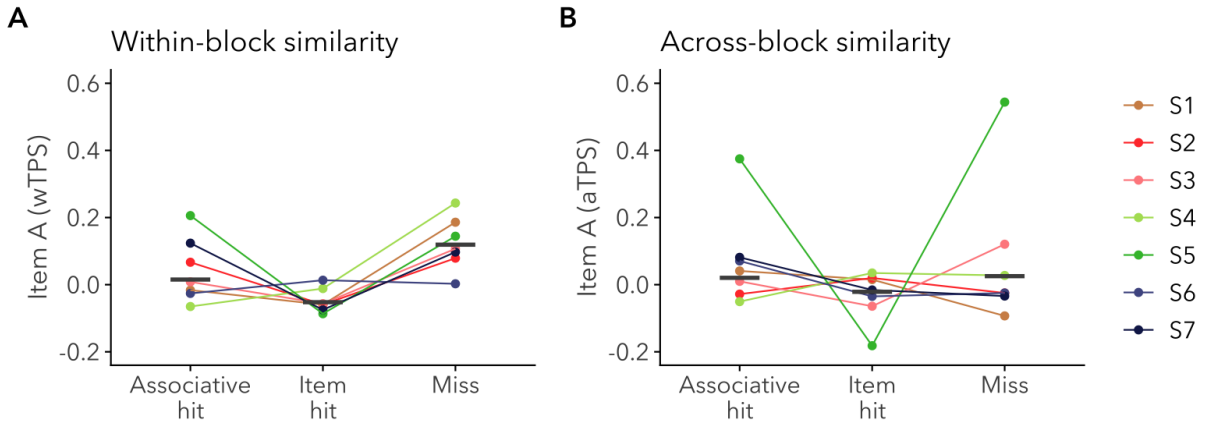
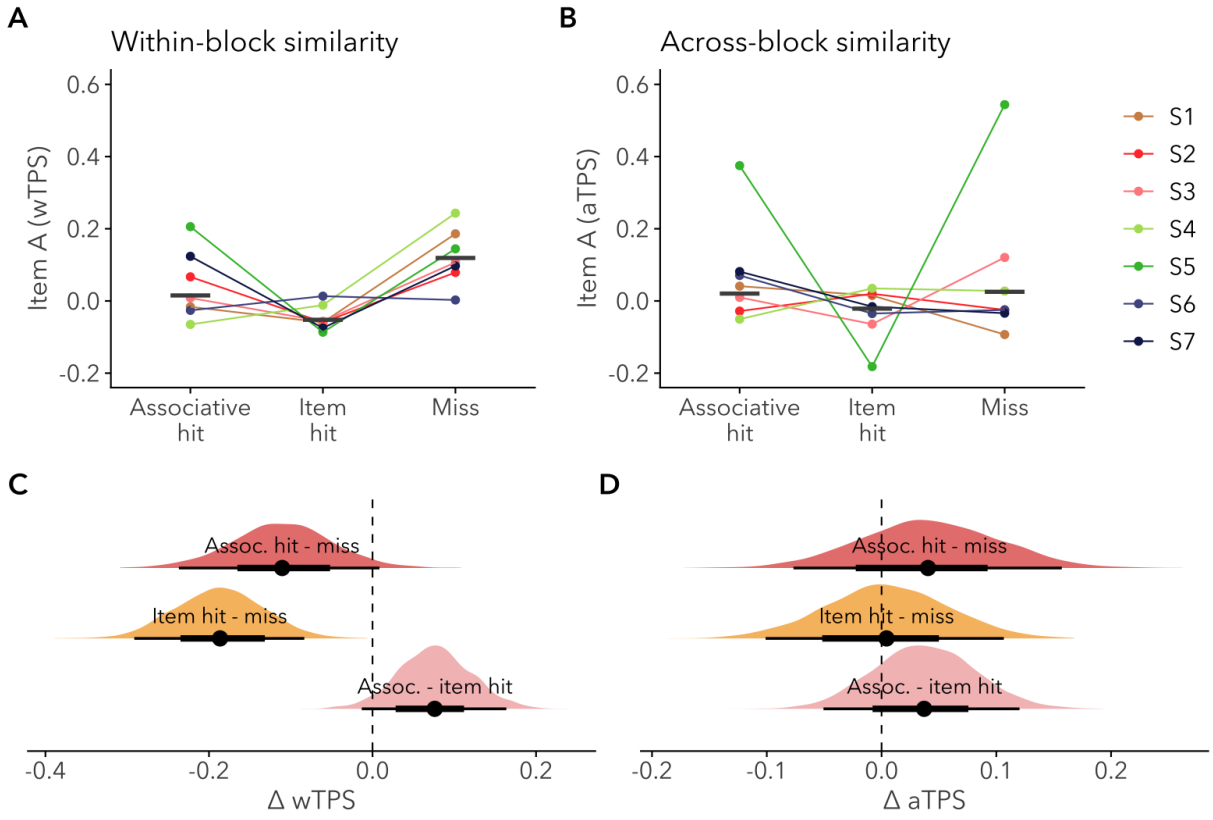


Fig S11





Interaction: TPS x memory outcome

```
## [1] "adapt_delta: 0.99"

## [1] "max_treedepth: 10"

##  Family: gaussian
##   Links: mu = identity
## Formula: similarity ~ type + trial_type_combo_alt + type:trial_type_combo_alt + cue1_category + HFB.r
##    Data: . (Number of observations: 5046)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 7)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.02      0.02     0.00     0.06 1.00     2652     2516
##
## ~subj:chan (Number of levels: 30)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.01      0.01     0.00     0.03 1.00     3561     2105
##
## Regression Coefficients:
##                                          Estimate Est.Error l-95% CI u-95% CI
## Intercept                                   -0.03      0.06    -0.14     0.08
```

```
## typewithin                                       0.09      0.06    -0.02     0.21
## trial_type_combo_altassoc\nhit                   0.01      0.06    -0.10     0.13
## trial_type_combo_altitemhit                     -0.03      0.05    -0.13     0.07
## cue1_categoryBuilding                            0.07      0.04    -0.00     0.14
## cue1_categoryWord                                0.04      0.04    -0.03     0.11
## cue1_categoryNumber                              0.07      0.04    -0.01     0.15
## HFB.mean                                         0.03      0.05    -0.08     0.14
## typewithin:trial_type_combo_altassoc\nhit       -0.10      0.08    -0.25     0.05
## typewithin:trial_type_combo_altitemhit          -0.12      0.07    -0.27     0.01
##                                              Rhat Bulk_ESS Tail_ESS
## Intercept                                    1.00     2500     2895
## typewithin                                   1.00     2184     2836
## trial_type_combo_altassoc\nhit               1.00     2602     3391
## trial_type_combo_altitemhit                  1.00     2454     2925
## cue1_categoryBuilding                        1.00     4481     3073
## cue1_categoryWord                            1.00     4269     3259
## cue1_categoryNumber                          1.00     3766     2880
## HFB.mean                                     1.00     7922     2773
## typewithin:trial_type_combo_altassoc\nhit    1.00     2364     2786
## typewithin:trial_type_combo_altitemhit       1.00     2348     2857
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.93      0.01     0.92     0.95 1.00     8146     2918
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

effect	component	term	estimate	std.error	conf.low	conf.high
fixed	cond	(Intercept)	-0.033	0.055	-0.142	0.077
fixed	cond	typewithin	0.093	0.060	-0.023	0.211
fixed	cond	trial_type_combo_altassoc hit	| 0.01	| 0.05	| -0.09	| 0.13
fixed	cond	trial_type_combo_altitemhit	-0.030	0.052	-0.133	0.071
fixed	cond	cue1_categoryBuilding	0.069	0.038	-0.005	0.142
fixed	cond	cue1_categoryWord	0.041	0.038	-0.034	0.115
fixed	cond	cue1_categoryNumber	0.073	0.041	-0.009	0.153
fixed	cond	HFB.mean	0.031	0.053	-0.077	0.137
fixed	cond	typewithin:trial_type_combo_altassoc hit	| -0.09	| 0.07	| -0.24	| 0.05
fixed	cond	typewithin:trial_type_combo_altitemhit	-0.124	0.071	-0.267	0.015

```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
## similarity ~ type + trial_type_combo_alt + type:trial_type_combo_alt + cue1_category + HFB.mean + (1
```

```
## similarity ~ type + trial_type_combo_alt + cue1_category + HFB.mean + (1 | subj/chan)
```

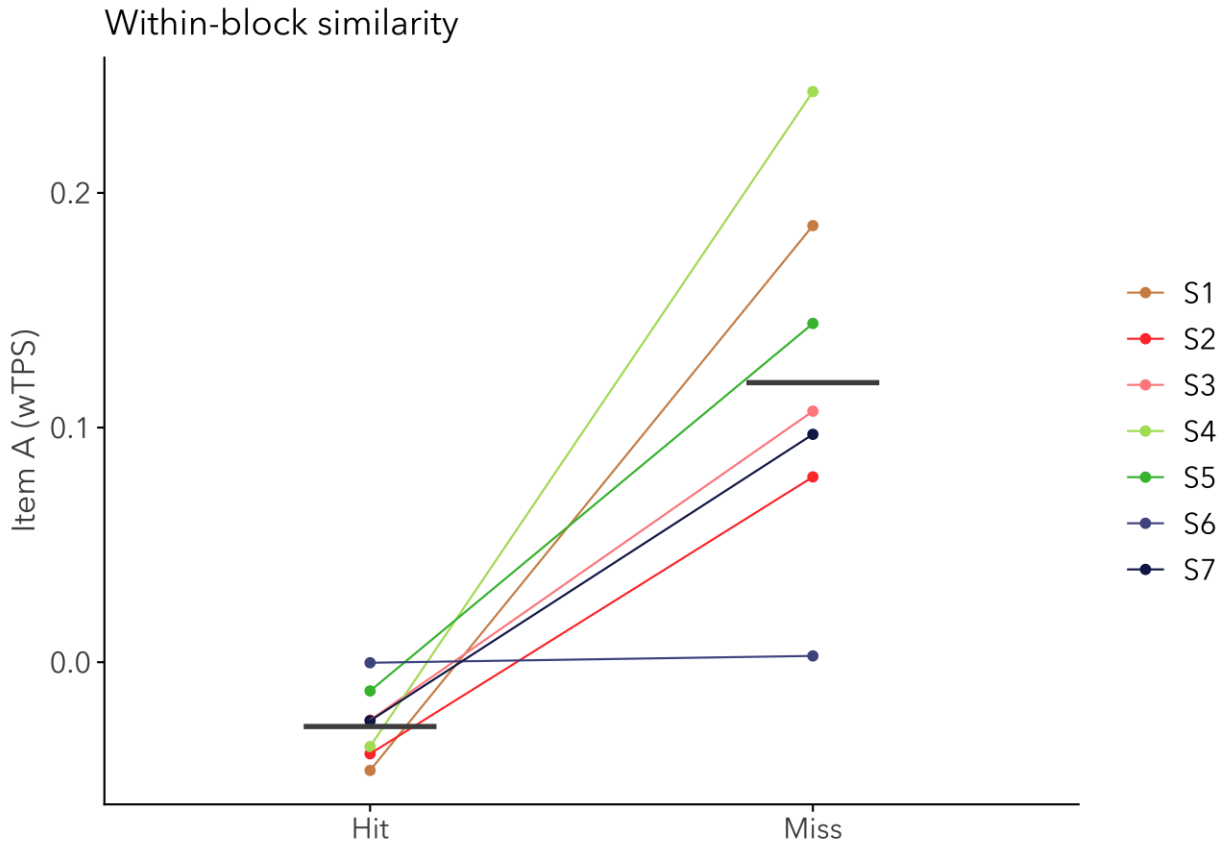
```
##             elpd_diff se_diff elpd_loo se_elpd_loo p_loo   se_p_loo looic
## mod.simple     0.0       0.0 -6819.1     41.4        10.5     0.2  13638.3
```

```
## mod           -0.4       1.8 -6819.6     41.4        12.5     0.2  13639.1
##            se_looic
## mod.simple    82.8
## mod           82.7

## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'mod.simple' is the best model (ELPD =
## -6819.14), followed by 'mod' (diff-ELPD = -0.41 +- 1.77, p = 0.814)

##  trial_type_combo_alt_pairwise type_pairwise   estimate lower.HPD upper.HPD
##  miss - assoc\nhit             across - within  -0.0986    -0.254    0.0472
##  miss - item hit               across - within  -0.1246    -0.271    0.0091
##  assoc\nhit - item hit         across - within  -0.0256    -0.144    0.0996
##
## Results are averaged over the levels of: cue1_category
## Point estimate displayed: median
## HPD interval probability: 0.95
```

~ **Fig S12A**




```
## [1] "adapt_delta: 0.99"

## [1] "max_treedepth: 10"
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: enc_corr_c1 ~ 1 + cue1_category + trial_type_agg + HFB.mean + (1 | subj/chan)
##    Data: . (Number of observations: 2520)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 7)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.03      0.02     0.00     0.09 1.00     2322     2074
##
## ~subj:chan (Number of levels: 30)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.02      0.01     0.00     0.05 1.00     3341     1820
##
## Regression Coefficients:
##                       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## Intercept                -0.05      0.04    -0.12     0.02 1.00     3122
## cue1_categoryBuilding     0.10      0.05    -0.00     0.20 1.00     3872
## cue1_categoryWord        -0.02      0.05    -0.12     0.08 1.00     3480
## cue1_categoryNumber       0.00      0.06    -0.10     0.11 1.00     2607
## trial_type_aggmiss        0.16      0.05     0.06     0.27 1.00     4700
## HFB.mean                  0.04      0.08    -0.11     0.20 1.00     7486
##                       Tail_ESS
## Intercept                 3029
## cue1_categoryBuilding     3290
## cue1_categoryWord         2971
## cue1_categoryNumber       3087
## trial_type_aggmiss        3243
## HFB.mean                  2967
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.93      0.01     0.91     0.96 1.00     8063     2783
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).

## [1] "adapt_delta: 0.99"

## [1] "max_treedepth: 10"

## enc_corr_c1 ~ 1 + cue1_category + trial_type_agg + HFB.mean + (1 | subj/chan)

## enc_corr_c1 ~ 1 + cue1_category + HFB.mean + (1 | subj/chan)

##            elpd_diff se_diff elpd_loo se_elpd_loo p_loo   se_p_loo looic
## wTPS.A.mod     0.0       0.0 -3406.0     29.7         8.6     0.2   6812.0
## mod.simple    -3.9       3.0 -3409.9     29.7         7.7     0.2   6819.8
##            se_looic
## wTPS.A.mod    59.3
## mod.simple    59.5
```

```
## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'wTPS.A.mod' is the best model (ELPD =
## -3406.00), followed by 'mod.simple' (diff-ELPD = -3.92 +- 3.03, p = 0.195)
```

Table 108: study A emmeans

trial_type_agg	estimate	lower.HPD	upper.HPD
hit	-0.030	-0.082	0.021
miss	0.133	0.038	0.227

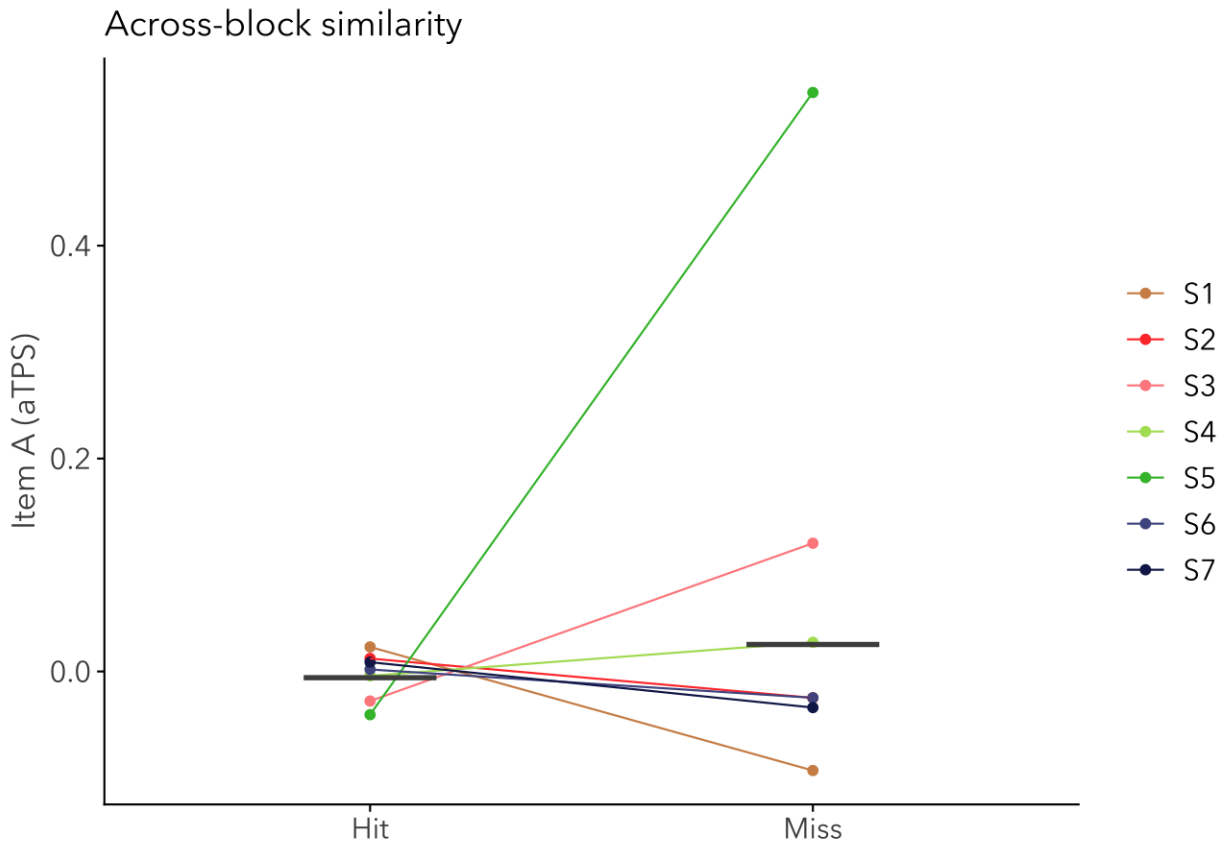
Table 109: study A emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
trial_type_agg	hit - miss	0	-0.163	-0.266	-0.065

Table 110: study A emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
trial_type_agg	miss - hit	0	0.163	0.065	0.266

~ **Fig S12B**



```
## [1] "adapt_delta: 0.8"
```

```
## [1] "max_treedepth: 10"
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: enc_corr_c1_future ~ 1 + cue1_category + trial_type_agg + HFB.mean + (1 | subj/chan)
##    Data: . (Number of observations: 2526)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 7)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.02      0.02     0.00     0.08 1.00     2273     2171
##
## ~subj:chan (Number of levels: 30)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.02      0.01     0.00     0.05 1.00     3587     2204
##
## Regression Coefficients:
##                       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## Intercept                -0.07      0.04    -0.15     0.00 1.00     3851
## cue1_categoryBuilding     0.04      0.05    -0.06     0.14 1.00     4844
## cue1_categoryWord         0.13      0.05     0.03     0.23 1.00     4393
## cue1_categoryNumber       0.13      0.06     0.02     0.24 1.00     3921
## trial_type_aggmiss       -0.02      0.05    -0.12     0.08 1.00     5124
## HFB.mean                  0.03      0.08    -0.12     0.18 1.00     8223
##                       Tail_ESS
## Intercept                 2731
## cue1_categoryBuilding     3431
## cue1_categoryWord         3334
## cue1_categoryNumber       3286
## trial_type_aggmiss        2670
## HFB.mean                  2889
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.93      0.01     0.91     0.96 1.00     6771     2735
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
## [1] "adapt_delta: 0.8"
```

```
## [1] "max_treedepth: 10"
```

```
## enc_corr_c1_future ~ 1 + cue1_category + trial_type_agg + HFB.mean + (1 | subj/chan)
```

```
## enc_corr_c1_future ~ 1 + cue1_category + HFB.mean + (1 | subj/chan)
```

```
##             elpd_diff se_diff elpd_loo se_elpd_loo p_loo   se_p_loo looic
## mod.simple     0.0       0.0 -3414.5     28.8         7.8     0.2   6829.0
## aTPS.A.mod    -0.8       0.3 -3415.3     28.8         8.7     0.2   6830.7
##             se_looic
## mod.simple    57.7
## aTPS.A.mod    57.7

## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'mod.simple' is the best model (ELPD =
## -3414.49), followed by 'aTPS.A.mod' (diff-ELPD = -0.84 +- 0.33, p = 0.011)
```

Table 111: study A emmeans

trial_type_agg	estimate	lower.HPD	upper.HPD
hit	0.006	-0.042	0.059
miss	-0.010	-0.102	0.082

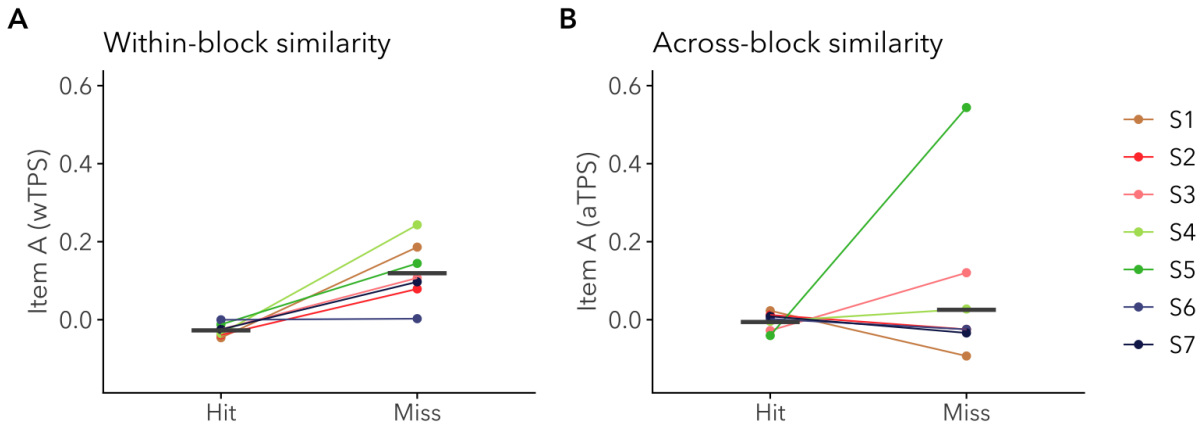
Table 112: study A emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
trial_type_agg	hit - miss	0	0.017	-0.082	0.12

Table 113: study A emmeans

term	contrast	null.value	estimate	lower.HPD	upper.HPD
trial_type_agg	miss - hit	0	-0.017	-0.12	0.082

Fig S12



Interaction: TPS x memory outcome

```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: similarity ~ type + trial_type_agg + type:trial_type_agg + cue1_category + HFB.mean + (1 | s
##    Data: . (Number of observations: 5046)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~subj (Number of levels: 7)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.02      0.02     0.00     0.06 1.00     2438     2610
##
## ~subj:chan (Number of levels: 30)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.01      0.01     0.00     0.03 1.00     3720     2418
##
## Regression Coefficients:
##                               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## Intercept                        -0.04      0.05    -0.14     0.07 1.00     2866
## typewithin                        0.09      0.06    -0.03     0.21 1.00     2717
## trial_type_aggHit                -0.02      0.05    -0.11     0.08 1.00     3077
## cue1_categoryBuilding             0.07      0.04    -0.00     0.14 1.00     4012
## cue1_categoryWord                 0.05      0.04    -0.02     0.13 1.00     3968
## cue1_categoryNumber               0.07      0.04    -0.01     0.15 1.00     3712
## HFB.mean                          0.03      0.05    -0.07     0.14 1.00     5749
## typewithin:trial_type_aggHit     -0.12      0.07    -0.25     0.02 1.00     2604
##                               Tail_ESS
## Intercept                         2724
## typewithin                        2992
## trial_type_aggHit                 3089
## cue1_categoryBuilding             3478
## cue1_categoryWord                 3288
## cue1_categoryNumber               3211
## HFB.mean                          3158
## typewithin:trial_type_aggHit      2661
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.93      0.01     0.92     0.95 1.01     7125     2721
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

effect	component	term	estimate	std.error	conf.low	conf.high
fixed	cond	(Intercept)	-0.035	0.054	-0.142	0.067

fixed	cond	typewithin	0.094	0.061	-0.028	0.212
fixed	cond	trial_type_aggHit	-0.015	0.051	-0.112	0.084
fixed	cond	cue1_categoryBuilding	0.072	0.037	-0.001	0.144
fixed	cond	cue1_categoryWord	0.053	0.037	-0.021	0.128
fixed	cond	cue1_categoryNumber	0.069	0.041	-0.012	0.149
fixed	cond	HFB.mean	0.034	0.054	-0.072	0.144
fixed	cond	typewithin:trial_type_aggHit	-0.115	0.068	-0.248	0.016

```
## [1] "adapt_delta: 0.99"
```

```
## [1] "max_treedepth: 10"
```

```
## similarity ~ type + trial_type_agg + type:trial_type_agg + cue1_category + HFB.mean + (1 | subj/chan
```

```
## similarity ~ type + trial_type_agg + cue1_category + HFB.mean + (1 | subj/chan)
```

```
##            elpd_diff se_diff elpd_loo se_elpd_loo p_loo   se_p_loo looic
## mod            0.0       0.0 -6819.6     41.4        10.8     0.2  13639.2
## mod.simple    -0.4       1.7 -6820.0     41.4         9.7     0.2  13639.9
##            se_looic
## mod           82.7
## mod.simple    82.8
```

```
## The difference in predictive accuracy, as indexed by Expected Log Predictive
## Density (ELPD-LOO), suggests that 'mod' is the best model (ELPD = -6819.60),
## followed by 'mod.simple' (diff-ELPD = -0.37 +- 1.72, p = 0.831)
```

```
##  trial_type_agg_pairwise type_pairwise   estimate lower.HPD upper.HPD
##  Miss - Hit              across - within   -0.116    -0.249    0.0141
##
## Results are averaged over the levels of: cue1_category
## Point estimate displayed: median
## HPD interval probability: 0.95
```